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Prompt-engineering-based
Hallucination Estimation for large
language models

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Abstract

Large Language Models (LLMs) have achieved remarkable performance across a wide range of natural language processing tasks. However, LLMs often generate factually incorrect or fabricated information, a phenomenon commonly referred to as hallucination. While existing research has primarily focused on hallucination detection, effective mitigation strategies—particularly in open-domain question answering settings where explicit grounding is limited—remain challenging.

This project presents a multi-signal framework for hallucination detection and estimation, coupled with detector-guided mitigation strategies, including re-prompting, candidate filtering, and abstention. The proposed system integrates multiple complementary signals, including LLM-as-a-judge evaluation, natural language inference (NLI) for factual consistency, and SelfCheck-based response consistency, to produce a unified hallucination score. Based on this score, a detector-conditioned re-prompting strategy is applied to selectively revise responses or abstain when confidence is low. The framework is evaluated on multiple benchmark datasets, including TruthfulQA, FEVER, and Natural Questions, using both performance and stability metrics.

Experimental results show that the approach is most effective on evidence-grounded datasets such as FEVER, while gains on more open-ended benchmarks are limited. In particular, notable reductions in hallucination rates are observed on evidence-based datasets such as FEVER. However, performance improvements are less consistent on open-ended datasets, highlighting the challenges of mitigating hallucinations in the absence of explicit grounding.

Overall, this work highlights the importance of integrating multiple evaluation signals with adaptive mitigation strategies and shows that the effectiveness of hallucination mitigation is strongly dependent on the availability of grounding evidence.

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Chapter 1

Introduction

1.1 Background

In recent years, large language models (LLMs) have demonstrated outstanding performance in various natural language processing tasks, including question answering, text summarization, language inference, knowledge retrieval, and dialogue generation [1]. These models are capable of generating fluent and context-coherent responses, thus finding widespread application in research and practical applications, such as clinical decision support systems [2]. Despite these advances, LLMs are known to generate factually incorrect or fabricated information that is not supported by the input or external knowledge, a phenomenon often referred to as “hallucination” [3]. These hallucinations can occur even when model outputs appear linguistically plausible, making them difficult to detect and potentially misleading to users. As a result, they can degrade system performance and fail to meet user expectations [4].

1.2 Motivation

Hallucinations in large language models pose a serious challenge to their reliability and practical deployment. In real-world applications, users often perceive model outputs as reliable sources of information, making hallucinated outputs highly misleading in real-world use. These errors can lead not only to distorted information dissemination, decreased user trust, and reduced system performance, but also to erroneous task execution in LLM-based agent systems, further impacting overall system reliability [5]. In high-risk applications such as medical decision support, while LLM-assisted tools can effectively support goal elicitation, clarification, and revision, the lack of human interaction, personalized patient contextual information, and the significant risk of overload due to unconstrained information access pose significant challenges [6]. These issues can amplify uncertainty and potential harm, limiting their safe application as independent asynchronous tools in real-world scenarios.

The causes of hallucinations are multifaceted, including both data-related and model-related factors. Training data may contain quality issues, including noise, bias, outdated information, subjective content, and unverified facts, leading to inaccurate or unsubstantiated model outputs [7]. Furthermore, LLMs are often poorly calibrated and prone to overconfidence, assigning excessively high confidence levels to erroneous outputs [8]. In open-domain environments, this problem is further exacerbated by a lack of reliable underlying information, allowing models to generate seemingly plausible but actually fabricated explanations [9]. These challenges collectively underscore the need for more robust methods to reliably detect, estimate and mitigate hallucinations.

Despite the growing interest in hallucination detection, existing methods remain limited in several aspects. Many hallucination evaluation methods rely on single-model or single-signal strategies, such as confidence estimation, natural language inference (NLI), or response consistency checks [4]. While these approaches provide advantages in handling lexical variations or detecting simple inconsistencies, they are prone to error propagation from underlying models, struggle to pinpoint hallucinated substrings at a fine-grained level, and often fail to fully capture the diversity (e.g., intrinsic vs. extrinsic) and complexity of hallucinations [4]. Therefore, these methods are often unstable and lack robustness across different tasks and datasets.

Furthermore, while detection methods have been extensively studied, effective mitigation strategies remain relatively scarce. Existing mitigation methods are often heuristic or poorly integrated with detection mechanisms, limiting their effectiveness in reducing hallucinatory outputs. This challenge is particularly pronounced in knowledge-intensive task environments such as open-domain question answering, because reliable contextual information is limited and pre-trained language models struggle to access and manipulate knowledge accurately, are prone to hallucinations, and fail to provide reliable sources for predictions [9]. These limitations underscore the need to develop a more comprehensive and integrated framework that combines detection, estimation, and mitigation measures to improve the factual reliability of LLM-generated outputs.

1.3 Research Gap

Despite recent progress in hallucination detection and mitigation, several key limitations remain. Firstly, many existing hallucination detection methods tend to rely on a single signal, such as model uncertainty estimation, Natural Language Entailment (NLI), or consistency among sampled responses. This dependence on a single source of information limits their robustness across different tasks, models, and datasets [10]. Secondly, research on hallucination detection metrics and mitigation strategies has largely proceeded in parallel. The decoupling of these two processes results in a lack of cohesive intervention strategies, where detection results are not effectively integrated into the generation loop to facilitate real-time mitigation [4]. Thirdly, existing hallucination evaluation metrics and mitigation methods often exhibit limited cross-dataset generalizability. While performing well on specific benchmarks, these approaches struggle to transfer effectively to other datasets or unseen domains [4].

These limitations highlight the need for more integrated and robust approaches. In particular, there is a need for multi-signal frameworks that not only improve the reliability of hallucination detection but also enable effective, detector-guided mitigation. To address this gap, this project proposes a unified framework that integrates multi-signal evaluation with detector-conditioned intervention strategies, including re-prompting, candidate filtering, and abstention mechanisms.

1.4 Aims and Objectives

This project aims to develop a unified and robust framework for hallucination detection, estimation, and mitigation in large language models (LLMs). Specifically, it addresses the limitations of existing single-signal methods and investigates how the effectiveness of hallucination mitigation varies across tasks with different levels of grounding availability. To achieve this goal, the project follows these objectives:

- To design a multi-signal hallucination detection mechanism that integrates complementary evaluation signals, including:
 1. an LLM-as-a-judge evaluation component
 2. a SelfCheckGPT-style response consistency layer based on contradiction esti-

mation across multiple generated outputs

3. an entailment-based fact verification module using natural language inference (NLI)

- To develop a quantitative hallucination estimation method that aggregates multiple evaluation signals and provides stable and reproducible measurements, while enabling the analysis of variability and consistency across different datasets through repeated sampling and statistical analysis (e.g., mean and variance).
- To implement a detector-conditioned mitigation strategy that forms a closed-loop pipeline, incorporating techniques such as re-prompting, candidate filtering, and abstention, with response selection guided by hallucination risk scores.
- To evaluate the proposed framework across multiple benchmark datasets, including TruthfulQA, FEVER, and Natural Questions, using performance and stability metrics, and to analyse dataset-dependent performance and the role of grounding.
- To conduct ablation studies and threshold tuning to assess the contribution of individual components and optimise the overall performance of the proposed framework.

1.5 Contributions

This project makes the following contributions:

- It proposes a unified multi-signal framework that integrates complementary evaluation signals for hallucination detection, estimation, and mitigation, enhancing robustness and stability compared to single-signal approaches, while revealing trade-offs between accuracy and consistency.
- It introduces a quantitative and stability-aware hallucination estimation method that leverages repeated sampling and statistical analysis to produce more reliable and reproducible measurements.
- It develops a detector-conditioned mitigation strategy that forms a closed-loop pipeline, enabling dynamic response revision, filtering, and selection based on hallucination risk.

- It provides a comprehensive cross-dataset evaluation on TruthfulQA, FEVER, and Natural Questions, demonstrating the effectiveness and limitations of the proposed approach in both grounded and open-domain settings.
- It offers empirical insights into the behaviour of multi-signal hallucination detection, including component contributions, stability characteristics, and challenges in open-domain scenarios.

Chapter 2

Literature Review

This chapter provides an overview of existing research related to hallucination in large language models, with a particular focus on detection, estimation, and mitigation approaches. The aim is to analyse the strengths and limitations of current methods and to identify gaps that motivate the proposed framework. This review is structured as follows: Section 2.1 introduces the concept and types of hallucinations in large language models. Section 2.2 examines the underlying causes of hallucination, including both data-related and model-related factors. Section 2.3 examines existing hallucination detection methods, including confidence-based, entailment-based, and consistency-based approaches. Section 2.4 discusses hallucination estimation and uncertainty quantification techniques. Section 2.5 presents mitigation strategies, including prompt-based and retrieval-augmented approaches. Section 2.6 highlights reproducibility challenges in hallucination research. Finally, Section 2.7 summarises the key limitations of existing work and identifies the research gap addressed in this project.

2.1 Concept and Types of Hallucination in LLMs

Hallucinations in Large Language Models (LLMs) refer to the generation of content that is factually inaccurate or contains fabricated information. While LLMs can generate fluent and contextually coherent responses, such outputs may lack factual grounding. These errors are difficult to detect, mislead users, and potentially undermine their reliability [3].

Existing literature has proposed different classification methods to describe hallucination phenomena. One widely accepted method is to divide hallucinations into intrinsic and extrinsic types [11]. Intrinsic hallucinations occur when the generated content contradicts the provided input or context [12]. Conversely, extrinsic hallucinations occur when the model introduces external information that the input does not support, and this information may be fabricated or unverifiable [12]. This distinction highlights that hallucinations can stem from inconsistencies with the input or from a lack of reliable evidence. In ad-

dition, hallucinations can also be divided into factuality hallucinations and faithfulness hallucinations. Factuality hallucinations emphasize that the generated content is inconsistent with real-world facts, even if it doesn't contradict the input [11]. In contrast, faithfulness hallucinations refer to generated content lacking support from the given input or source context, or being inconsistent with it [11]. This perspective emphasizes that hallucinations can stem from a lack of factual accuracy or a failure to faithfully follow the provided input, further illustrating the multifaceted nature of hallucinations. Beyond this binary classification, hallucinations can also be understood in relation to knowledge grounding and verification. In knowledge-intensive tasks, such as open-domain question answering, the lack of explicit external evidence often leads to hallucinations, making it difficult to verify the correctness of generated answers. This further increases the risk of producing seemingly reasonable but factually incorrect outputs [13].

In summary, hallucination in language generation models large language models (LLMs) is a multifaceted phenomenon stemming from limitations in language generation and the knowledge grounding base. This complexity necessitates robust methods for detecting, estimating, and mitigating hallucination, which will be explored in the following chapters.

2.2 Causes of Hallucination

Hallucinations in large language models (LLMs) arise from a combination of data-related, model-related, and task-specific factors. These factors interact in complex ways, contributing to the generation of outputs that are fluent but factually incorrect.

From a data perspective, source-reference divergence in training data is a significant cause of hallucinations. When the target text contains information not supported by the input, the model learns generation patterns that do not fully rely on the source text [4]. This bias often stems from heuristic data collection processes or inherent characteristics of certain natural language generation tasks. Furthermore, large-scale training corpora may contain noise, biases, duplicate content, or unverified information. Consequently, the model may internalize and reproduce these defects, leading to inaccurate or unsubstantiated outputs [4]. These data-related issues collectively contribute to the generation of hallucinatory content in natural language models. More broadly, they reflect underlying data quality challenges, where inconsistencies between source and target, as well as noisy or biased

training corpora, can systematically influence model behaviour.

From a modeling perspective, Large Language Models (LLMs) typically suffer from poor calibration and are prone to overconfidence, assigning excessively high confidence levels to incorrect predictions [14]. This phenomenon is closely related to the model’s autoregressive training objective. LLMs primarily generate output by predicting the next most likely token, which makes the model more inclined during training to generate fluent, coherent, and confident outputs, rather than rigorously ensuring factual accuracy or estimating inherent uncertainty [15].

From a task and environment perspective, hallucinations are particularly prevalent in open-domain and knowledge-intensive settings. In these scenarios, the model primarily relies on parametric knowledge learned during pre-training and cannot acquire explicitly provided supporting evidence. The lack of reliable external evidence makes it extremely difficult to verify the correctness of the generated answers, which significantly increases the risk of producing seemingly reasonable but actually wrong answers.[16]. This challenge is even more pronounced in long-form content generation tasks, as maintaining factual consistency becomes increasingly difficult with longer outputs [17].

Overall, these factors not only explain the prevalence of hallucination, but also highlight the limitations of existing approaches that rely on single-source verification or surface-level evaluation signals. Hallucinations in LLMs result from the combined influence of imperfect training data, limitations in model calibration and objectives, and the inherent challenges of knowledge-intensive tasks. These insights motivate the need for more robust and integrated approaches to effectively detect, estimate, and mitigate hallucinations, as discussed in the following sections.

2.3 Hallucination Detection Methods

A wide range of approaches has been proposed to detect hallucinations in large language models (LLMs). Existing methods can be broadly categorised based on the source of information used for evaluation, including internal signal-based methods, external verification-based methods, and model-based evaluation methods. Internal signal-based methods rely on the model’s own uncertainty or output variability, such as uncertainty-based and

consistency-based approaches. External verification-based methods assess factual alignment using external evidence, including entailment-based and retrieval-based approaches. In addition, model-based evaluation methods utilise large language models as evaluators, commonly referred to as LLM-as-a-judge methods. While these approaches differ in their underlying assumptions, required resources, and robustness across tasks, each captures only partial aspects of hallucination. This section reviews these methods and provides a critical analysis of their strengths and limitations.

2.3.1 Internal signal-based methods

Uncertainty-based methods

Uncertainty-based methods detect hallucinations by estimating the model’s uncertainty in its generated outputs. A common approach is to analyse internal probability signals, such as token-level probabilities, conditional sequence probabilities (log-likelihood), or entropy measures, under the assumption that higher uncertainty (or lower confidence) is associated with an increased risk of unreliable or hallucinated content [18].

In practice, these methods are widely adopted due to their simplicity and computational efficiency. They are typically unsupervised and training-free, relying solely on the model’s internal output without requiring additional models, external knowledge bases, or retrieval mechanisms. Therefore, they provide a lightweight and practical solution for hallucination detection, especially suitable for resource-constrained environments [18].

However, uncertainty estimation based solely on model confidence is fundamentally limited by the poor calibration of large language models. Prior research has shown that LLMs often exhibit overconfidence, producing plausible but incorrect outputs with high probability [19]. As a result, internal confidence signals cannot reliably reflect factual correctness, making overconfident hallucinations difficult to detect. Furthermore, these methods do not incorporate external evidence or semantic consistency, limiting their effectiveness in complex or knowledge-intensive tasks.

In this work, uncertainty-based signals are not explicitly used. Instead, we focus on a multi-signal evaluation framework that combines semantic consistency (SelfCheckGPT), NLI-based factuality, and LLM-as-a-judge. This design addresses the limitations of

confidence-based methods by introducing external and cross-sample validation signals, enabling more reliable hallucination detection in scenarios where model confidence is poorly calibrated.

Consistency-based methods

Consistency-based methods detect hallucinations by analyzing the agreement among multiple sampled responses generated for the same input. The underlying assumption is that when a model possesses correct knowledge, different sampled outputs tend to be consistent, whereas hallucinated content is more likely to diverge or contradict across samples [10].

In practice, these methods generate multiple candidate responses through stochastic sampling and evaluate their consistency using techniques such as semantic similarity, contradiction detection, natural language inference (NLI), or answer aggregation. For example, SelfCheckGPT [10] estimates hallucination risk by identifying contradictions among sampled responses without relying on external knowledge sources. Similarly, self-consistency [20] improves reasoning performance by selecting the most consistent answer across multiple sampled reasoning paths.

These approaches provide a lightweight and training-free mechanism for detecting hallucinations, enabling semantic validation through cross-sample agreement without requiring additional supervision or retrieval.

However, consistency-based methods also have several limitations. First, they require multiple sampling and forward passes, increasing computational cost and inference latency [4]. Second, they assume that correct answers are consistent across samples, which may not hold when multiple valid answers exist [21]. More importantly, consistency does not guarantee factual correctness, as models may produce highly consistent but incorrect responses due to systematic biases or flawed parametric knowledge [22].

In this work, consistency-based signals are incorporated as one component of a multi-signal hallucination detection framework. Specifically, SelfCheckGPT is used to capture cross-sample consistency, but is complemented by NLI-based factuality and LLM-as-a-

judge evaluation. This design addresses the key limitation of pure consistency methods—namely, their inability to ensure factual correctness—by integrating external semantic validation signals and enabling more robust hallucination detection.

2.3.2 External / verification-based methods

Entailment-based methods (NLI)

Entailment-based methods use Natural Language Inference (NLI) to verify whether generated content is logically supported by references or evidence, thereby detecting hallucinations. The core idea is to model hallucination detection as a textual entailment problem: if the generated answer is implied by the reference text, it is considered reliable; if it contradicts the reference or lacks supporting evidence, it is considered a hallucination [23].

In practice, these methods typically employ pre-trained NLI models (e.g., models trained on MultiNLI) to compute entailment, contradiction, and neutral probabilities between reference text and generated answers [24]. Entailment scores are used to measure factual consistency, while contradiction signals help identify and penalise incorrect or unsupported content [25]. Prior work has shown that NLI-based approaches can effectively detect factual inconsistencies across tasks such as summarization and question answering [26].

A key advantage of entailment-based methods is their ability to capture semantic relationships beyond surface-level matching, enabling robust detection of factual inconsistencies. Compared to uncertainty-based or consistency-based methods, NLI explicitly evaluates the alignment between generated content and reference knowledge, making it particularly suitable for knowledge-intensive tasks.

However, these methods also have limitations. Their effectiveness depends on the quality and coverage of reference evidence, which may be incomplete or noisy in real-world scenarios [23]. In addition, NLI models themselves are imperfect and may produce erroneous judgments in complex contexts.

In this project, entailment-based signals play a central role in the hallucination detection

framework. Unlike standard NLI-based approaches, we extend this paradigm by incorporating both positive (correct-answer entailment) and negative (incorrect-answer entailment and contradiction) evidence, enabling more fine-grained discrimination of factual correctness. Furthermore, NLI signals are combined with consistency-based evaluation (SelfCheckGPT) and LLM-as-a-judge scoring to form a multi-signal detector, mitigating the limitations of relying on NLI alone and improving overall robustness.

Retrieval-based methods

Retrieval-based methods aim to reduce hallucinations by grounding model outputs in external knowledge sources, such as documents or knowledge bases. By retrieving relevant evidence and verifying generated content against it, these methods improve factual reliability and reduce reliance on parametric knowledge [9].

In practice, retrieval-based approaches typically combine a retrieval module (e.g., BM25 or dense retrievers) with a generation or verification model, either by incorporating retrieved evidence during generation (e.g., Retrieval-Augmented Generation) or by performing post-hoc verification [27].

A key advantage of these methods is their ability to incorporate up-to-date external knowledge and provide explicit evidence for model outputs, thereby improving both factual accuracy and interpretability.

However, their effectiveness is highly dependent on the quality of the retrieval component. Missing or irrelevant evidence can lead to incorrect predictions, while multi-stage retrieval pipelines introduce additional system complexity and computational cost [9].

In this work, we do not employ retrieval-based grounding. Instead, we focus on a fully model-based, multi-signal detection and mitigation framework that operates without external knowledge sources. This design enables a fair comparison across datasets such as TruthfulQA, FEVER, and Natural Questions, where retrieval conditions are not uniformly available, and highlights the effectiveness of detector-guided correction without reliance on external evidence.

LLM-as-a-Judge methods

LLM-as-a-Judge methods utilize large language models as evaluators to assess the factual correctness of generated content. Instead of relying solely on internal confidence signals or external retrieval, these approaches reformulate hallucination detection as a reasoning-based evaluation task, where the model explicitly determines whether a given answer is truthful and semantically consistent with the reference [28].

In practice, evaluation is guided through carefully designed prompts that incorporate the input question, generated answer, and optionally reference answers or evidence. The model then produces discrete judgments (e.g., true/false) or scalar scores reflecting factual correctness [29]. This allows the evaluator to leverage its implicit knowledge and reasoning capabilities, enabling flexible and task-agnostic assessment across diverse settings.

A key advantage of LLM-as-a-Judge methods is their ability to capture semantic correctness beyond surface-level matching. Unlike uncertainty-based approaches, they do not depend on internal probability estimates, and unlike retrieval-based methods, they do not require access to external knowledge sources. As a result, they are particularly suitable for open-ended tasks such as TruthfulQA, where semantic correctness and misconception rejection are critical.

However, these methods also exhibit important limitations. First, the reliability of the evaluation depends on the capability and calibration of the evaluation model itself, which may still be biased or prone to hallucinations [30]. Second, evaluation results can be sensitive to prompt design and exhibit instability under different prompting or sampling conditions [31]. Third, LLM-based evaluation introduces additional computational overhead, especially when multiple candidate answers are assessed.

In this work, we address these limitations by integrating LLM-as-a-Judge as one component within a multi-signal hallucination detector, rather than relying on it as a standalone decision mechanism. Specifically, the judge output is combined with NLI-based factuality signals and self-consistency measures, and further calibrated using a learned scoring function. This design improves robustness against individual signal bias and enables more stable and reliable hallucination detection across datasets.

Furthermore, the judge signal is not only used for detection but also plays a role in the mitigation pipeline, where it contributes to candidate filtering and reranking. This integration allows the system to leverage semantic evaluation while mitigating its inherent instability through multi-signal aggregation and calibration.

2.4 Hallucination Mitigation Methods

Hallucination mitigation methods aim to reduce the generation of factually incorrect or unsupported content by large language models, complementing detection approaches by actively improving output reliability. Unlike detection methods, which identify hallucinations after generation, mitigation methods focus on preventing or correcting such errors during or after the generation process.

2.4.1 Prompt-based approaches

Prompt-based mitigation methods aim to reduce hallucinations by influencing the model’s generative behaviour through carefully designed input instructions. These approaches do not modify model parameters but instead guide the model to produce more cautious, structured, and evidence-aware responses. Prior work has shown that well-designed prompts, including few-shot examples and instruction tuning, can significantly improve performance without additional training [32].

In particular, prompting strategies that encourage uncertainty awareness, such as allowing the model to abstain or avoid unsupported claims, have been shown to reduce hallucinations by promoting conservative generation [8]. Similarly, reasoning-oriented prompting techniques, such as Chain-of-Thought (CoT), guide the model to decompose complex problems into intermediate steps, thereby improving factual consistency and reasoning accuracy across a range of tasks [33].

Despite their effectiveness, prompt-based methods are inherently heuristic and highly sensitive to prompt design. The same prompt may yield significantly different performance across models and tasks, and even small variations in prompt wording can lead to unstable outputs [31]. Moreover, such methods do not fundamentally address the underlying causes of hallucination, as generation remains dependent on the model’s internal paramet-

ric knowledge [21]. As a result, prompt-based techniques alone cannot guarantee factual correctness, especially in knowledge-intensive or open-domain settings.

These limitations motivate more structured and adaptive prompting strategies that integrate external evaluation signals into the generation process. Instead of relying on static, one-shot prompts, recent approaches explore multi-stage and feedback-driven prompting mechanisms, where model outputs are iteratively refined or selectively corrected. In this work, this idea is further extended by combining prompt-based correction with a detector-guided framework, where different prompting strategies (e.g., mild editing, strict correction, and abstention) are dynamically applied based on the estimated hallucination risk. This enables more targeted and reliable mitigation compared to conventional prompt-based methods.

2.4.2 Retrieval-augmented approaches

Retrieval-augmented approaches aim to mitigate hallucinations by grounding model outputs in external knowledge sources, such as documents, knowledge bases, or online corpora. Instead of relying solely on parametric knowledge, these methods incorporate retrieved evidence to improve factual accuracy and reliability [9].

In practice, retrieval-augmented methods typically consist of two main stages: (1) retrieving relevant documents using sparse or dense retrieval techniques (e.g., BM25 or neural retrievers such as DPR), and (2) conditioning the generation process on the retrieved context to produce evidence-grounded outputs [27]. By explicitly integrating external information, these approaches significantly improve performance in knowledge-intensive tasks such as open-domain question answering and fact verification [34]. In addition, grounding model outputs in retrieved evidence enhances interpretability, as predictions can be traced back to supporting documents.

However, retrieval-augmented approaches also have several limitations. Their effectiveness is highly dependent on the quality and relevance of the retrieved evidence; missing or noisy retrieval results can lead to incorrect or misleading outputs [4], [9], [27]. Furthermore, retrieval systems may introduce irrelevant information that interferes with reasoning and increases the risk of inconsistent predictions. These approaches also involve multi-stage pipelines (retrieval, ranking, and generation), which increase system complexity and com-

putational cost.

These limitations motivate alternative approaches that do not rely on external retrieval. In this work, we adopt a training-free, model-based framework that operates without access to external knowledge sources. Instead, hallucination detection and mitigation are achieved through a combination of internal consistency signals (SelfCheckGPT), NLI-based factuality evaluation, and LLM-as-a-Judge. This design enables effective hallucination mitigation in closed-book and resource-constrained settings, while avoiding the dependency on retrieval quality and external knowledge availability.

2.4.3 Post-hoc correction methods

Post-hoc correction methods aim to reduce hallucinations by iteratively improving or correcting the model output after the initial generation. Unlike prompt-based or retrieval-augmented methods, these methods do not directly intervene in the generation process but introduce additional feedback and editing steps to improve the quality and factual consistency of the generated response. For example, Self-Refine [35] uses the same language model to first generate an initial output, then produce feedback on that output, and iteratively refine it, significantly improving output quality without additional training. In practice, post-hoc correction methods typically adopt a multi-stage process: the language model first generates an initial response, which is then iteratively improved through evaluation and modification steps. A mainstream strategy is self-correction, which prompts the same model to re-evaluate its output and identify potential factual errors, inconsistencies, or logical issues. This process often combines external feedback signals (e.g., task success metrics or unit test results) for consistency checks with internal reasoning to generate reflective feedback, ultimately enabling explicit verification and correction of the output [10], [35], [36]. Existing research shows that iterative improvement and self-correction mechanisms can help models detect and correct their own errors, thereby significantly improving the factual reliability of the output. In particular, Self-Refine [35] employs self-feedback and iterative refinement, while SelfCheckGPT [10] uses multi-sampling consistency checks and internal feedback loops to identify inconsistencies and reduce hallucinations. These methods are especially effective when external knowledge is limited or lacking, as they primarily rely on the model’s own internal reasoning ability and multi-round self-assessment process.

However, post-hoc correction methods also have several important limitations. First, their effectiveness is essentially limited by the capabilities of the underlying model, as both generation and error correction depend on the same or similar model representations and reasoning patterns, which can easily lead to the retention or even reinforcement of systematic errors [35], [36]. In addition, such methods usually incur high computational overhead, and the quality of self-feedback is difficult to guarantee when external knowledge is absent or the base model is weak [10]. These limitations highlight the need to develop a more robust multi-signal detection and intervention framework.

Overall, post-hoc correction methods provide a flexible and model-agnostic approach to improving factual consistency by introducing an additional verification stage after generation. However, their reliance on internal model capabilities and iterative refinement may limit their effectiveness, particularly when the base model exhibits systematic errors or lacks sufficient knowledge.

In contrast, this work extends the idea of post-hoc correction by incorporating a multi-signal hallucination detector to guide the correction process. Instead of relying solely on self-feedback, the proposed framework integrates NLI-based factuality signals, LLM-as-a-judge evaluation, and self-consistency to evaluate and rerank multiple candidate answers. This detector-conditioned correction mechanism enables more reliable identification and mitigation of hallucinations, addressing the limitations of purely self-refinement-based approaches.

2.4.4 Training-based methods (RLHF / fine-tuning)

Training-based methods aim to mitigate hallucinations by directly modifying model parameters through additional training procedures, such as supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF). These approaches improve alignment and response quality by training on curated datasets or optimizing model behavior with human preference signals [32]. As a result, they can enhance the overall reliability and helpfulness of model outputs across a wide range of tasks.

However, such methods have several limitations. First, alignment does not guarantee

factual correctness, and models may still produce plausible but incorrect responses, a phenomenon known as “alignment without truthfulness” [32]. Second, these approaches rely heavily on high-quality annotated data and incur substantial computational and human annotation costs [37]. Finally, they require access to model parameters, making them less flexible for black-box or deployment settings.

In contrast, this work focuses on a training-free, post-hoc framework that operates without modifying model parameters. Instead, hallucination detection and mitigation are achieved through multi-signal evaluation, including NLI-based factuality, LLM-as-a-judge, and self-consistency. This design enables effective hallucination mitigation in a plug-and-play manner, making it more practical for real-world applications where model retraining is not feasible.

2.5 Reproducibility Challenges in Hallucination Research

For large language models (LLMs), reproducibility has emerged as a critical challenge in hallucination research. Although deterministic decoding settings (e.g., fixed temperature and random seed) are commonly employed, the inherent stochasticity of the underlying generation process can still lead to significant output variability. Consequently, repeated runs of the same model on identical inputs may produce different responses, resulting in inconsistency in hallucination detection and evaluation [10]. A primary source of this variability lies in the probabilistic nature of language generation. Even under constrained decoding conditions, the outputs of Large Language Models (LLMs) are sampled from a probability distribution over possible continuations. Consequently, minor perturbations, such as changes in the sampling seed, adjustments to decoding parameters, or even subtle fluctuations in the underlying computational environment, can lead to significant discrepancies in both the generated content and downstream evaluation results [31]. This inherent uncertainty arising from probabilistic sampling contributes to the model’s diminished robustness, particularly when processing edge cases or out-of-distribution inputs.

Furthermore, hallucination detection methods often exhibit instability. Prompt-based evaluation approaches are highly sensitive to prompt engineering, where minor changes in wording or format can significantly alter the model’s judgment. Moreover, LLM-as-a-judge methods are frequently affected by position bias, verbosity bias (favoring longer

answers), and scoring inconsistencies, further weakening the reliability of the evaluation results [28]. These issues are exacerbated by cross-dataset variability: detection algorithms that perform well on evidence-retrieval tasks often struggle to generalize effectively to more adversarial or open-ended tasks, reflecting the limited robustness of current hallucination detection approaches across diverse evaluation settings.

Motivated by these limitations, this project incorporates a stability-aware evaluation framework that complements traditional accuracy metrics with statistical measures such as mean performance and variance across repeated runs. This enables a more robust and reproducible assessment of hallucination detection and mitigation performance across different datasets and experimental settings.

2.6 Summary

In summary, existing research has explored a wide range of approaches for hallucination detection and mitigation in large language models, including prompt-based methods, retrieval-augmented generation, post-hoc correction, and training-based alignment. While these methods have demonstrated effectiveness under specific conditions, they often exhibit limitations in robustness, stability, and generalization across tasks and datasets.

In particular, many approaches rely on single evaluation signals, and detection and mitigation are typically treated as separate processes. Moreover, as discussed in Section 2.5, hallucination evaluation itself is inherently unstable due to stochastic generation, prompt sensitivity, and model biases. These challenges are further exacerbated in cross-dataset scenarios, where performance varies significantly between evidence-grounded and open-domain tasks.

This review identifies three key limitations in existing work:

- Over-reliance on single-signal detection methods
- Lack of integration between detection and mitigation
- Limited robustness and reproducibility across datasets

To address these challenges, this work proposes a multi-signal, detector-guided hallu-

ination mitigation framework that integrates semantic evaluation (LLM-as-a-Judge), entailment-based verification (NLI), consistency-based signals (SelfCheckGPT), and additional auxiliary signals.

By combining these complementary signals within a unified scoring and calibration framework, the system estimates hallucination risk and dynamically applies mitigation strategies. In particular, prompt-based correction is used as an intervention mechanism, where different prompts (e.g., mild editing, strict correction, and abstention) are adaptively selected based on the predicted risk level.

Together with stability-aware evaluation, the proposed approach aims to achieve more robust, generalizable, and reproducible hallucination detection and mitigation across diverse tasks.

Chapter 3

Methodology and System design

This chapter introduces the method framework and system design developed for hallucination detection and mitigation in large language models. Existing hallucination detection methods typically rely on a single signal, such as model confidence, entailment scores, or self-consistency. However, such approaches suffer from well-known limitations, including instability across prompts, sensitivity to model bias, and limited effectiveness in cases where outputs are logically coherent but factually incorrect.

Compared to existing approaches, the proposed framework provides a unified multi-signal and closed-loop solution that jointly models detection, estimation, and mitigation, leading to improved robustness and stability across diverse datasets.

In addition, many prior works treat hallucination detection as a purely post-hoc evaluation task, without integrating it into the generation process. To address these limitations, this project proposes a closed-loop hallucination detection, estimation, and mitigation framework. Rather than operating in isolation, detection is integrated with generation through a feedback mechanism, where detection signals guide subsequent mitigation and answer refinement.

The framework combines multiple complementary signals to evaluate factual reliability, including semantic judgment, NLI-based factual consistency, and self-consistency analysis. These signals are aggregated into a unified scoring function, which is then used to trigger threshold-based mitigation strategies, such as answer rewriting, candidate filtering, reranking, and abstention. A key feature of the framework is the detector-conditioned mitigation mechanism, where the level of intervention is dynamically determined based on the estimated reliability of each answer.

While the overall architecture is general, the system incorporates lightweight dataset-

specific adaptations (e.g., prompt design, parsing, and output constraints) to accommodate different task formulations. This ensures flexibility across diverse evaluation settings without requiring model retraining.

In addition, this chapter presents the threshold selection and calibration procedures used to convert detection scores into actionable decisions, as well as the evaluation protocols designed to ensure robustness and reproducibility of experimental results. Together, these components form a coherent system aimed at reducing hallucinations while maintaining answer quality. This design is motivated by the observation that hallucination is a multi-faceted phenomenon, involving semantic inconsistency, factual contradiction, and instability under stochastic generation. Therefore, multiple complementary signals are required to provide a robust and generalisable estimation of answer reliability.

3.1 Problem Formulation

This project addresses the problem of hallucination detection, estimation, and mitigation in large language models (LLMs) through a prompt-engineering-based framework. Given an input query (which may be a claim or a question), an optional supporting context (depending on the task formulation), and a model-generated answer, the objective is to evaluate the factual reliability of the answer. This evaluation is performed both as a binary classification task (hallucinated vs. non-hallucinated) and as a continuous scoring problem that estimates the degree of factual correctness. Based on this assessment, the system applies mitigation strategies, such as answer rewriting, extractive correction, or abstention, to reduce hallucination and improve overall answer reliability.

Formally, let q denote the input claim or question, e the associated evidence (if possible), and a the answer generated by a language model. The task is to estimate a dataset-dependent reliability score $s^{(d)}$ defined as:

$$\begin{aligned}
s^{(d)} = \mathcal{C}^{(d)} \Big(& w_1^{(d)} f_{\text{judge}}(q, a) \\
& + w_2^{(d)} f_{\text{NLI}}^{(d)}(\tilde{e}^{(d)}, a) \\
& + w_3^{(d)} f_{\text{consistency}}(\{a_i\}_{i=1}^N) \\
& + w_4^{(d)} f_{\text{lexical}}^{(d)}(a, r) \\
& - \lambda^{(d)} \mathbf{1}_{\text{contra}}^{(d)} f_{\text{contra}}(\tilde{e}^{(d)}, a) \Big)
\end{aligned} \tag{1}$$

where,

$$\tilde{e}^{(d)} = \begin{cases} e, & \text{if explicit evidence is available (e.g., FEVER)} \\ c(q), & \text{if context is constructed from documents (e.g., NQ)} \\ \emptyset, & \text{otherwise (e.g., TruthfulQA)} \end{cases} \tag{2}$$

Here, f_{judge} denotes a semantic correctness signal produced by an external language model evaluator. $f_{\text{NLI}}^{(d)}$ represents a dataset-specific NLI-based semantic consistency function: it operates as an evidence verification signal in evidence-grounded tasks (e.g., FEVER), as a semantic alignment signal against reference answers in open-ended settings (e.g., TruthfulQA), and as an auxiliary consistency signal complementing lexical matching in extractive QA tasks (e.g., Natural Questions). $f_{\text{consistency}}$ measures cross-sample agreement across multiple generated answers, implemented using a SelfCheckGPT-style NLI-based consistency estimator. $f_{\text{lexical}}^{(d)}$ denotes a lexical similarity function (e.g., exact match or token-level F1) between the generated answer and the reference answer, where r denotes the reference answer associated with the input query when available. In span-based QA datasets such as Natural Questions, lexical overlap serves as a primary correctness signal. $f_{\text{contra}}^{(d)}$ denotes a contradiction-based penalty derived from NLI signals. The term $\mathbf{1}_{\text{contra}}^{(d)}$ is a dataset-dependent indicator function that controls whether the contradiction penalty is activated, enabling stronger enforcement in evidence-grounded datasets (e.g., FEVER) and weaker or auxiliary usage in other settings (e.g., Natural Questions and TruthfulQA). The weights $w_i^{(d)}$ and $\lambda^{(d)}$ are dataset-dependent coefficients that determine the relative importance of each signal. $\mathcal{C}^{(d)}(\cdot)$ denotes an optional calibration or normalization function that maps the aggregated score into a bounded reliability range, and may be applied selectively depending on the dataset.

Not all components in Equation 1 are equally active or equally weighted across datasets. Instead, the formulation represents a unified scoring framework in which different signals are selectively emphasized depending on dataset characteristics. This formulation therefore defines a unified scoring framework, where each dataset corresponds to a specific instantiation with different dominant signals.

Different datasets (e.g., FEVER, TruthfulQA, and Natural Questions) exhibit varying characteristics in terms of supervision signals and answer formats. Consequently, the proposed framework adopts dataset-specific configurations of scoring components rather than enforcing a uniform instantiation. In evidence-grounded datasets such as FEVER, explicit evidence is available, and the NLI-based signal together with contradiction penalties plays a central role in verifying factual consistency. In contrast, TruthfulQA does not provide explicit supporting evidence; therefore, the NLI component primarily serves as a semantic alignment signal with reference answers, while contradiction penalties are applied more conservatively. In span-based question answering tasks such as Natural Questions, lexical overlap (e.g., exact match and token-level F1) becomes the dominant correctness signal, and NLI-based contradiction is used only as an auxiliary filtering mechanism.

Furthermore, calibration or normalization functions $\mathcal{C}^{(d)}(\cdot)$ are applied selectively depending on the dataset and experimental setting, rather than being universally enforced. Although a unified formulation is presented in Equation 1 for clarity, not all components are equally active across datasets. Instead, the framework is instantiated in a dataset-aware manner, where different signals are emphasized or down-weighted according to task-specific constraints. This design enables flexible adaptation to heterogeneous QA settings while maintaining a consistent underlying scoring principle.

The score $s^{(d)}$ represents a reliability measure indicating the degree of factual correctness of the answer. A higher score corresponds to higher confidence in factual correctness, while a lower score indicates a higher risk of hallucination.

Based on this score, a decision function is defined to classify the answer as reliable or

hallucinated using a threshold $\tau^{(d)}$, resulting in a binary decision:

$$\hat{y} = \begin{cases} \text{reliable,} & \text{if } s^{(d)} \geq \tau^{(d)} \\ \text{hallucinated,} & \text{otherwise} \end{cases} \quad (3)$$

The threshold $\tau^{(d)}$ is tuned on validation data and fixed during test-time evaluation. In practice, the decision process is further extended to a multi-threshold mechanism, where multiple confidence levels (e.g., low-confidence and high-confidence thresholds) are used to guide subsequent mitigation actions.

While this decision mechanism follows a unified structure across datasets, its behaviour is dataset-dependent, as different grounding settings and answer formats (e.g., evidence-based verification in FEVER, multi-reference answers in TruthfulQA, and span-based answers in Natural Questions) influence the effectiveness of different intervention strategies.

Unlike conventional approaches that treat hallucination detection as a standalone post-hoc classification problem, this project extends the formulation to a closed-loop framework. Specifically, the detection score is further used to guide a mitigation function:

$$a' = g^{(d)}(q, \tilde{e}^{(d)}, a, s^{(d)}) \quad (4)$$

where a' is a refined or alternative answer generated to reduce hallucination risk.

In practice, the mitigation function $g^{(d)}(\cdot)$ is implemented as a structured, detector-conditioned decision process based on the reliability score $s^{(d)}$. This enables adaptive intervention strategies at different confidence levels while maintaining consistency between detection and mitigation.

For low-confidence answers, multiple candidate answers are generated to improve reliability. The form of candidate generation is dataset-dependent. In span-based QA settings such as Natural Questions, extractive candidate answers are generated directly from the available context. In contrast, for datasets without strict extractive constraints (e.g., TruthfulQA and FEVER), candidate answers are obtained through alternative genera-

tion or selection strategies consistent with the task formulation. These candidates are subsequently filtered using NLI-based consistency and contradiction-aware constraints, and further evaluated using the proposed multi-signal scoring mechanism. In addition, cross-sample consistency (SelfCheck-style) is incorporated during reranking to improve robustness. The final answer is selected through a group-based reranking strategy that prioritizes factual consistency and agreement across multiple signals.

For higher-confidence answers, the system explicitly preserves the original output to avoid unnecessary modification and prevent performance regression. In uncertain cases, conservative fallback strategies may be applied instead of aggressive intervention.

Although the overall mitigation framework is unified, its behaviour varies across datasets due to differences in grounding signals and answer formats. For example, FEVER relies heavily on evidence-based verification and contradiction filtering, TruthfulQA emphasizes semantic alignment with reference answers, while Natural Questions prioritizes extractive span selection and lexical matching.

Overall, this formulation defines hallucination detection, estimation, and mitigation as a unified, score-driven process that integrates continuous reliability estimation, binary classification, and adaptive intervention. By combining multi-signal scoring with a structured decision mechanism, the framework provides a principled foundation for improving both detection robustness and mitigation effectiveness.

Building on this formulation, the next section presents the overall system architecture, illustrating how the proposed components are integrated into a complete hallucination detection and mitigation pipeline.

3.2 System Overview

This section presents an overview of the proposed framework for hallucination detection, estimation, and mitigation in large language models. As illustrated in Figure 1, the system is designed as a closed-loop pipeline that integrates answer generation, reliability estimation, and corrective intervention into a unified process. Unlike conventional approaches that treat hallucination detection as a post-hoc classification task, the proposed

framework uses reliability estimation to actively guide the generation process.

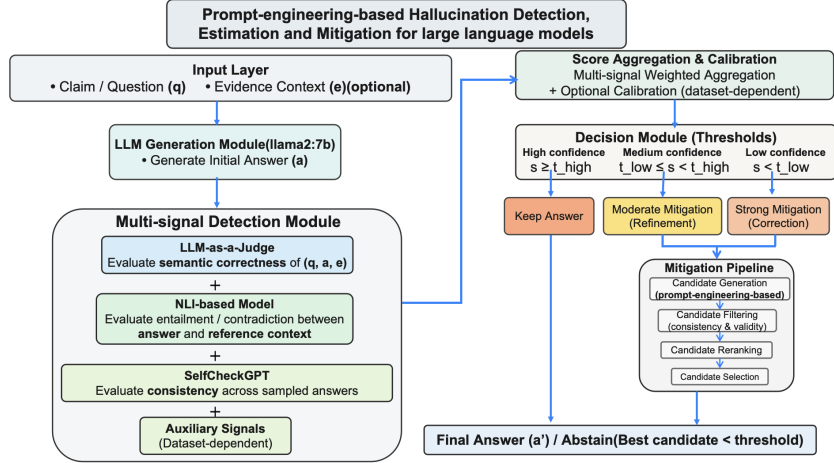


Figure 1: Overview of the proposed prompt-engineering-based framework for hallucination detection, estimation, and mitigation. The system operates as a closed-loop pipeline that integrates answer generation, multi-signal reliability estimation, score aggregation, and threshold-based decision-making. Based on the estimated reliability, the system either accepts the initial answer or applies mitigation through candidate generation, filtering, and reranking.

Given an input claim or question q and an optional evidence context e (if available), the system first generates an initial answer a using a large language model. This answer is then analysed by a multi-signal detection module, which estimates factual reliability from complementary perspectives. These signals include semantic correctness, factual consistency, contradiction-aware signals, and cross-sample consistency, allowing the system to capture different types of hallucination behaviour.

The outputs of these signals are aggregated into a unified reliability score through a weighted combination and calibration process. This score provides a continuous estimate of factual correctness and serves as the control signal for subsequent decision-making.

Based on the estimated reliability, a decision module determines whether the answer should be accepted or further refined. High-confidence answers are preserved, while lower-confidence answers trigger mitigation procedures of varying intensity. This results in a score-conditioned intervention mechanism that adapts the level of correction to the estimated hallucination risk.

When mitigation is required, the system generates multiple candidate answers and ap-

plies a structured selection process to identify the most reliable one. This process includes consistency-based filtering and reliability-aware reranking. The filtering criteria are dataset-aware and may rely on evidence when available (e.g., FEVER and Natural Questions), or on internal consistency signals when external grounding is not provided (e.g., TruthfulQA). If no candidate meets the required reliability criteria, the system may abstain from producing a final answer.

Overall, this design establishes a closed-loop framework in which hallucination detection, estimation and mitigation are tightly coupled. By using reliability estimation to guide corrective interventions, the system not only identifies hallucinated outputs but also actively improves them while maintaining answer quality. This integration distinguishes the proposed approach from conventional post-hoc detection methods.

3.3 Datasets

To comprehensively evaluate the effectiveness, robustness, and generalisation ability of the proposed hallucination detection, estimation and mitigation framework, three widely used benchmark datasets are employed in this study: TruthfulQA, FEVER, and Natural Questions (NQ). These datasets are selected to represent different types of question-answering scenarios and hallucination characteristics, ranging from adversarial questions to evidence-based verification and open-domain extractive QA.

3.3.1 TruthfulQA

TruthfulQA is designed to evaluate whether large language models produce truthful answers to questions that are prone to common misconceptions. Unlike standard QA datasets, many questions in TruthfulQA are intentionally constructed such that commonly memorised or intuitive responses are incorrect.

This dataset is particularly suitable for hallucination evaluation because:

- It explicitly targets model-generated misinformation
- It provides both correct and incorrect reference answers
- It requires semantic understanding rather than surface-level matching

In this project, TruthfulQA is used to evaluate hallucination behaviour in open-ended generative settings where no explicit supporting evidence is available. This makes it a representative benchmark for assessing the ability of the proposed framework to detect and mitigate hallucinations in ungrounded scenarios, where answers rely entirely on the model’s internal knowledge.

3.3.2 FEVER

FEVER is a fact verification dataset consisting of claims labelled as *SUPPORTED*, *REFUTED*, or *NOT ENOUGH INFORMATION*, along with corresponding evidence retrieved from Wikipedia.

Unlike standard question-answering datasets, FEVER focuses on claim verification rather than answer generation. Each claim is associated with one or more pieces of evidence, enabling explicit reasoning about factual consistency between the claim and the supporting information.

This dataset is particularly suitable for hallucination evaluation because:

- It provides explicit grounding through annotated evidence
- It enables clear verification of factual consistency and contradiction
- It supports structured evaluation through discrete labels

In this project, FEVER is used to evaluate hallucination detection and mitigation in evidence-grounded settings, where factual correctness can be directly assessed with respect to external evidence. This allows the framework to be tested in scenarios where reliable grounding signals are available, in contrast to ungrounded settings such as TruthfulQA.

3.3.3 Natural Questions (NQ)

Natural Questions is a large-scale open-domain question-answering dataset consisting of real user queries paired with answers derived from Wikipedia documents. Answers are typically short spans extracted directly from the source text.

Compared to TruthfulQA and FEVER, Natural Questions represents a more realistic and practical QA scenario, characterised by:

- Naturally occurring user questions
- Context-dependent, extractive answers
- Long and potentially noisy documents

This dataset is particularly suitable for hallucination evaluation because it provides implicit grounding through document context, while still requiring the model to accurately locate relevant information within complex and noisy inputs.

In this project, Natural Questions is used to evaluate hallucination behaviour in partially grounded settings, where relevant evidence is available but not explicitly structured as verification signals. This setting lies between fully ungrounded generation (e.g., TruthfulQA) and explicitly evidence-grounded verification (e.g., FEVER), enabling the assessment of how well the proposed framework adapts to intermediate levels of grounding.

Overall, Natural Questions emphasises the importance of precise information extraction and robustness to noisy context, making it an effective benchmark for evaluating hallucination in realistic QA scenarios.

The combination of these three datasets enables a comprehensive evaluation across diverse settings: (1) adversarial generation without grounding (TruthfulQA), (2) evidence-based verification (FEVER), and (3) extractive QA with local context (Natural Questions). This design ensures that the proposed framework is not overfitted to a single task type and provides insights into its stability and generalisation across different hallucination scenarios.

To address the inherent stochasticity of large language models, a controlled experimental protocol is adopted. For each dataset, fixed subsets of samples are constructed, and multiple independent runs are conducted using different random seeds. This protocol approximates the expected performance under stochastic generation and provides a more reliable estimate compared to single-run evaluation, reducing sensitivity to sampling variance. This enables the measurement of both average performance and variability, thereby providing a more reliable and reproducible evaluation of hallucination detection and mitigation performance.

In terms of data preparation, all datasets are obtained from the HuggingFace Datasets library and subsequently preprocessed into local JSON files to facilitate efficient loading and reproducibility.

Due to the large scale of the FEVER and Natural Questions datasets, it is computationally expensive to perform full-scale experiments, particularly under the multi-run evaluation setting. Therefore, subset construction is applied in a two-stage manner.

First, intermediate data pools are created. For the FEVER dataset, stratified sampling is applied by grouping the data according to labels (SUPPORTED, REFUTED, NOT ENOUGH INFORMATION) and randomly selecting an equal number of samples from each class, resulting in a balanced subset of 3,000 instances. For the Natural Questions dataset, simple random sampling is applied to select 2,000 instances, preserving the natural distribution of question types and answer structures.

Second, during experimentation, smaller subsets are further sampled from these intermediate data pools in each run. This enables a Monte Carlo-style evaluation protocol, where different subsets are used across runs to assess both performance and stability.

Overall, this design strikes a balance between computational efficiency and experimental validity, while preserving the diversity and complexity of the original datasets and ensuring consistent and reproducible evaluation.

3.4 Input Preparation & Answer Generation

3.4.1 Context Construction

Before answer generation, a context construction step is applied for datasets that provide supporting information. This step aims to create a grounded input for the generation model by incorporating relevant contextual information when available.

For **Natural Questions**, where answers correspond to spans within long documents, a local context window is constructed around the reference answer. The system first iden-

tifies the position of the answer within the source document using annotated spans or matching heuristics. A fixed-size window of tokens surrounding the answer span is then extracted to form the context. This approach reduces noise from long documents while preserving relevant information for extractive answer generation.

If the answer span cannot be reliably located, a fallback strategy is applied, where a pre-defined portion of the document (e.g., the first segment of tokens) is used as the context. This ensures that a valid input is always available for subsequent processing.

For **FEVER**, where each claim is associated with annotated evidence sentences, the provided evidence is directly used as the context. Since the dataset already supplies concise and structured supporting information, no additional context construction is required.

For **TruthfulQA**, no external context is provided. Therefore, answer generation is performed without additional grounding, allowing the system to evaluate hallucination under fully ungrounded conditions.

Unlike retrieval-based approaches, this design relies solely on dataset-provided information, ensuring that context construction is deterministic and reproducible. This avoids variability introduced by external retrieval systems and enables controlled evaluation across datasets.

3.4.2 Prompting Strategy

To ensure controlled and consistent answer generation across different datasets, prompt engineering is employed as the primary mechanism for guiding the behaviour of the language model. Instead of relying on model fine-tuning, task-specific prompts are designed to impose structural and semantic constraints on the generated outputs.

For **Natural Questions**, a strictly constrained extractive prompting strategy is adopted. The model is explicitly instructed to produce an answer that corresponds to a contiguous span from the provided context. The prompt enforces hard constraints, including copying only words that appear in the context and prohibiting paraphrasing or the introduction of external knowledge. This effectively restricts the output space to grounded evidence

and significantly reduces hallucination.

For **FEVER**, the generation process follows a verification-oriented setting, where the model produces answers based on the input claim and associated evidence. Unlike the extractive setting, the generation remains largely free-form, while factual consistency is primarily enforced through downstream verification signals such as NLI-based entailment and contradiction detection, rather than strict prompt constraints.

For **TruthfulQA**, no external context is provided, and prompts are designed for open-ended generation. The model is required to produce answers based solely on its internal knowledge, enabling the evaluation of hallucination behaviour in the absence of explicit grounding signals.

Across all datasets, a structured output format is enforced to ensure consistency and facilitate downstream processing. In particular, the model is instructed to produce answers following a standardised prefix (e.g., **FINAL_ANSWER:**), allowing reliable extraction and evaluation of the generated content.

This prompt-based design enables flexible yet controlled generation behaviour, while maintaining a unified framework that can be adapted to heterogeneous dataset settings.

3.4.3 Answer Generation Process

The answer generation process serves as the entry point of the hallucination detection and mitigation pipeline. Given an input question or claim, the system generates candidate answers using a large language model (LLM), specifically **llama2:7b**, under a controlled prompting setting. The generation strategy is adapted according to dataset characteristics to ensure consistency with task requirements.

Dataset-specific generation settings

The generation behaviour varies across datasets depending on the availability of grounding signals.

For **Natural Questions**, answer generation follows a strictly constrained extractive setting. A context window is constructed from the source document, and the model is instructed to produce an answer that corresponds to a contiguous span within the context. This constraint limits the output space to grounded content and reduces the likelihood of hallucination.

For **TruthfulQA**, no external context is provided, and answers are generated in an open-ended manner. This setting allows the model to rely entirely on its internal parametric knowledge, making it suitable for evaluating hallucination under unconstrained generation conditions.

For **FEVER**, answer generation is performed using a claim-based input with associated evidence provided as context. The generation remains free-form, while factual correctness is primarily enforced in downstream stages through NLI-based verification rather than strict constraints during generation.

Controlled decoding

To balance reproducibility and task-appropriate variability, decoding strategies are configured in a dataset-dependent manner.

For **TruthfulQA**, moderately stochastic decoding (e.g., temperature ≈ 0.3 , top- $p \approx 0.9$) is adopted to encourage diversity across generated answers. This variability is important for exposing inconsistencies and enabling self-consistency analysis.

For **FEVER**, stochastic decoding is used (e.g., temperature ≈ 0.6 , top- $p \approx 0.9$) to allow moderate variation in generated answers. This helps the model explore different evidence-consistent formulations, while factual correctness is enforced in downstream verification stages rather than through deterministic generation.

For **Natural Questions**, deterministic decoding (temperature = 0) is adopted. Since answers are expected to match short extractive spans, eliminating randomness improves answer precision and stability.

Multi-sample generation

Instead of producing a single answer, the system generates multiple candidate answers for

each input:

$$A = \{a_1, a_2, \dots, a_n\}, \quad n = 5$$

Each answer is obtained through an independent decoding run. The use of multiple samples captures residual variability in model outputs, even under partially deterministic settings.

The role of multi-sample generation differs across datasets.

For **TruthfulQA**, it plays a central role in hallucination detection by enabling self-consistency analysis, where agreement between multiple generated answers serves as an indicator of reliability. The use of moderately stochastic decoding encourages diversity, which is essential for exposing inconsistent or contradictory responses.

For **FEVER**, although stochastic decoding is applied (e.g., temperature ≈ 0.6), multi-sample generation is not primarily intended to maximise diversity. Instead, it provides robustness against variability in model outputs, while factual correctness is enforced through downstream evidence-based verification and NLI-based consistency checking.

For **Natural Questions**, where deterministic decoding is used (temperature = 0), multi-sample generation mainly serves to reduce sensitivity to occasional generation noise rather than to introduce diversity. Since answers are expected to match short extractive spans, excessive variation is neither necessary nor desirable.

Consistency estimation

To quantify agreement among generated answers, a SelfCheck-style mechanism is applied, where each answer is compared against others using NLI-based scoring. This produces a consistency signal that reflects cross-sample agreement, which is later incorporated into the multi-signal hallucination detection framework.

Caching and reproducibility.

To improve efficiency and ensure reproducibility, generated answers are cached and reused across experiments. Once a set of answers is produced for a given input, it is stored and subsequently reused to avoid redundant model calls.

Discussion

Overall, the answer generation process is treated as a controlled and analyzable stage rather than a purely black-box component. By combining dataset-aware prompting, controlled decoding, and multi-sample generation, the system ensures that generated answers are both representative of model behaviour and suitable for downstream reliability estimation and mitigation.

3.4.4 Output Parsing and Normalisation

The raw outputs produced by large language models are not always directly suitable for evaluation. Depending on the dataset and prompt format, the model may generate additional text such as explanations, formatting artifacts, placeholder expressions, or partially malformed responses. To ensure reliable downstream scoring, a post-processing step is applied to convert raw model outputs into a standardised answer representation. Although this step serves a unified objective across datasets, its parsing rules are dataset-specific.

Dataset-Specific Answer Extraction

For **TruthfulQA**, the system expects model outputs to follow a structured format of the form `FINAL_ANSWER: <answer>`, as explicitly enforced during prompting. When this pattern is present, the text following the marker is extracted as the final answer using a regular expression-based parser.

If the expected prefix is not found, the entire output is retained after whitespace trimming as a fallback. Although this fallback mechanism allows imperfectly formatted outputs to be processed, the system is designed to strongly encourage the use of the `FINAL_ANSWER` structure, which provides a consistent and reliable interface for downstream evaluation. This design ensures that open-ended answers in TruthfulQA can be normalised into a comparable format while maintaining robustness to minor deviations in output structure.

For **FEVER**, the model output is required to follow a structured verification format containing an explicit label field (e.g., `LABEL: SUPPORTED`), optionally accompanied by a concise justification.

The parsing process is designed to robustly extract and normalise the predicted label using a combination of pattern matching and fallback heuristics, ensuring resilience to formatting variations. Since the predicted label constitutes the primary evaluation target, it is treated as the core representation for downstream scoring.

To improve consistency and prevent prompt leakage or verbose generation, the output is further normalised into a controlled format that retains only the essential label and a single-sentence explanation. This design enforces structural regularity while minimising noise in subsequent evaluation and mitigation stages.

For **Natural Questions**, the system extracts answers from a structured `FINAL_ANSWER:` field, ensuring consistency with the extractive nature of the task.

Since the expected outputs correspond to short answer spans, only the first line of the extracted content is retained. This design removes any additional generated text, preventing it from interfering with lexical evaluation metrics such as exact match and token-level F1.

As a result, the evaluation focuses strictly on the predicted answer span, improving robustness and alignment with dataset annotations.

Cleaning and Surface Normalisation

After answer extraction, lightweight cleaning is applied to improve consistency across outputs. This includes trimming whitespace, collapsing repeated spaces, removing surrounding quotation marks, and eliminating trailing punctuation when appropriate. The cleaning process is intentionally conservative, so that semantically meaningful content is not removed.

In **Natural Questions**, additional normalisation is applied to remove placeholder-like expressions and extraction artifacts (e.g., span markers or template fragments) that may appear in malformed model outputs.

Furthermore, the extracted answer is cleaned and reduced to a single-line span before

evaluation. This is particularly important because the final answer is evaluated using lexical overlap measures, such as exact match and token-level F1, which are highly sensitive to formatting inconsistencies.

By eliminating extraneous tokens and enforcing a clean answer representation, the system ensures that evaluation focuses strictly on the predicted answer span, thereby improving robustness and metric reliability.

Functional Role Across Datasets

Although output parsing is applied in all three datasets, its role differs across tasks.

In **TruthfulQA**, parsing mainly serves to isolate concise answer content from potentially verbose open-ended generations, thereby improving comparability across samples.

In **FEVER**, parsing is primarily label-oriented, ensuring that verification outputs are mapped consistently to the predefined classes **SUPPORTED**, **REFUTED**, and **NOT ENOUGH INFO**.

In **Natural Questions**, parsing and normalisation are especially important because the evaluation depends strongly on lexical precision, and even small surface-level deviations may affect correctness scores.

Invalid Outputs

During post-processing, outputs that remain empty, malformed, or non-informative after parsing are marked as invalid for subsequent handling. The strategies used to manage such cases, including fallback mechanisms and answer replacement rules, are described separately in the next subsection.

Overall, this stage transforms heterogeneous raw generations into stable and comparable answer representations, thereby providing a clean and reliable foundation for downstream evaluation and hallucination detection.

3.4.5 Invalid Output Handling and Fallback Strategy

Despite structured prompting and post-processing, language models may still produce outputs that are unusable for evaluation. Typical failure cases include empty responses, semantically uninformative outputs (e.g., “unknown”), or outputs that fail to satisfy task-

specific structural requirements.

To address this, the system applies a lightweight validation and fallback strategy. Instead of performing aggressive correction, invalid outputs are detected using simple heuristics (e.g., emptiness, minimal length, or predefined invalid tokens), and are handled in a task-aware manner. The primary objective is to ensure that all samples remain evaluable, while preserving the original semantic behaviour of the model wherever possible.

Dataset-Specific Handling

The handling strategy is adapted to each dataset according to its task formulation:

- **TruthfulQA:**

As an open-ended generation task without external grounding, TruthfulQA prioritises preserving the model’s original outputs. The system first attempts to extract answers using structured patterns (e.g., `FINAL_ANSWER:`); if extraction fails, the raw output is retained with only lightweight normalisation.

In addition, candidate answers are implicitly filtered using simple validity constraints (e.g., minimum length, avoidance of overly long or explanatory responses, and basic semantic plausibility checks via a lightweight judge model). This prevents clearly malformed or low-quality outputs from propagating to later stages.

Abstention-like responses (e.g., “I do not know”) are explicitly detected and normalised into a consistent representation. Importantly, abstention is treated as a valid outcome rather than an error, and is incorporated into downstream scoring and decision-making.

No semantic rewriting or correction is applied at this stage. All processing remains limited to structural parsing, lightweight filtering, and normalisation, ensuring that the evaluation reflects the model’s original behaviour.

- **FEVER:**

For FEVER, outputs are required to conform to a fixed set of discrete labels

(SUPPORTED, REFUTED, NOT ENOUGH INFO).

To ensure validity, model outputs are first processed using robust label extraction and normalisation. When the label cannot be reliably parsed, a conservative fallback (e.g., NOT ENOUGH INFO) is applied.

Beyond output normalisation, the system incorporates a closed-loop mitigation mechanism. Candidate answers are generated and filtered using a combination of multi-signal factuality checks, including NLI-based consistency scoring and contradiction detection.

For high-risk cases, a stricter rewriting strategy is applied, where relevant evidence spans are explicitly extracted and used to constrain the generation process. This ensures that revised answers remain grounded in the provided evidence and reduces the likelihood of unsupported claims.

Additionally, an abstention policy is employed when no sufficiently reliable answer can be identified, prioritising hallucination reduction over coverage.

This design ensures that outputs remain valid categorical predictions while actively improving factual consistency through evidence-grounded correction.

- **Natural Questions:**

In the extractive QA setting of Natural Questions, stricter validity constraints are imposed, as answers are expected to correspond closely to short spans grounded in the provided context.

When model outputs are invalid or unusable, a fallback span selection mechanism is applied. In this case, a plausible answer span is heuristically selected directly from the context, for example by locating question-related terms or using a local fallback segment when no reliable output is produced.

In the subsequent mitigation stage, candidate answers are further filtered to ensure that retained spans remain explicitly grounded in the context. This includes checking whether candidate answers appear as contiguous spans within the context, thereby reducing out-of-context or hallucinated outputs.

Design Rationale

This design ensures that all generated samples are valid and comparable across datasets while respecting task-specific constraints. Crucially, invalid output handling is restricted to structural validation and fallback selection, without introducing semantic modifications at the initial generation stage. This maintains a clear separation between generation, validation, and subsequent correction, allowing downstream hallucination detection and mitigation to operate on unbiased model outputs.

3.4.6 Justification of Design Choices

The design of the initial answer generation module is guided by the need to balance factual grounding, controllability, stability, and compatibility with downstream hallucination detection. Rather than treating generation as an isolated component, it is explicitly designed to support a multi-stage, evaluation-driven pipeline.

(1) Grounding vs. hallucination risk

To reduce hallucination, grounding is incorporated in a dataset-aware manner, depending on the availability of reliable context and task formulation.

In extractive question answering settings such as *Natural Questions*, prompts are designed to encourage the model to produce answers that align with spans from the provided context. In addition, heuristic fallback strategies and later-stage filtering further promote context-grounded outputs. This combination reduces the likelihood of unsupported or fabricated content.

In contrast, for tasks without strict extractive requirements (e.g., *TruthfulQA* and *FEVER*), grounding is not enforced during initial generation. For *FEVER*, although evidence is available, imposing extractive constraints at generation time may conflict with its

classification-oriented objective. Instead, grounding is enforced indirectly during downstream processing through NLI-based verification and evidence-aware filtering.

This dataset-aware strategy ensures that grounding is applied when beneficial, while preserving task-appropriate generation behaviour in other settings.

(2) Prompt-based control

The system adopts prompt engineering rather than model fine-tuning as the primary control mechanism. Task-specific prompts impose lightweight structural constraints (e.g., extractive-style answers or standardised formats such as `FINAL_ANSWER:`), which improves output consistency across heterogeneous datasets.

Compared to fine-tuning, this approach avoids additional training overhead, improves reproducibility, and enables rapid adaptation to different tasks and datasets.

(3) Stability and controlled variability

Low-temperature decoding is used to improve output stability by reducing stochastic variation. However, due to inherent randomness in large language models, deterministic decoding alone does not eliminate variability.

To capture residual uncertainty, multiple answers are generated for each question and retained for downstream analysis. This enables the use of self-consistency signals, where disagreement among sampled answers provides a useful indicator of potential hallucination.

(4) Compatibility with detection signals

The generation module is explicitly designed to align with the downstream multi-signal hallucination detection framework. In particular:

- Structured outputs enable reliable parsing and normalisation
- Multiple sampled answers support self-consistency estimation
- Clean and concise answers improve NLI-based factual verification
- Standardised formats ensure consistent LLM-as-a-judge evaluation

This tight integration ensures that detection signals can be applied consistently without requiring complex post-processing.

(5) Robustness through controlled fallback

To ensure robustness, lightweight fallback strategies are applied to handle invalid or unusable outputs (e.g., empty or malformed responses). These mechanisms prioritise maintaining structurally valid outputs while avoiding semantic alterations that could bias evaluation.

In extractive settings, fallback operations select plausible spans directly from the context using simple heuristics. In non-extractive settings, outputs are normalised and retained without introducing additional content.

This conservative design ensures that all samples remain usable for evaluation while minimising unintended modifications to model-generated answers.

Summary

Overall, the generation module is not treated as an isolated component, but as a tightly integrated part of a hallucination-aware pipeline. Its design prioritises producing structured and consistent outputs that are well-suited for downstream evaluation.

By enabling reliable application of multi-signal hallucination detection and supporting uncertainty estimation through multi-sample generation, it provides a stable foundation for both hallucination detection and mitigation.

3.4.7 Summary

In summary, the initial answer generation module adopts a controlled and dataset-aware generation strategy tailored to different task formulations. Extractive prompting is applied in settings where context grounding is required (e.g., *Natural Questions*), while structured non-extractive generation is used for open-ended and classification-style tasks such as *TruthfulQA* and *FEVER*.

The module combines prompt-based constraints, low-temperature decoding, and multi-

sample generation to produce consistent and diverse candidate answers. Lightweight post-processing ensures that outputs remain structurally valid and suitable for further analysis.

These design choices enable the generation module to produce reliable candidate answers that can be directly used for downstream hallucination detection, scoring, and mitigation.

3.5 Hallucination Detection Model

This section presents the proposed hallucination detection model, which formulates hallucination detection as a multi-signal factual reliability estimation problem. Instead of relying on a single confidence score or isolated metric, the model integrates multiple complementary signals to capture different aspects of hallucination behaviour.

In particular, the framework combines semantic evaluation from LLM-as-a-judge, NLI-based signals capturing both entailment and contradiction, and self-consistency signals derived from multiple sampled answers (SelfCheckGPT). The NLI component is further refined through a contradiction-aware scoring scheme, where contradiction probabilities are explicitly penalised to produce a conservative estimate of factual reliability. These signals provide complementary perspectives on hallucination behaviour, capturing semantic correctness, factual consistency, and response stability, and enable more robust and reliable detection compared to single-signal approaches.

The individual signals are aggregated into a unified scoring function, which produces a continuous reliability score for each generated answer. A decision threshold is then applied to classify answers as correct or hallucinated. The threshold is tuned on a validation set and fixed during test evaluation to ensure fair and consistent comparison.

The remainder of this section is organised as follows. Section 3.5.1 defines the detection problem, Section 3.5.2 introduces the multi-signal framework, Sections 3.5.3, 3.5.4, and 3.5.5 describe the individual signals and score aggregation process, and Section 3.5.6 presents dataset-specific adaptations.

3.5.1 Problem Formulation

This project formulates hallucination detection as a probabilistic reliability estimation problem over generated answers. Given a question q , a model-generated answer a , and an optional reference or evidence signal r , the objective is to estimate whether the answer is factually correct or hallucinated.

Formally, the task is defined as learning a scoring function:

$$f(q, a, r) \rightarrow s \in [0, 1] \quad (5)$$

where s represents the estimated factual reliability of the answer. A higher score indicates higher confidence in correctness, while a lower score suggests potential hallucination. A threshold τ is applied to convert the score into a binary decision (hallucinated vs. non-hallucinated).

This formulation is designed to generalise across datasets with heterogeneous structures and supervision signals. Due to differences in annotation schemes and task definitions, hallucination is operationalised differently across datasets, as described below:

- **Natural Questions (NQ):** Each instance consists of a question, an associated context, and a short reference answer.

The hallucination detection task is defined based on span-level factual alignment, where the correctness of a generated answer is evaluated with respect to the reference answer. To reflect the extractive nature of the dataset, correctness is determined using a hierarchy of lexical matching criteria, including exact matching and token-level overlap.

This design reflects the extractive nature of the dataset, where correct answers are expected to correspond closely to textual spans in the source document. Unlike other datasets, semantic NLI models are not used for defining correctness, ensuring consistency with the original annotation protocol.

- **FEVER:** Each instance consists of a claim, associated evidence sentences, and a

veracity label (*SUPPORTED*, *REFUTED*, or *NOT ENOUGH INFO*).

The detection task is defined based on evidence consistency, where a generated answer is considered factually correct if it is consistent with the provided evidence and does not contradict it. Correctness is therefore determined by alignment with the ground-truth label and the absence of contradiction with respect to the evidence, as assessed by a natural language inference (NLI) model.

This formulation frames hallucination as evidence-inconsistent or unsupported generation. A contradiction-aware mechanism is incorporated to penalise outputs that conflict with the available evidence, improving robustness against factually incorrect predictions.

- **TruthfulQA:** Each instance contains a question, multiple correct reference answers, a best answer, and multiple incorrect answers reflecting common misconceptions.

The detection task is defined based on semantic alignment, where a generated answer is considered reliable if it aligns with truthful references while avoiding alignment with known incorrect or misleading responses. Accordingly, hallucination is defined as either deviation from truthful answers or the generation of content that reinforces common misconceptions.

Across all datasets, hallucination is defined as the generation of incorrect, unsupported, or contradictory information relative to available reference signals. Due to the heterogeneity of dataset structures, a unified binary label is constructed for evaluation, while the underlying supervision signals differ across tasks.

This formulation motivates a multi-signal detection approach, where complementary signals (e.g., semantic entailment, contradiction detection, and cross-sample consistency) are integrated to capture different aspects of factual correctness. This design enables a more robust and generalisable hallucination detection framework across diverse task settings.

3.5.2 Multi-signal Detection Framework

To improve robustness across heterogeneous tasks, the proposed detector does not rely on a single confidence signal. Instead, it estimates answer reliability by combining multiple complementary signals extracted from the generated answer under the task context. This choice is motivated by prior work showing that individual signals capture only partial aspects of factual correctness, while their combination can improve robustness across tasks. Given a question q , a generated answer a , and task-dependent reference information r , the framework derives a signal vector:

$$\mathbf{x}(q, a, r) = [x_1, x_2, \dots, x_k] \quad (6)$$

where each component represents a different aspect of factual reliability, such as semantic plausibility, contradiction risk, lexical agreement, or cross-sample consistency.

Rather than enforcing a single fixed scoring function across all tasks, the framework adopts a shared multi-signal principle with flexible aggregation. The final reliability score is computed as:

$$s = h(\mathbf{x}(q, a, r)), \quad (7)$$

where $h(\cdot)$ denotes a task-dependent aggregation function. This design reflects the fact that the three datasets differ substantially in task formulation, supervision structure, and evaluation protocol.

This formulation reflects the diversity of task structures and supervision signals across datasets. While the underlying signals remain consistent, the aggregation strategy may vary depending on the availability of reference answers, evidence, or semantic constraints.

This design allows the detector to integrate complementary signals capturing semantic correctness, logical consistency, and generation stability, forming a unified yet adaptable hallucination detection framework. Instead of fine-tuning a task-specific model, this work adopts a prompt-based and post-hoc detection strategy. This design improves generality across heterogeneous datasets and avoids reliance on large amounts of annotated training data.

3.5.3 Individual Signals

To estimate the factual reliability of generated answers, the proposed framework primarily relies on three complementary signals. Each signal captures a distinct aspect of reliability, and together they contribute to a more robust and stable assessment.

LLM-as-a-Judge

The first signal is a judge-based evaluation obtained from an external large language model (LLM). Given a question, a generated answer, and task-specific reference information (e.g., ground-truth answers, labels, or evidence), the judge model assesses whether the generated answer is factually correct and consistent with the reference.

Formally, this signal is defined as a binary score:

$$s_{\text{judge}} \in \{0, 1\} \quad (8)$$

where $s_{\text{judge}} = 1$ indicates that the generated answer is considered correct and consistent, and $s_{\text{judge}} = 0$ otherwise.

This component provides a high-level semantic correctness signal, complementing the more structured verification methods used in other components.

NLI-based Factual Verification

The second signal evaluates factual consistency between the available evidence and the generated answer using a Natural Language Inference (NLI) model. Across datasets, this signal is computed using pretrained NLI models that estimate the probabilities of entailment and contradiction between a premise and a hypothesis.

Formally, given an evidence text e and a hypothesis h derived from the model output, the NLI model produces two probabilities:

$$p_{\text{ent}}, \quad p_{\text{con}} \quad (9)$$

corresponding to entailment and contradiction, respectively.

To obtain a robust factual consistency score, we adopt a contradiction-aware formulation:

$$s_{\text{fact}} = \max(p_{\text{ent}} - \alpha \cdot p_{\text{con}}, 0) \quad (10)$$

where α is a hyperparameter controlling the penalty for contradiction. This formulation explicitly suppresses answers that are contradicted by the evidence, making it more sensitive to hallucinated content than standard entailment-only scoring.

The construction of the hypothesis depends on the task formulation and is adapted to align with the structure of the generated answer and the available reference information.

This signal provides a principled mechanism for grounding model outputs in available evidence while remaining flexible across different task settings.

Self-consistency

The third signal captures the stability of model outputs under stochastic generation. For each question, multiple answers are sampled from the model, and their mutual consistency is evaluated using a SelfCheckGPT mechanism instantiated with an NLI-based backend.

Formally, given a set of sampled answers $\{a_1, \dots, a_n\}$, we compute an inconsistency score using a pretrained SelfCheck model. For each answer a_i , we compare it against the remaining answers $\{a_j\}_{j \neq i}$ and estimate its inconsistency with respect to the group:

$$\text{SC}(a_i) = \text{SelfCheck}(a_i, \{a_j\}_{j \neq i}) \quad (11)$$

The overall inconsistency is obtained by averaging over all sampled answers:

$$\text{SelfCheck}(\{a_1, \dots, a_n\}) = \frac{1}{n} \sum_{i=1}^n \text{SC}(a_i) \quad (12)$$

and the self-consistency signal is defined as:

$$s_{\text{sc}} = 1 - \text{SelfCheck}(\{a_1, \dots, a_n\}) \quad (13)$$

where higher values indicate stronger agreement across generations.

In practice, answer representations are normalized to reduce superficial variation and ensure that consistency is evaluated at the semantic level.

This signal reflects the model’s internal uncertainty: unreliable outputs tend to vary across samples, resulting in lower consistency, while stable and well-grounded answers exhibit stronger agreement. In implementation, this signal is computed using a SelfCheckGPT-style NLI-based model, where each answer is compared pairwise with the remaining sampled answers. The final consistency score is obtained by averaging the pairwise inconsistency scores, providing a robust estimate of agreement across multiple generations.

3.5.4 Score Aggregation and Calibration

While individual signals capture complementary aspects of factual reliability, none of them alone is sufficient to robustly detect hallucinations across diverse tasks. Prior work has shown that single-signal approaches, such as standalone NLI or self-consistency, often fail under distributional variation or task-specific ambiguity. Therefore, a multi-signal aggregation strategy is adopted to combine their strengths in a principled manner.

Given the signal vector $\mathbf{x}(q, a, r) = [s_{\text{judge}}, s_{\text{fact}}, s_{\text{sc}}]$, the final reliability score is computed as a weighted combination:

$$S(a) = w_j \cdot s_{\text{judge}} + w_f \cdot s_{\text{fact}} + w_s \cdot s_{\text{sc}} \quad (14)$$

where w_j, w_f, w_s are non-negative weights satisfying $w_j + w_f + w_s = 1$. In practice, these weights are manually specified based on prior intuition about the relative importance of each signal and remain fixed during both validation and test evaluation. Ablation studies are conducted to assess the contribution of individual components.

This formulation reflects the complementary roles of the signals: the judge signal provides high-level semantic assessment, the NLI-based signal enforces evidence-grounded factuality, and the self-consistency signal captures model uncertainty through agreement across sampled generations. By combining these signals, the framework achieves greater robustness than any individual component.

A threshold-based decision rule is applied to convert the continuous score into a binary prediction:

$$\hat{y} = \begin{cases} \text{factual} & \text{if } S(a) \geq \tau \\ \text{hallucinated} & \text{otherwise} \end{cases} \quad (15)$$

where the threshold τ is tuned on a validation set. This design follows standard practice in classification tasks, ensuring that the decision boundary is adapted to the score distribution of the model.

However, raw scores produced by the aggregation function are not guaranteed to be well-calibrated probabilities. In particular, different signals may operate on different scales, and their weighted combination may not reflect true likelihoods of hallucination. To address this, a post-hoc calibration step is applied.

Specifically, isotonic regression is used to learn a monotonic mapping:

$$P(\text{hallucination} \mid S(a)) = f_{\text{iso}}(S(a)) \quad (16)$$

which transforms the raw score into a calibrated probability. Compared to parametric alternatives such as Platt scaling, isotonic regression makes fewer assumptions about the underlying score distribution and has been shown to perform well in settings with heterogeneous signals.

This calibrated probability improves interpretability and enables more consistent decision-making across datasets and experimental settings. It also facilitates downstream analysis, such as threshold selection, risk-aware evaluation, and mitigation strategies.

Overall, the combination of weighted aggregation and probabilistic calibration provides a principled and flexible mechanism for hallucination detection, balancing accuracy, robustness, and interpretability.

3.5.5 Auxiliary Mechanisms

In addition to the primary signals and aggregation mechanism, the proposed framework incorporates several auxiliary components to further enhance robustness and reliability. These mechanisms operate at different stages of the pipeline, complementing the core multi-signal estimation.

First, a contradiction-aware filtering mechanism is introduced to explicitly suppress factually inconsistent outputs. An independent NLI model is used to estimate contradiction between the generated answer and the available reference (e.g., evidence or context). Candidate answers with high contradiction scores are downweighted or discarded, providing a strong constraint against hallucinated content.

Second, a lightweight candidate filtering strategy is applied to remove structurally unreliable outputs. This includes excessively verbose responses, underspecified answers, or outputs containing unnecessary reasoning patterns that do not contribute to factual correctness. This step improves the overall quality of candidates prior to scoring and aggregation.

Third, a group-based reranking mechanism is employed when multiple candidate answers are available. By leveraging agreement across candidate outputs, the framework prioritises answers that are both factually consistent and mutually supported, improving robustness against stochastic generation variability.

Fourth, an abstention-aware control mechanism is incorporated for open-ended question answering settings. Instead of forcing a potentially incorrect prediction, the system allows abstention when confidence is insufficient. This design is particularly important for tasks where incorrect answers are more harmful than missing answers.

Finally, these auxiliary mechanisms are integrated in a modular manner, allowing flexible adaptation across different datasets and task formulations. While the core detection framework remains unchanged, the use and importance of each mechanism may vary depending on the characteristics of the task, as further discussed in Section 4.5.6.

Overall, these components complement the primary signals by enforcing structural and logical constraints, resulting in a more reliable and stable hallucination detection process.

3.5.6 Dataset-specific Adaptation

While the proposed framework adopts a unified multi-signal detection strategy, the characteristics of different datasets require task-specific adaptations in how signals are interpreted and applied. In particular, differences in task formulation, supervision signals, and answer structure necessitate tailored implementations to ensure effective hallucination detection.

For **TruthfulQA**, which is an open-ended question answering task, hallucination detection focuses on distinguishing truthful responses from common misconceptions. In this setting, the judge signal serves as a primary semantic evaluator, assessing alignment with correct knowledge while avoiding agreement with known misconceptions. The self-consistency signal further captures instability across multiple generated responses, reflecting model uncertainty. Since answers are free-form and may include additional correct information, the framework emphasises semantic correctness rather than exact matching.

In addition, a calibrated hallucination probability is estimated using a logistic regression layer, which integrates multiple signals and provides a more stable basis for decision-making. Abstention is explicitly incorporated as a detector-controlled mechanism, where low-confidence or high-risk predictions are replaced with abstained responses. This process is further regulated by a utility-aware strategy to ensure that abstention is only selected when it improves overall answer reliability.

For **FEVER**, a fact verification task, the objective is to determine whether a claim is supported, refuted, or lacks sufficient evidence. In this setting, the NLI-based factuality signal becomes the dominant component, as the task is inherently framed as an evidence-grounded entailment problem between the claim and the provided evidence.

The framework employs both detector-level NLI scoring and an independent GOLD NLI model for contradiction filtering, providing a strong constraint against evidence-

inconsistent predictions. Candidate answers that exhibit high contradiction are explicitly filtered out, rather than merely downweighted, enforcing strict evidence consistency.

In addition, a label-group-based reranking mechanism is applied to avoid mixing conflicting predictions (e.g., SUPPORTED, REFUTED, and Not Enough Information), thereby improving robustness within the structured output space. Compared to open-ended QA tasks, the answer space in FEVER is more constrained, allowing consistency-based strategies to be more effective.

While the “Not Enough Information” label naturally represents uncertainty within the task formulation, it also aligns with the framework’s abstention behaviour under low-confidence conditions.

For **Natural Questions**, an extractive question answering task, answers are expected to be grounded in the provided context, typically as short spans. In this setting, lexical overlap signals (e.g., token-level F1) become a primary component, enforcing alignment between generated answers and reference spans.

Unlike FEVER, the NLI-based signal is not the dominant component, but is used as a contradiction-aware constraint with explicit penalty on conflicting predictions. This helps suppress factually inconsistent outputs without overriding span-level matching.

The framework further enforces extractive constraints by restricting candidate answers to spans derived from the context, ensuring that predictions remain grounded and do not introduce unsupported information.

Self-consistency still contributes to robustness by capturing variability across sampled answers, although its impact is reduced compared to open-ended settings. Unlike TruthfulQA, explicit abstention is disabled in the current configuration, as answers are expected to be extractable from the context.

Overall, these adaptations demonstrate that while the core multi-signal framework remains consistent, the relative importance and interpretation of each signal are adjusted

according to the task characteristics. This design enables the framework to generalise across diverse settings while maintaining strong performance in each specific scenario.

3.6 Threshold Selection

To convert the continuous scoring function into binary hallucination predictions, we select decision thresholds on a held-out validation set.

For each scoring variant, we perform a grid search over thresholds in $[0, 1]$ and select the value that maximizes the F1 score for hallucination detection. The selected threshold is then fixed and applied to the test set without further tuning.

Formally, given a score $s(x)$ for an answer x , we predict it as non-hallucinated if $s(x) \geq t^*$, and hallucinated otherwise, where t^* is the optimal threshold determined on the validation set.

It is important to note that the definition of the scoring function $s(x)$ varies across datasets, reflecting differences in task structure and supervision signals. For open-ended QA tasks such as TruthfulQA, $s(x)$ is derived from a calibrated hallucination probability combined with heuristic signals. For fact verification tasks such as FEVER, $s(x)$ is primarily based on NLI-driven factual consistency with evidence. For extractive QA tasks such as Natural Questions, $s(x)$ integrates lexical overlap, consistency, and contradiction-aware signals. Despite these differences, the same threshold selection procedure is consistently applied across all datasets.

3.6.1 Summary

Overall, the hallucination detection model can be viewed as a two-stage process. First, multiple complementary signals are extracted and aggregated into a unified score $s(x)$ through a combination of heuristic aggregation and probabilistic calibration. Second, a threshold-based decision rule is applied to obtain the final prediction.

This formulation provides a flexible and modular framework, where different signals and calibration strategies can be incorporated while maintaining a consistent decision mechanism across datasets.

The proposed framework introduces additional computational overhead due to multi-sample generation and NLI-based evaluation. However, this trade-off is necessary to improve robustness, and the modular design allows flexible adjustment of the number of samples and signals depending on resource constraints.

3.7 Mitigation Strategy

This section presents a closed-loop mitigation framework designed to reduce hallucinations in large language model (LLM) outputs. Given an initial answer generated by the model, the proposed approach first estimates its factual reliability using a multi-signal hallucination detector, and then applies a detector-conditioned correction strategy to improve the answer.

The overall framework integrates three key components: (1) candidate generation, where multiple revised answers are produced using controlled prompting strategies; (2) candidate filtering and reranking, where unreliable or low-quality candidates are removed and the remaining ones are ranked based on calibrated factuality scores; and (3) an abstention mechanism, which allows the system to return a “do not know” response when all candidate answers are deemed unreliable.

A central feature of the proposed method is the use of a calibrated hallucination probability to guide the mitigation process. Based on this score, the system dynamically decides whether to retain the original answer, apply mild correction, perform more aggressive rewriting, or abstain entirely. This enables a principled trade-off between answer correctness and informativeness.

Furthermore, the mitigation strategy is adapted to different dataset characteristics, allowing it to handle diverse task types such as open-ended question answering, fact verification, and extractive answering. The following subsections describe each component of the framework in detail.

3.7.1 Problem Formulation

Given a question q and an initial model-generated answer a , the objective of the mitigation module is to reduce hallucinations by transforming a into a more factually reliable answer a' , or abstaining when a reliable answer cannot be produced.

To support this process, we define a scoring function $f(q, a)$ that estimates the factual reliability of an answer. The function outputs an *acceptability score* $\text{ok_score} \in [0, 1]$, where higher values indicate higher confidence in correctness.

The acceptability score is computed by combining a calibrated hallucination estimate with auxiliary multi-signal evidence:

$$\text{ok_score}(q, a) = \alpha \cdot (1 - p_{\text{hall}}(q, a)) + (1 - \alpha) \cdot s_{\text{aux}}(q, a) \quad (17)$$

where $p_{\text{hall}}(q, a)$ denotes the estimated probability that the answer is hallucinatory, and $s_{\text{aux}}(q, a)$ represents an auxiliary reliability score derived from multiple complementary signals. The parameter α controls the relative importance of calibrated estimation versus auxiliary signals.

The hallucination probability p_{hall} is obtained via a calibration model (logistic regression), which takes as input features derived from multiple signals. These signals include:

- **LLM-as-a-Judge**: a semantic assessment of answer correctness,
- **NLI-based consistency**: entailment and contradiction signals computed between the answer and available references or alternative answers,
- **Self-consistency**: agreement across multiple independently generated answers.

The auxiliary score s_{aux} aggregates these signals into a continuous estimate of answer reliability.

The mitigation problem can be formulated as selecting an output a' from a candidate set $\mathcal{C}$ (including the original answer and its rewrites), or producing an abstention response, such that:

$$a' = \arg \max_{a \in \mathcal{C} \cup \{\text{abstain}\}} \text{ok_score}(q, a) \quad (18)$$

subject to risk-aware constraints that allow abstention when all candidates are deemed unreliable.

In practice, this optimisation is implemented as a structured decision process: candidate answers are generated conditionally based on the estimated risk of the initial answer, filtered using validity and consistency constraints, and reranked according to their acceptability scores, with abstention applied when confidence remains insufficient.

This formulation provides a unified framework for answer correction, candidate selection, and abstention, and is consistently applied across different datasets and task settings.

3.7.2 Overall Mitigation Framework

The proposed mitigation framework adopts a structured, closed-loop process to improve the factual reliability of model-generated answers while controlling hallucination risk. Given a question q and an initial answer a , the framework aims to produce a refined answer a' or abstain when a reliable response cannot be ensured.

The overall process consists of three main stages:

Stage 1: Risk Assessment The initial answer a is evaluated under a unified acceptability estimation framework, which approximates the underlying objective of selecting factually reliable responses. This produces an *ok_score* that reflects the estimated reliability of the answer. The scoring integrates multiple complementary signals, including semantic consistency, entailment-based factuality, and cross-sample agreement. Based on this score, the answer is categorised into different risk levels, which determine whether further mitigation is necessary.

Remark: While the overall framework remains consistent, the specific instantiation of the scoring function varies across datasets. For example, *TruthfulQA* emphasises calibrated hallucination estimation, whereas *FEVER* and *Natural Questions* place greater emphasis on evidence alignment and answer-span correctness.

Stage 2: Candidate Generation If the initial answer is deemed uncertain, a set of alternative candidate answers $\mathcal{C}$ is constructed through controlled transformations of the original response. These transformations aim to preserve correct information while correcting factual inaccuracies and reducing hallucination risk.

Candidate filtering is applied to remove low-quality or inconsistent outputs. The generation strategy is adapted to the task formulation: extractive-style refinement is emphasised in *Natural Questions*, while more flexible transformations are used in *TruthfulQA* and *FEVER*.

Stage 3: Candidate Selection and Abstention All candidates are re-evaluated under the same acceptability estimation framework. The final output a' is selected by optimising the acceptability objective under risk-aware constraints. Additional consistency signals may be incorporated to improve robustness.

If no candidate satisfies the required reliability criteria, the system may abstain from answering. This decision is governed by risk-aware policies and dataset-specific thresholds, enabling a controlled trade-off between answer accuracy and coverage.

Overall, this framework provides a practical approximation of the optimisation objective defined in the previous section. By combining risk-aware decision-making, controlled transformation, and calibrated scoring, it enables a unified and robust approach to hallucination mitigation across heterogeneous tasks.

3.7.3 Candidate Generation

Given an initial answer a , the candidate generation stage constructs a set of alternative responses $\mathcal{C}$ intended to improve factual reliability while preserving useful information. At a high level, this stage does not rely on unrestricted regeneration; instead, it applies controlled candidate construction strategies that are adapted to the structure of each task.

For generative tasks such as *TruthfulQA* and *FEVER*, candidate answers are mainly produced by transforming the initial answer through prompt-based correction. Multiple editing strategies with different intervention strengths are used. Mild rewriting aims to

preserve as much of the original answer as possible while correcting clearly problematic content, whereas stricter rewriting performs stronger factual correction and encourages shorter, more conservative outputs. In *TruthfulQA*, an additional intermediate edit-based strategy is used to balance faithfulness to the original wording and factual correction. In *FEVER*, the strict strategy can further incorporate evidence-span extraction, so that the rewritten answer is more tightly constrained by the provided evidence and the required label format.

For extractive tasks such as *Natural Questions*, candidate generation follows a different mechanism. Instead of free-form rewriting, the system primarily constructs candidates by extracting short answer spans from the local context. Candidate spans are generated using overlap-based heuristics between the question and context, and the resulting answers are biased towards short, contiguous, context-grounded spans. This makes the generation process more consistent with the short-answer nature of the task.

To improve robustness, multiple candidates are constructed for each question, producing a candidate set $\mathcal{C}$. Candidate filtering is then applied to remove low-quality outputs. The exact filtering criteria vary slightly across datasets, but generally exclude empty, malformed, excessively long, weakly informative, or inconsistent responses. For example, *TruthfulQA* removes answers with undesirable reasoning-style patterns or implausible content, *FEVER* discards candidates that fail label parsing or contradict the evidence under NLI-based checks, and *Natural Questions* filters candidates using context-groundedness and answer-length constraints.

Overall, this stage produces a diverse but controlled set of candidate answers that serves as the basis for the subsequent reranking and risk-aware selection procedure.

3.7.4 Candidate Filtering and Reranking

After candidate generation, the framework applies a two-stage selection process consisting of candidate filtering and risk-aware reranking, in order to identify the most reliable final answer. This procedure is shared across all datasets, while allowing dataset-specific constraints to be incorporated where necessary.

Candidate Filtering

The initial candidate set $\mathcal{C}$ may contain low-quality or unreliable responses. Therefore, a filtering step is first applied to remove candidates that are unlikely to be useful. The filtering procedure follows a unified design across datasets, with additional task-specific constraints.

First, invalid candidates are removed, including empty responses, malformed outputs, or answers that fail to follow the required output format (e.g., missing the `FINAL_ANSWER` field). Second, length-based constraints are applied to discard overly short or excessively long answers, which are empirically associated with low information content or increased hallucination risk. Third, heuristic patterns associated with unreliable generation are filtered out. In generative settings such as *TruthfulQA*, answers containing unnecessary reasoning-style expressions (e.g., “because”, “therefore”) are removed, as such patterns are often correlated with hallucinated explanations.

In addition, lightweight semantic validation is performed using an auxiliary judge model. Candidates that are deemed implausible, misleading, or inconsistent with the question are discarded. This judge operates without access to ground-truth references and provides a coarse but efficient filtering signal.

Beyond these shared criteria, dataset-specific constraints are enforced:

- For *FEVER*, candidates must produce valid labels (e.g., `SUPPORTED`, `REFUTED`, or `NOT ENOUGH INFO`) and remain consistent with the provided evidence. Outputs that fail label parsing or contradict evidence are removed.
- For *Natural Questions*, which is extractive in nature, candidate answers are required to be grounded in the given context. In strict mitigation settings, span-based constraints are applied to ensure that answers correspond to plausible text spans from the context.

Candidate Grouping

After filtering, the remaining candidates are grouped based on their semantic type. Specifically, candidates are divided into two groups: an *answer* group and an *abstention* group. The latter includes cautious fallback responses such as “I do not know” or semantically

equivalent variants.

This grouping prevents direct competition between informative answers and abstention responses, as they follow different optimization objectives: the former aims to maximize correctness, while the latter prioritizes risk avoidance.

Risk-aware Reranking

Within each group, candidates are evaluated using a composite reliability score derived from multiple complementary signals. In the implemented framework, this score combines a calibrated hallucination probability with heuristic reliability components.

The hallucination probability is estimated using a learned calibration model based on features including:

- LLM-as-a-Judge outputs
- NLI-based semantic consistency (entailment and contradiction signals)
- self-consistency across sampled answers (SelfCheck)
- auxiliary heuristic features (e.g., contradiction flags and agreement scores)

These features are combined through a logistic regression calibrator trained on validation data, producing a probabilistic estimate of hallucination risk. This calibrated probability is further integrated with heuristic acceptability scores to form the final reliability score. Reranking is first performed independently within each group. For each group, the candidate with the lowest estimated hallucination probability (or highest reliability score) is selected as the group representative. The final decision is then made by comparing the best candidates across groups.

Abstention Control

To further control hallucination risk, an abstention mechanism is applied. If the best candidate remains insufficiently reliable, the system may output an abstention response instead of returning a potentially incorrect answer.

This decision is governed by validation-tuned thresholds, including:

- lower and upper reliability thresholds that determine whether mitigation is triggered, and
- a separate abstention threshold that controls when fallback responses are preferred.

In addition, utility-aware control can be optionally applied to balance informativeness and safety, preventing unnecessary abstention when a sufficiently informative answer is available.

Summary

Overall, this filtering and reranking procedure enables the system to systematically reduce hallucination risk while preserving useful information. By combining multi-signal reliability estimation, dataset-specific constraints, and calibrated decision thresholds, the framework achieves a robust balance between correctness, reliability, and answer coverage across different

3.7.5 Detector-Conditioned Policy

To further reduce hallucination while preserving informative answers, we introduce a detector-conditioned mitigation policy that dynamically adjusts the correction strategy based on the estimated reliability of the initial model output. This policy follows a unified high-level framework across all datasets (TruthfulQA, FEVER, and Natural Questions), while allowing dataset-specific implementations.

Reliability Estimation

Given a question q and an initial answer a_0 , the system computes a reliability score $s(a_0)$ using a multi-signal hallucination detector. The score is derived from a weighted combination of three components:

- LLM-as-a-Judge evaluation
- NLI-based semantic consistency between the answer and reference or evidence
- self-consistency across multiple sampled answers

This produces an aggregated acceptability score used directly for decision-making. Additionally, a calibration model (e.g., isotonic regression) is trained on validation data to map scores to hallucination probabilities, which are used for analysis and evaluation rather than primary decision control.

Policy Decision

Based on the estimated reliability score, the system selects one of three strategies using two thresholds, t_{low} and t_{high} , tuned on validation data:

- **Low-risk region** ($s(a_0) \geq t_{\text{high}}$): The answer is considered sufficiently reliable and is returned unchanged.
- **Medium-risk region** ($t_{\text{low}} \leq s(a_0) < t_{\text{high}}$): A correction strategy is applied, where multiple candidate answers are generated through controlled rewriting. Depending on dataset configuration, this may use mild rewriting or stricter evidence-grounded rewriting.
- **High-risk region** ($s(a_0) < t_{\text{low}}$): A stricter correction strategy is applied. A larger set of candidate answers is generated using aggressive rewriting prompts, optionally constrained by evidence or extracted spans to reduce hallucination risk.

Candidate Generation

For medium- and high-risk cases, multiple candidate answers are generated via prompt-based rewriting. The generation strategy varies across datasets:

- In *TruthfulQA*, candidates are generated using multiple prompting strategies (e.g., mild rewriting, strict rewriting, and answer editing).
- In *FEVER*, candidates are generated in a label-constrained format and may incorporate evidence span extraction to ensure grounding.
- In *Natural Questions*, candidates may include extractive answers derived from context spans as well as rewritten responses.

Candidate Filtering and Reranking

Generated candidates are processed through a filtering and reranking pipeline. This includes:

- **Filtering:** removing invalid, empty, or semantically inconsistent candidates. In particular, NLI-based checks are used to discard candidates that contradict the reference or evidence.

- **Grouping:** candidates may be grouped according to task structure (e.g., FEVER labels such as SUPPORTED / REFUTED / NOT ENOUGH INFO) to avoid inconsistent comparisons.
- **Reranking:** candidates are evaluated using the same detector score $s(\cdot)$, and the highest-scoring candidate is selected.

This ensures consistency between detection and mitigation.

Abstention Mechanism

If the best candidate remains insufficiently reliable (i.e., below an abstention threshold), the system may abstain from answering. The form of abstention depends on the dataset:

- In *TruthfulQA*, abstention is expressed as natural language responses (e.g., “I do not know”).
- In *FEVER*, abstention is implemented explicitly as a special output token (e.g., ABSTAIN), which is distinct from the standard label *NOT ENOUGH INFO*.
- In *Natural Questions*, it is not enabled in the current configuration. The system is designed to always produce an answer span or candidate, following an extractive answering strategy.

This decision is governed by validation-tuned thresholds and acts as a safety mechanism to prevent high-risk outputs.

Dataset-Specific Adaptations. While the overall framework is shared, each dataset introduces specific adaptations:

- In *TruthfulQA*, the policy focuses on correcting misleading or misconception-based answers using diverse prompting strategies and candidate reranking.
- In *FEVER*, the policy operates over structured label outputs and enforces strict consistency with evidence through NLI-based filtering and label-group reranking.
- In *Natural Questions*, the policy emphasizes extractive consistency by incorporating context-based answer spans and prioritizing answers grounded in the provided passage.

Summary

Overall, the detector-conditioned policy provides a unified yet adaptable mitigation framework. By combining multi-signal reliability estimation, threshold-based strategy selection, and dataset-specific candidate generation and filtering, the system effectively reduces hallucination while maintaining informative and task-consistent responses across diverse evaluation settings.

3.7.6 Dataset-specific Mitigation Strategies

Although the overall mitigation framework is unified, the implementation of the detector-conditioned policy is adapted to the characteristics of each dataset to ensure effectiveness across different task formulations.

TruthfulQA

TruthfulQA focuses on open-ended question answering where hallucinations often arise from common misconceptions or fabricated facts. In this setting, mitigation is primarily performed through controlled rewriting. Given an initial answer, the system generates multiple candidate responses using prompt-based rewriting strategies with varying strictness levels (e.g., mild editing and stricter correction prompts). These candidates are evaluated using the hallucination detector, which incorporates LLM-as-a-Judge, NLI-based semantic signals, and self-consistency across sampled answers.

Since no explicit ground-truth evidence is provided, the mitigation process relies entirely on model-internal consistency and semantic plausibility rather than external grounding. Candidate filtering is performed using lightweight heuristics (e.g., removing overly long, overly short, or reasoning-heavy answers) combined with a reference-free judge that eliminates candidates likely to contain misconceptions.

Candidates are further grouped into two categories: answer candidates and abstention candidates (e.g., “I do not know”). Reranking is performed across these groups using the calibrated detector score, and a utility-aware abstention mechanism may select abstention when all answer candidates remain unreliable. This enables the system to reduce hallucinations while maintaining informative responses when possible.

FEVER

FEVER is a fact verification task with structured labels (*SUPPORTED*, *REFUTED*, *NOT ENOUGH INFO*) and associated evidence. The mitigation strategy is therefore designed to enforce strict consistency with the provided evidence.

During candidate generation, both mild and strict rewriting prompts are employed. The strict strategy optionally incorporates a span extraction step, where relevant text segments are copied verbatim from the evidence and used to constrain the generated answer.

Candidate filtering is more aggressive than in other datasets. Candidates are discarded if they violate NLI consistency constraints (i.e., entailment lower than contradiction) or if they contradict the evidence according to a separate GOLD NLI model. If all candidates are filtered out, the system falls back to the unfiltered set to avoid degeneration.

Furthermore, candidates are grouped by predicted label, and reranking is performed within each group to avoid mixing inconsistent labels during self-consistency evaluation.

Abstention is explicitly supported via a dedicated token (e.g., **ABSTAIN**). Rather than requiring all candidates to be invalid, abstention is triggered when the highest-scoring candidate falls below a predefined confidence threshold, as determined by the detector score. This allows the system to avoid making unsupported claims when evidence is insufficient or conflicting.

paragraphNatural Questions

Natural Questions is an extractive question answering task where answers are expected to correspond to short spans within a given context. Consequently, the mitigation strategy emphasizes extractive consistency rather than free-form generation. Candidate answers are primarily derived from context spans, and strict correction prioritizes selecting short, context-aligned answers.

Unlike FEVER, no hard evidence-based contradiction filtering is applied; instead, contradiction signals are incorporated as soft penalties within the NLI-based scoring function. Unlike TruthfulQA, abstention is not enabled in the current configuration, and the system is required to always produce an answer, with fallback mechanisms ensuring answer

coverage.

While extractive span candidates are prioritized, limited short-form rewriting is also allowed as a supplementary candidate generation strategy. This design reflects the extractive nature of the task, where selecting a plausible context-aligned span is generally preferred over abstaining.

Summary

These dataset-specific adaptations ensure that the mitigation strategy aligns with the underlying task structure: open-ended semantic reasoning in TruthfulQA, evidence-grounded verification in FEVER, and extractive answering in Natural Questions. By tailoring candidate generation, filtering, and abstention behavior accordingly, the system aims to improve hallucination mitigation, although the effectiveness varies across datasets and task settings.

3.8 Summary

In this section, we presented a closed-loop framework that integrates answer generation, hallucination detection, estimation and mitigation in a unified pipeline. The approach first produces candidate answers using a base language model, including both direct outputs and sampled variations, followed by a multi-signal hallucination detector that combines LLM-as-a-judge, NLI-based factuality scoring, self-consistency, lexical overlap, and additional auxiliary signals. These signals are aggregated into a continuous confidence score to assess answer reliability.

Based on the detector output, a detector-conditioned mitigation strategy is applied, which dynamically selects between preserving the original answer, performing mild rewriting, or enforcing stricter correction through candidate generation and reranking. Importantly, the mitigation strategy is adapted to the characteristics of each dataset, including evidence-grounded verification in FEVER, extractive span selection in Natural Questions, and open-ended reasoning in TruthfulQA.

Overall, this design enables a flexible and modular framework that not only detects hallucinations but also actively mitigates them. While the effectiveness varies across datasets,

the results highlight the importance of dataset-aware mitigation strategies and demonstrate that hallucination reduction remains a task-dependent challenge in large language models.

Chapter 4

Implementation and Evaluation

This chapter presents the technical implementation of the proposed hallucination detection, estimation and mitigation framework, along with the experimental setup used for evaluation. The implementation details describe how the proposed methodology is realized in practice, including system architecture, model integration, and pipeline design. Following this, the experimental setup is introduced, covering the datasets, models, and evaluation metrics used in the study. Finally, the evaluation protocol is described, including data splitting, threshold tuning, and the experimental procedures used to ensure reliable and reproducible results.

4.1 System Implementation

4.1.1 Overall Pipeline

The proposed system is implemented as a closed-loop pipeline that integrates answer generation, hallucination detection, and mitigation in a unified workflow. This design ensures that the model not only produces answers but also evaluates and refines them based on estimated factual reliability.

Given an input question, the system first generates one or more candidate answers using a base language model. Depending on the dataset and configuration, this may include direct model outputs, sampled responses for self-consistency evaluation, or extractive span candidates derived from the provided context (e.g., in Natural Questions).

Each candidate answer is then evaluated by a multi-signal hallucination detector. The detector integrates several complementary signals, including LLM-as-a-judge for semantic correctness, NLI-based entailment and contradiction scores for factual consistency, self-consistency scores across multiple sampled answers, and lexical overlap with reference answers. These heterogeneous signals are aggregated into a continuous confidence score

representing the estimated reliability of each candidate.

Based on the detector output, a mitigation module is applied to refine the answer. The mitigation strategy is detector-conditioned and dynamically adjusts its behavior according to the confidence score. High-confidence answers are preserved, while lower-confidence answers trigger additional correction steps such as candidate regeneration, extractive span selection, or reranking based on the aggregated score.

Importantly, the pipeline is designed to support multiple datasets with different task characteristics. For FEVER, the system emphasizes evidence-grounded verification and strict consistency filtering. For Natural Questions, the pipeline prioritizes extractive span selection from the context. For TruthfulQA, the system allows more flexible answer generation while relying on the detector to assess semantic correctness. Despite these variations, all datasets share the same unified pipeline structure, ensuring consistency in detection and mitigation while allowing task-specific adaptations.

Finally, the refined answer is selected as the system output. The proposed system is implemented as a closed-loop pipeline that integrates answer generation, hallucination detection, and mitigation in a unified workflow. This design ensures that the model not only produces answers but also evaluates and refines them based on estimated factual reliability. As illustrated in Figure 1 in the System Overview, the overall pipeline follows a closed-loop structure integrating generation, detection, and mitigation.

4.1.2 Hallucination Detection Implementation

The hallucination detection component is implemented as a unified multi-signal framework that evaluates the factual reliability of candidate answers using multiple complementary signals. This design is shared across all three datasets (TruthfulQA, FEVER, and Natural Questions), while allowing dataset-specific adaptations in label definition and evaluation criteria.

For each candidate answer, several signals are computed. First, an LLM-as-a-judge module is employed to assess semantic correctness by comparing the generated answer with the reference answer, producing a binary or soft correctness judgment. Second, a nat-

ural language inference (NLI) model is used to compute entailment and contradiction probabilities between the reference text and the generated answer, capturing fine-grained factual consistency. Third, a self-consistency signal is obtained using multiple sampled answers, where disagreement across samples indicates a higher likelihood of hallucination. In practice, multiple answers (e.g., $N = 5$) are generated for each question, and pairwise NLI comparisons are performed to estimate agreement across answers, where disagreement across samples indicates a higher likelihood of hallucination. In addition, lexical overlap is measured using token-level F1 scores between the generated answer and the reference answer, providing a lightweight similarity signal.

These signals are aggregated into a continuous confidence score using a weighted combination. The weights used for signal aggregation are manually specified based on empirical observations of signal reliability and remain fixed during both validation and test-time evaluation. Rather than tuning weights directly, ablation studies are conducted to assess the contribution of individual signals. Specifically, the detector integrates semantic judgment, NLI-based factuality, self-consistency, and lexical similarity into a unified scoring function, with additional penalties applied to contradiction signals to discourage factually incorrect answers. The resulting score represents the estimated reliability of each candidate and is used both for hallucination classification and for downstream mitigation. These signals are complementary in nature: the LLM-as-a-judge captures high-level semantic correctness, NLI models provide fine-grained factual consistency, self-consistency reflects answer stability, and lexical overlap offers a lightweight similarity measure. Their combination enables more robust hallucination detection than any single signal alone.

While the detector architecture remains consistent across datasets, the definition of correctness and hallucination varies depending on the task. In TruthfulQA, hallucination is defined based on semantic alignment with known correct answers and contradiction with incorrect ones. In FEVER, the detector operates at the claim level, where predictions must be consistent with both the ground-truth label and the supporting evidence. In Natural Questions, correctness is determined using exact match and token-level F1 with short reference answers, reflecting the extractive nature of the task.

Overall, this implementation provides a flexible yet consistent detection mechanism that

can be applied across different question answering settings, while adapting to dataset-specific evaluation criteria.

4.1.3 Mitigation Strategy Implementation

The key idea of the mitigation strategy is to treat hallucination correction as a constrained search problem, where candidate answers are generated, filtered, and reranked based on a unified reliability score. This design allows the system to balance correction, diversity, and safety (via abstention) in a principled manner. The mitigation module is implemented as a detector-conditioned post-processing pipeline that operates on cached model outputs and integrates candidate generation, filtering, reranking, and abstention decisions. While the overall structure is shared across datasets, the specific implementation details are adapted to each task.

Pipeline Overview

For each input instance, the system first obtains an initial answer and its sampled variants (cached for efficiency). A multi-signal hallucination detector is then applied to compute a reliability score for each candidate. Based on the score of the initial answer, the system determines the mitigation level using two thresholds (t_{low} , t_{high}), which are tuned on a validation split and fixed during test-time evaluation.

If mitigation is triggered, additional candidate answers are generated using prompt-based rewriting or task-specific strategies (e.g., extractive span selection in Natural Questions). All candidates are then passed through a filtering and reranking pipeline, and the final answer is selected based on the highest acceptability score. If no candidate meets the required threshold, the system may abstain when enabled for the dataset configuration.

TruthfulQA Implementation

For TruthfulQA, candidate answers are generated using multiple prompt templates with varying levels of strictness (e.g., mild rewriting and strict correction prompts). These prompts are executed through an external LLM interface, producing multiple rewritten candidates.

Candidate filtering removes outputs that are empty, excessively long, or contain unde-

sirable reasoning patterns. In addition, a semantic judge model is used to eliminate implausible responses.

Candidates are first grouped into answer and abstention categories, and reranking is performed within each group before selecting the best candidate across groups. Reranking is based on a calibrated reliability score, which combines a calibrated hallucination probability with heuristic signals derived from judge outputs, NLI-based consistency scores, and self-consistency signals.

Abstention (e.g., “I do not know”) is enabled and triggered when all answer candidates fall below a predefined reliability threshold.

FEVER Implementation

For the FEVER dataset, the mitigation process is designed for structured label prediction, where candidate answers are generated using prompt templates that enforce explicit outputs (SUPPORTED, REFUTED, NOT ENOUGH INFO).

In strict mitigation mode, evidence sentences are incorporated into the prompt, and a two-stage process is optionally applied: relevant spans are first extracted from the evidence and then used to constrain answer generation. This design ensures that the generated responses remain grounded in the provided evidence.

Candidate filtering is more aggressive compared to other datasets. Outputs are discarded if: (1) the predicted label cannot be parsed, (2) the answer violates NLI consistency constraints (entailment score lower than contradiction), or (3) the answer contradicts the evidence according to a separate GOLD NLI model.

If all candidates are filtered out, a fallback mechanism is applied to retain unfiltered candidates, preventing empty outputs.

After filtering, candidates are grouped by predicted label, and reranking is performed within each group to avoid mixing inconsistent predictions. The final answer is selected based on the highest reliability score, which integrates multiple signals including LLM-

as-a-Judge, NLI-based factuality, and self-consistency.

The mitigation strategy is detector-conditioned: the strength of rewriting (e.g., mild vs. strict and the number of candidates generated) is dynamically determined based on the hallucination risk score of the base answer.

An abstention mechanism is implemented via a decision threshold, where the system outputs a dedicated token (e.g., **ABSTAIN**) when no candidate satisfies the reliability requirement.

Natural Questions Implementation

For Natural Questions, the mitigation strategy follows a primarily extractive paradigm. Candidate answers are constructed from spans extracted directly from the provided context, rather than relying solely on free-form generation.

Span candidates are generated using heuristic matching between the question and the context, based on token-level overlap. When model outputs are invalid or empty, additional fallback spans are selected from the context to ensure answer availability.

Most candidates are context-grounded spans. However, a small number of short rewritten answers may also be introduced as auxiliary candidates under relaxed constraints.

Candidate filtering enforces span validity for extractive answers by requiring them to appear as contiguous segments in the context. In contrast, short rewritten answers are retained with looser constraints to maintain diversity.

Unlike FEVER, contradiction signals are not used as a hard rejection rule. Instead, contradiction is incorporated as a penalty term within the scoring function, allowing soft constraint-based optimization.

Reranking is performed using a combination of lexical overlap (token-level F1), self-consistency (SelfCheckGPT), and NLI-based factuality signals. The candidate with the highest aggregated score is selected as the final answer.

Abstention is disabled in the current configuration to ensure full answer coverage during evaluation, and the system always outputs a valid answer span, even in cases where the confidence of the selected candidate is relatively low.

Implementation Details

The mitigation pipeline is implemented in Python and operates on precomputed answer caches to ensure efficiency and reproducibility. All candidate scoring and filtering steps are deterministic given the cached inputs, while candidate generation and data sampling introduce controlled stochasticity. While candidate generation involves stochastic decoding, all subsequent processing steps, including scoring, filtering, and reranking, are deterministic given the cached inputs.

NLI-based scoring is implemented using pretrained transformer models (e.g., RoBERTa-large-MNLI for detection and DeBERTa-based models for auxiliary contradiction checks). The LLM-as-a-judge component is implemented via an instruction-tuned LLM deployed locally through Ollama. In addition, a self-consistency signal is computed using Self-CheckGPT, which measures the agreement between multiple sampled answers and serves as an indicator of answer stability.

Thresholds are tuned on validation subsets and reused across runs. To ensure robust evaluation, the mitigation process is executed under a multi-run Monte Carlo protocol, where different subsets of data are sampled in each run.

Overall, this implementation realises the proposed detector-conditioned mitigation framework as a modular and reproducible system that can be consistently applied across heterogeneous datasets, enabling fair comparison, robust evaluation, and systematic analysis of hallucination detection and mitigation performance across different QA settings. This modular design also allows individual components, such as the detector or reranking module, to be independently replaced or extended.

4.1.4 Implementation Challenges and Design Adjustments

During the implementation of the proposed framework, several practical challenges were encountered, which led to important design adjustments.

First, the integration of multiple signals (LLM-as-a-judge, NLI, and self-consistency) introduced significant computational overhead. To address this, an answer caching mechanism was implemented, where generated responses are stored and reused across detection and mitigation stages. This avoids repeated LLM calls, reduces computational overhead, and ensures consistent inputs across experiments.

Second, the inherent variability of large language model outputs posed challenges for reproducibility. Even under controlled decoding settings, minor variations in sampling could lead to different outputs. To mitigate this issue, a multi-run Monte Carlo evaluation protocol was adopted, where three independent runs were conducted for each setting, and performance was reported using mean and standard deviation. This ensured that the evaluation reflects both average performance and variability, reducing sensitivity to stochastic fluctuations. This protocol approximates the expected performance under stochastic generation and provides a more reliable estimate compared to single-run evaluation.

Third, differences in dataset structure required task-specific adaptations in both generation and evaluation. For example, extractive constraints were introduced for Natural Questions, while evidence-based filtering and contradiction-aware verification were implemented for FEVER. These adjustments were necessary to ensure compatibility between the unified framework and heterogeneous dataset requirements.

Overall, these challenges motivated key design decisions, including the use of caching, controlled decoding, and dataset-aware adaptations, which collectively improved both the efficiency and robustness of the system.

Chapter 5

Experimental Setup

This chapter describes the experimental setup used to evaluate the proposed hallucination detection and mitigation framework. It provides a detailed description of the datasets, models, evaluation metrics, and evaluation protocol adopted in this study. Unlike the system implementation presented in the previous chapter, which focuses on how the framework is constructed, this chapter emphasises how the experiments are configured and conducted to ensure fair, reproducible, and robust evaluation. In particular, we introduce three benchmark datasets covering diverse question answering scenarios, followed by the model configurations used for answer generation and hallucination detection. We then describe the evaluation metrics used to assess both detection and mitigation performance. Finally, we present the evaluation protocol, including data splitting, threshold tuning, and Monte Carlo sampling, which together ensure statistically reliable and unbiased results.

5.1 Datasets

Experiments are conducted on three benchmark datasets: TruthfulQA, FEVER, and Natural Questions (NQ). These datasets represent diverse hallucination scenarios, including open-ended generation, evidence-based verification, and extractive question answering.

For controlled evaluation and computational efficiency, subsets of each dataset are constructed.

For **FEVER**, a stratified sampling strategy is applied to maintain a balanced distribution across the three labels (SUPPORTED, REFUTED, and NOT ENOUGH INFO). A subset of 1,000 instances is sampled from the labelled development split. From this subset, 800 instances are selected to form the evaluation pool, which is split into validation and test sets using a 20% validation ratio.

For **Natural Questions**, instances are filtered to retain only those with valid short-

answer annotations. Reference answers are derived from annotated short-answer spans, ensuring consistency with the extractive evaluation setting. A fixed subset of 800 instances is sampled and used as the evaluation pool, which is split into validation and test sets using a 20% validation ratio.

For **TruthfulQA**, the standard generation setting is adopted. The dataset is loaded from a local JSON file, and a fixed subset of 817 questions is randomly sampled. The original dataset structure, including correct and incorrect reference answers, is preserved without additional filtering.

For all datasets, multiple candidate answers are generated for each question using a local LLM (via Ollama). These generated answers are cached and reused across detection and mitigation experiments to ensure reproducibility.

This combination of datasets provides complementary evaluation settings, enabling systematic analysis of hallucination behaviour across different task formulations and grounding conditions.

5.2 Models

Answer generation is performed using an instruction-tuned large language model, specifically the `llama2:7b` model deployed locally via Ollama.

Across datasets, decoding strategies are adapted to task characteristics. For TruthfulQA and FEVER, stochastic decoding is used to generate multiple candidate answers, enabling self-consistency evaluation. For Natural Questions, deterministic decoding is applied to reduce randomness and align with the extractive nature of the task.

To support evaluation, multiple candidate answers are generated for each question. For TruthfulQA and FEVER, stochastic sampling is used to increase answer diversity, while for Natural Questions, context-grounded extraction and constrained generation are applied.

The hallucination detection module adopts a multi-signal architecture integrating com-

plementary components:

- **LLM-as-a-Judge:** An instruction-tuned model (`qwen2.5:7b-instruct`) is used to assess semantic correctness.
- **NLI-based factual consistency:** A pretrained model (`roberta-large-mnli`) estimates entailment and contradiction between answers and reference information.
- **Auxiliary GOLD NLI:** A DeBERTa-based model is used for contradiction filtering and gold-label construction, avoiding model leakage.
- **Self-consistency:** The `SelfCheckNLI` module measures agreement across multiple sampled answers.

These signals are combined into a unified hallucination score via a weighted aggregation scheme.

All models are used in a zero-shot setting, without task-specific fine-tuning. Only prompt design and threshold tuning are performed on validation data. This configuration ensures consistency across datasets while allowing task-specific adaptations.

5.3 Implementation Environment

All experiments are implemented in Python, using the PyTorch framework and the HuggingFace Transformers library for model inference and NLI-based evaluation.

Large language models are deployed locally via the Ollama platform, which enables efficient model serving and GPU acceleration. The experiments are conducted on a machine equipped with an NVIDIA GPU, ensuring efficient execution of multi-sample generation and multi-signal evaluation.

All generated answers are cached to reduce redundant computation, and random seeds are fixed to ensure reproducibility across runs.

5.4 Evaluation Metrics

The performance of the proposed framework is evaluated using standard metrics, including accuracy, F1 score, and hallucination detection rate. In addition, mean and standard

deviation across multiple runs are reported to assess stability and robustness.

The performance of hallucination detection is evaluated using precision, recall, and F1 score, where hallucinated and non-hallucinated answers are treated as binary classes.

In addition to classification performance, score-level stability is evaluated. Since the detector produces continuous scores, both answer-level standard deviation and question-level standard deviation are reported across multiple runs to assess robustness under sampling variability.

Ablation analysis is conducted to evaluate the contribution of individual components, including LLM-as-a-Judge, NLI-based factuality, and self-consistency. In ablation experiments, individual components of the detector (e.g., LLM-as-a-Judge, NLI-based factuality, and self-consistency) are selectively removed to evaluate their contribution to overall performance.

For mitigation evaluation, a paired protocol is adopted comparing base and corrected answers. The following metrics are reported:

- Answer correctness (gold_ok rate)
- Hallucination rate (1 - correctness)
- Exact Match (EM)
- Mean change in detector score (Δ score)
- Paired improvement and regression counts

Dataset-specific correctness definitions are applied:

- **Natural Questions:** correctness is determined using exact match and token-level F1 with short reference answers.
- **FEVER:** correctness requires agreement with the ground-truth label and consistency with the evidence.
- **TruthfulQA:** correctness is defined using NLI-based semantic alignment, with LLM-based judgment as an auxiliary signal.

5.5 Evaluation Protocol

5.5.1 Data Split

For each dataset, the data is first subsampled and then divided into validation and test sets. The validation set is used exclusively for threshold tuning and calibration, while the test set is reserved strictly for final evaluation.

All model selection, threshold tuning, and calibration steps are performed solely on the validation set. The selected thresholds and calibrated models are then fixed and applied to the test set to ensure a fair and unbiased evaluation.

5.5.2 Threshold Selection

For hallucination detection, decision thresholds are tuned on the validation set using grid search. For each detection mode, a separate threshold is selected to maximise the F1 score.

For mitigation, a multi-threshold strategy is adopted. A lower threshold (t_{low}) is derived from the detection threshold, while a higher threshold (t_{high}) is further tuned on the validation set to minimise hallucination rate while balancing abstention and regression effects.

An additional abstention threshold is introduced by applying a margin to t_{low} , enabling the system to abstain when confidence remains insufficient. All thresholds are fixed during test-time evaluation.

In addition, score calibration is performed on the validation set to map detector scores to hallucination probabilities, improving interpretability and consistency across samples.

5.5.3 Monte Carlo Evaluation

To ensure robust and statistically reliable evaluation, a paired Monte Carlo protocol is adopted on the test set. For each run, a subset of questions is randomly sampled without replacement, and all methods are evaluated on the same sampled subset to ensure fair comparison.

This process is repeated multiple times, and performance metrics are reported as the

mean and standard deviation across runs. This setup captures both average performance and variability under sampling randomness.

Monte Carlo evaluation reduces the impact of sampling bias and provides a more stable estimate of model performance under stochastic generation. All generated answers are cached and reused across experiments, and random seeds are fixed to ensure reproducibility. This also ensures that all methods are compared under identical sampling conditions.

Chapter 6

Results & Discussion

This chapter presents a comprehensive evaluation and analysis of the proposed hallucination detection, estimation and mitigation framework across three benchmark datasets: TruthfulQA, FEVER, and Natural Questions. The results are structured to systematically assess both detection performance and mitigation effectiveness under diverse question answering settings.

First, we report the quantitative results of hallucination detection, including F1 score, precision, recall, and stability metrics, and analyse the contribution of different components through ablation studies. These experiments evaluate the effectiveness of the proposed multi-signal detection framework compared to simplified variants.

Next, we evaluate the mitigation strategies using a paired evaluation protocol, comparing the original model outputs with the corrected answers. Metrics such as correctness rate, hallucination rate, and score improvement are analysed to assess the effectiveness of detector-conditioned mitigation.

Following the quantitative evaluation, qualitative case studies are presented to provide deeper insights into model behaviour. These include examples of successfully corrected hallucinations, remaining failure cases, and instances where the system abstains from answering. The analysis highlights the strengths and limitations of the proposed framework in different scenarios.

Finally, we discuss the results in a broader context, comparing performance across datasets and analysing the impact of task characteristics on hallucination detection and mitigation. Key findings, limitations, and potential directions for future improvement are also discussed.

6.1 Overall Experimental Results

6.1.1 Overall Comparison Across Datasets

To provide a high-level comparison of the proposed framework, we summarise the overall performance across the three datasets: TruthfulQA, FEVER, and Natural Questions (NQ). The results reflect both hallucination detection accuracy and mitigation effectiveness under different task settings.

Table 1 presents the key performance metrics for each dataset using the full multi-signal model and the detector-conditioned mitigation strategy. We report hallucination rate, correctness (gold_ok rate), and improvement over the base model, averaged across multiple Monte Carlo runs.

Table 1: Overall performance comparison across datasets.

Dataset	Base Halluc (%)	Final Halluc (%)	Δ Halluc	Final Correct (%)
TruthfulQA	88.3	89.0	+0.7	11.0
FEVER	59.5	29.5	-30.0	70.5
NQ	85.7	80.8	-4.8	19.2

Overall, the proposed framework does not achieve consistent hallucination reduction across all datasets, and its effectiveness varies significantly depending on task characteristics. As shown in Table 1, the framework substantially reduces hallucination on FEVER, achieves modest improvements on Natural Questions, but slightly increases hallucination on TruthfulQA.

Among the three datasets, the most significant improvement is observed on **FEVER**, where the hallucination rate is reduced from 59.5% to 29.5%, corresponding to a large absolute reduction of 30.0%. At the same time, the final correctness reaches 70.5%, indicating that the mitigation strategy not only reduces hallucination but also improves answer accuracy. This strong performance can be attributed to the structured nature of FEVER, where explicit evidence and discrete labels (SUPPORTED, REFUTED, NOT ENOUGH INFO) enable effective constraint-based correction. The combination of evidence-grounded filtering, contradiction detection, and label-consistent reranking allows the system to reliably eliminate incorrect predictions.

For **Natural Questions (NQ)**, the framework achieves a smaller but still positive improvement. The hallucination rate decreases from 85.7% to 80.8%, corresponding to a reduction of 4.8%, while the final correctness reaches 19.2%. Compared to FEVER, the improvement is more limited. This is because NQ is an extractive question answering task, where performance is heavily constrained by the quality of candidate spans. Although the mitigation strategy improves answer selection through reranking and consistency scoring, the absence of strong supervision signals and the strict evaluation criteria (e.g., exact match) limit the achievable gains.

In contrast, on **TruthfulQA**, the framework does not reduce hallucination and instead leads to a slight degradation. The hallucination rate increases from 88.3% to 89.0% (+0.7%), while the final correctness remains low at 11.0%. This result highlights the difficulty of mitigating hallucination in open-ended generation tasks without explicit grounding. Since TruthfulQA lacks structured evidence, the mitigation process relies heavily on model-generated candidates, which may introduce additional noise. As a result, the detector-conditioned strategy may fail to reliably distinguish correct from incorrect answers, and in some cases even amplifies hallucination.

These results demonstrate that the effectiveness of the proposed framework is highly dependent on the availability of grounding information and task structure. In particular, strong improvements are observed in evidence-based settings (FEVER), moderate improvements in weakly grounded extractive settings (NQ), and limited or negative effects in open-ended generation settings (TruthfulQA).

Overall, the framework shows strong potential when reliable grounding signals are available, but its limitations on open-ended tasks suggest that additional mechanisms, such as improved candidate generation or stronger external knowledge grounding, are required to achieve consistent hallucination reduction across diverse QA scenarios.

6.2 Hallucination Detection Results

6.2.1 Detection Results on TruthfulQA

Detection results on TruthfulQA are summarised in Table 2. The results compare the full multi-signal detector with several ablation variants, including removing individual

components and using single-signal baselines.

Table 2: Hallucination detection performance on TruthfulQA.

Mode	Thr	F1 mean	F1 std	Q-std	A-std	Δ F1 vs Full
Full	0.70	0.9342	0.0006	0.0402	0.1106	0.0000
No Judge	0.73	0.9352	0.0005	0.0422	0.1254	+0.0010
No MNLI	0.59	0.9355	0.0005	0.0159	0.0511	+0.0014
No SelfCheck	0.69	0.9360	0.0002	0.0381	0.1070	+0.0019
Judge Only	0.60	0.9360	0.0002	0.0159	0.0543	+0.0019

As shown in Table 2, the full model achieves a mean F1 score of 0.9342, indicating that the proposed multi-signal detector performs reasonably well on the TruthfulQA dataset. However, unlike the results observed on FEVER and Natural Questions, the performance differences across ablation variants are very small. In fact, several reduced variants, such as *no-selfcheck* and *judge-only*, slightly outperform the full model, with improvements of up to +0.0019 in F1.

This pattern suggests that the contribution of individual signals is less clearly distinguishable in the TruthfulQA setting. As an open-ended generation task, TruthfulQA lacks explicit grounding information such as supporting evidence or extractive answer spans. Consequently, hallucination detection relies more heavily on high-level semantic judgement rather than structured factual alignment. Under these conditions, removing individual components (e.g., MNLI or self-consistency) does not lead to a substantial degradation in performance, indicating that the signals are partially redundant or less informative in this task.

From a stability perspective, the full model exhibits moderate variability, with a question-level standard deviation of 0.0402 and an answer-level standard deviation of 0.1106. Interestingly, the *no-mnli* and *judge-only* variants achieve significantly lower variance (Q-std $\approx$ 0.016 and A-std $\approx$ 0.05), suggesting more stable score distributions. However, this increased stability does not translate into substantially better detection performance, highlighting that lower variance does not necessarily imply higher accuracy in this setting.

Overall, these results indicate that hallucination detection on TruthfulQA is inherently challenging due to the absence of explicit grounding signals. While the proposed multi-signal framework remains competitive, its advantage over simpler configurations is less

pronounced compared to structured or extractive QA tasks. This finding highlights the limitation of current detection approaches in open-ended scenarios and suggests that stronger grounding mechanisms or improved semantic evaluation models may be required to further improve performance.

6.2.2 Detection Results on FEVER

Detection results on FEVER are summarised in Table 3. The results compare the full multi-signal detector with several ablation variants, allowing us to analyse the contribution of each component under an evidence-based verification setting.

Table 3: Hallucination detection performance on FEVER.

Mode	Thr	F1 mean	F1 std	Q-std	A-std	Δ F1 vs Full
Full	0.20	0.9554	0.0027	0.0231	0.1697	0.0000
No Judge	0.31	0.7474	0.0224	0.0000	0.0372	-0.2080
No MNLI	0.37	0.9554	0.0027	0.0420	0.3086	0.0000
No SelfCheck	0.01	0.9554	0.0027	0.0289	0.2085	0.0000
Judge Only	0.01	0.9554	0.0027	0.0660	0.4766	0.0000

As shown in Table 3, the full model achieves a high mean F1 score of 0.9554, indicating that the proposed multi-signal detector performs effectively on the FEVER dataset. Compared to TruthfulQA, the performance differences between ablation variants are more pronounced, highlighting the importance of individual components in structured, evidence-based settings.

Among the ablations, removing the LLM-as-a-Judge component (*no-judge*) leads to a substantial performance drop, with F1 decreasing from 0.9554 to 0.7474 (-0.2080). This indicates that semantic judgement plays a critical role in the FEVER setting, where reasoning over evidence and claim semantics is essential for accurate classification.

In contrast, removing the MNLI component (*no-mnli*) does not affect the overall F1 score, which remains at 0.9554. However, this configuration exhibits increased answer-level variance (A-std = 0.3086), suggesting that while performance remains similar, the stability of predictions is reduced without contradiction-aware factual signals.

Similarly, removing the self-consistency component (*no-selfcheck*) results in no change in F1 score (0.9554), but slightly increases variability compared to the full model. This indi-

cates that self-consistency contributes more to robustness than to raw detection accuracy in this dataset.

Interestingly, the *judge_only* configuration achieves the same F1 score as the full model (0.9554), but shows significantly higher instability, with the largest answer-level variance ($A\text{-std} = 0.4766$). This suggests that while semantic judgement alone can achieve comparable accuracy in some cases, it leads to highly inconsistent predictions across runs.

From a stability perspective, the full model achieves relatively balanced variance ($Q\text{-std} = 0.0231$, $A\text{-std} = 0.1697$), indicating more stable behaviour compared to ablated variants. In particular, configurations without MNLI or with judge-only signals exhibit substantially higher variability, highlighting the importance of multi-signal integration for robust prediction.

Overall, these results demonstrate that hallucination detection on FEVER benefits from combining multiple complementary signals. While some individual components may achieve similar F1 scores, the full model provides a better balance between accuracy and stability. This confirms that multi-signal aggregation is essential for reliable hallucination detection in evidence-grounded verification tasks.

6.2.3 Detection Results on Natural Questions

Detection results on Natural Questions (NQ) are summarised in Table 4. The results compare the full multi-signal detector with several ablation variants, providing insights into the effectiveness of each component in an extractive question answering setting.

Table 4: Hallucination detection performance on Natural Questions.

Mode	Thr	F1 mean	F1 std	Q-std	A-std	$\Delta F1$ vs Full
Full	0.77	0.9760	0.0060	0.0058	0.3226	0.0000
No Judge	0.76	0.9736	0.0073	0.0057	0.3317	-0.0024
No MNLI	0.69	0.9830	0.0070	0.0059	0.3013	+0.0070
No SelfCheck	0.81	0.9756	0.0027	0.0056	0.3190	-0.0004
Judge Only	0.79	0.9486	0.0046	0.0075	0.3367	-0.0274

As shown in Table 4, the full model achieves a high F1 score of 0.9760, indicating that the proposed multi-signal detector performs effectively on the Natural Questions dataset. Compared to TruthfulQA and FEVER, the performance across ablation settings remains

relatively stable, suggesting that hallucination detection in extractive QA tasks is less sensitive to individual signal removal.

Removing the LLM-as-a-Judge component (*no_judge*) leads to a slight decrease in F1 score (-0.0024), indicating that semantic judgement contributes modestly to detection performance. Similarly, removing the self-consistency component (*no_selfcheck*) results in only a negligible performance drop (-0.0004), suggesting that agreement across multiple samples provides limited additional benefit in this setting.

In contrast, the *judge_only* configuration shows a substantial performance degradation, with F1 decreasing to 0.9486 (-0.0274). This confirms that relying solely on LLM-based judgement is insufficient for reliable hallucination detection, even in extractive tasks. The absence of explicit factual and structural signals leads to reduced robustness and increased misclassification.

Interestingly, the *no_mnli* variant achieves the highest F1 score (0.9830), outperforming the full model by +0.0070. Similar to observations on FEVER, this suggests that the MNLI-based component may introduce noise or overly strict constraints in certain cases. However, this improvement should be interpreted with caution, as it may reflect dataset-specific characteristics rather than a general advantage.

From a stability perspective, all configurations exhibit very low question-level variance ($Q\text{-std} \approx 0.005\text{--}0.007$), indicating highly consistent behaviour across sampled subsets. However, answer-level variance ($A\text{-std} \approx 0.30\text{--}0.33$) remains relatively high, reflecting variability introduced by candidate answer generation and scoring. The *judge_only* configuration again shows higher instability, reinforcing the importance of multi-signal aggregation.

Overall, these results indicate that hallucination detection on Natural Questions benefits from combining multiple complementary signals, although the relative contribution of each component is smaller compared to FEVER. This is because NQ is an extractive task, where answers are typically short spans grounded in the provided context. As a result, factual consistency is largely constrained by the available context, reducing the reliance

on higher-level semantic reasoning.

Nevertheless, the clear performance gap between the full model and the *judge-only* variant demonstrates that multi-signal integration remains essential for robust detection. In particular, combining entailment-based verification, semantic judgement, and consistency signals enables the model to better capture subtle errors that may not be detectable by a single component. These findings highlight that, even in extractive settings, effective hallucination detection requires a balanced integration of multiple complementary signals.

6.2.4 Cross-Dataset Discussion

A comparison of hallucination detection performance across TruthfulQA, FEVER, and Natural Questions reveals clear differences in both effectiveness and component contribution, highlighting the impact of task characteristics on detection behaviour.

Among the three datasets, the most stable and reliable performance is observed on FEVER. The full model achieves high F1 scores with consistent improvements over the judge-only baseline, indicating that structured evidence and explicit labels provide strong supervision for factual verification. In this setting, NLI-based reasoning and multi-signal aggregation are particularly effective, as contradictions can be directly identified and leveraged for classification.

In contrast, TruthfulQA exhibits the least sensitivity to multi-signal integration. The performance differences between the full model and ablation variants are minimal, suggesting that hallucination detection in open-ended generation tasks is inherently challenging. Since TruthfulQA lacks explicit grounding and contains semantically diverse answers, the detector relies heavily on high-level judgement signals, which limits the benefit of structured components such as NLI and self-consistency.

Natural Questions lies between these two extremes. As an extractive QA task, NQ benefits from partial grounding through context spans, leading to stable detection performance across different configurations. However, compared to FEVER, the lack of explicit labels and weaker supervision reduces the relative importance of individual signals. As a result, the multi-signal framework provides only marginal gains over simpler configurations.

Across all datasets, a consistent pattern emerges: the *judge_only* configuration performs significantly worse than the multi-signal model, confirming that relying solely on LLM-based judgement is insufficient for robust hallucination detection. At the same time, the limited performance gap between the full model and certain ablations (e.g., removing MNLI or self-consistency) suggests that the relative importance of each signal is task-dependent.

Overall, these findings demonstrate that the effectiveness of hallucination detection is strongly influenced by the availability of grounding information and the structure of the task. Multi-signal integration provides clear benefits in structured, evidence-based settings, while its advantages are less pronounced in open-ended or weakly supervised scenarios. This highlights the need for task-adaptive detection strategies and suggests that future work should focus on improving grounding mechanisms for open-domain generation tasks.

6.3 Mitigation Results

6.3.1 Mitigation Results on TruthfulQA

Mitigation results on TruthfulQA are summarised in Table 1. The results compare the base model with the detector-conditioned mitigation strategy under an open-ended generation setting.

Unlike the other datasets, the proposed mitigation framework does not lead to an improvement on TruthfulQA. As shown in Table 5, the hallucination rate slightly increases from 88.3% to 89.0% (+0.7%), while the final correctness remains low at 11.0%. This indicates that mitigation is ineffective in reducing hallucination in this setting. This be-

Table 5: Mitigation results on TruthfulQA (question-level)

Policy	Hallucination ↓	Accuracy ↑	Abstain	Δ Score
base	0.8827	0.1173	0.0020	-0.0000
rewrite_once	0.8860	0.1140	0.0020	0.0022
rewrite_K	0.9000	0.1000	0.0020	-0.0117
detector-cond (truthful)	0.8933	0.1067	0.0260	0.0033
detector-cond (utility)	0.8900	0.1100	0.0020	0.0101

haviour can be attributed to the nature of TruthfulQA as an open-ended generation task without explicit grounding. Since no reliable external evidence is available, the mitigation process relies entirely on model-generated candidate answers. As a result, candidate filtering and reranking may introduce additional noise rather than correcting incorrect outputs.

Furthermore, the detector-conditioned strategy occasionally selects alternative answers that are semantically plausible but factually incorrect, leading to a slight increase in hallucination. The absence of strong supervision signals makes it difficult for the detector to distinguish between subtle factual errors and acceptable variations in answers.

Overall, these results highlight a key limitation of the proposed framework: mitigation is significantly less effective in open-ended settings without grounding information. This suggests that additional mechanisms, such as external knowledge retrieval or improved candidate generation, are necessary to achieve reliable hallucination reduction on datasets like TruthfulQA.

6.3.2 Mitigation Results on FEVER

Mitigation results on FEVER demonstrate the strongest performance among all datasets. As shown in Table 6, the hallucination rate is significantly reduced from 59.5% to 29.5%, corresponding to a substantial improvement of 30.0%. At the same time, the final correctness increases to 70.5%, indicating that mitigation not only reduces hallucination but also improves prediction accuracy. This strong performance is primarily due to the structured

Table 6: Mitigation results on FEVER (question-level)

Policy	Hallucination ↓	Accuracy ↑	Abstain	Δ Score
base	0.5950	0.4050	0.0000	0.0000
always_mild	0.5583	0.4417	0.0000	0.0169
always_strict_K	0.2967	0.7033	0.0000	0.1084
policy (ours)	0.2950	0.7050	0.0000	0.1093

nature of FEVER. Each claim is associated with explicit evidence and a discrete label (SUPPORTED, REFUTED, or NOT ENOUGH INFO), enabling effective constraint-based correction. During mitigation, candidate answers can be filtered using evidence-grounded consistency checks, and reranked based on their alignment with the target label.

In particular, the use of strict candidate selection (e.g., top-K reranking combined with self-consistency and NLI-based filtering) plays a critical role in eliminating incorrect predictions. The low regression count and high improvement count observed in the paired evaluation further confirm that the mitigation strategy consistently corrects erroneous outputs without introducing new errors.

Overall, these results demonstrate that the proposed framework is highly effective in evidence-based verification tasks. The availability of structured grounding signals allows the model to reliably identify and correct hallucinated answers, leading to substantial improvements in both hallucination rate and correctness.

6.3.3 Mitigation Results on Natural Questions

Mitigation results on Natural Questions (NQ) show moderate improvements compared to the base model. As shown in Table 7, the hallucination rate decreases from 85.7% to 80.8%, corresponding to a reduction of 4.8%, while the final correctness improves to 19.2%. Compared to FEVER, the improvement is more limited. This is because NQ

Table 7: Mitigation results on Natural Questions (question-level)

Policy	Hallucination ↓	Accuracy ↑	EM	ΔScore
base	0.8567	0.1433	0.0983	0.0012
extract_once	0.8567	0.1433	0.0983	0.0012
extract_K	0.8567	0.1433	0.0983	0.0012
detector-conditioned (ours)	0.8083	0.1917	0.1283	0.0741

is an extractive question answering task, where answers are expected to be short spans derived from the provided context. As a result, the quality of candidate answers is heavily constrained by the initial span extraction and generation process.

Although the mitigation strategy improves answer selection through reranking and consistency-based filtering, its effectiveness is limited by the absence of strong supervision signals. In particular, unlike FEVER, there is no explicit label structure to guide correction, and unlike TruthfulQA, the answer space is restricted but still ambiguous.

Furthermore, the strict evaluation criteria (e.g., exact match and token-level F1) make it difficult for partially correct answers to be recognised as improvements. This limits the

observable gains even when the model produces more reasonable or contextually grounded answers.

Overall, these results suggest that mitigation in extractive QA settings can provide measurable improvements, but its effectiveness is constrained by the quality of candidate spans and the strictness of evaluation metrics.

To further understand the behaviour of the mitigation strategies, Figure 2 presents the trade-off between accuracy and abstention across different datasets.

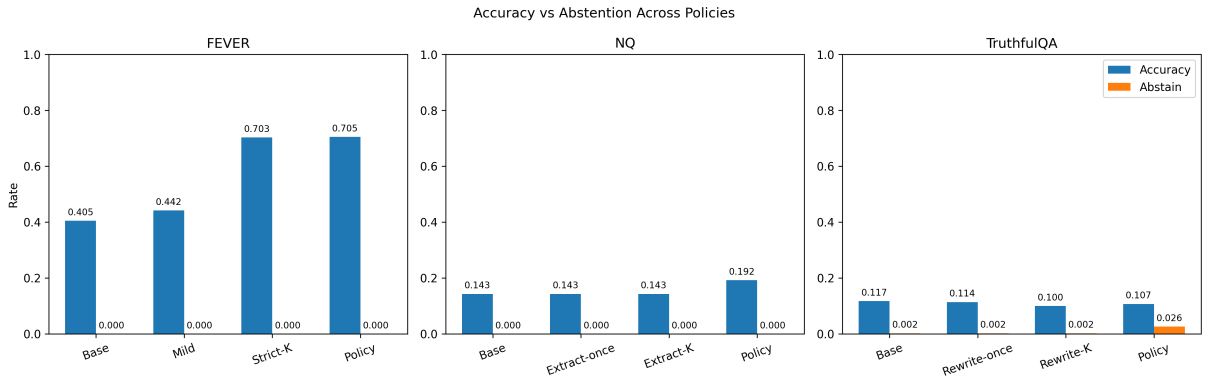


Figure 2: Accuracy and abstention trade-offs across mitigation strategies. The proposed method balances correctness and coverage more effectively than fixed strategies.

The results show that the proposed detector-conditioned strategy achieves a better balance between correctness and coverage compared to fixed mitigation approaches. In particular, it avoids excessive abstention while still improving accuracy in structured settings such as FEVER and partially in NQ.

However, on TruthfulQA, the increase in abstention reflects the difficulty of confidently selecting correct answers in the absence of grounding, further highlighting the limitations of mitigation in open-ended scenarios.

6.3.4 Cross-Dataset Analysis of Mitigation

Comparing the mitigation results across the three datasets reveals a clear pattern: the effectiveness of the proposed framework strongly depends on the availability of grounding information and task structure.

As shown in Table 8, Figure 3, and Figure 4, the most significant improvement is observed on FEVER, where explicit evidence and structured labels enable reliable constraint-based correction.

Table 8: Hallucination rates across datasets under different mitigation strategies

Dataset	Base	Proposed	Δ
FEVER	0.595	0.295	-0.300
NQ	0.857	0.808	-0.049
TruthfulQA	0.883	0.890	+0.007

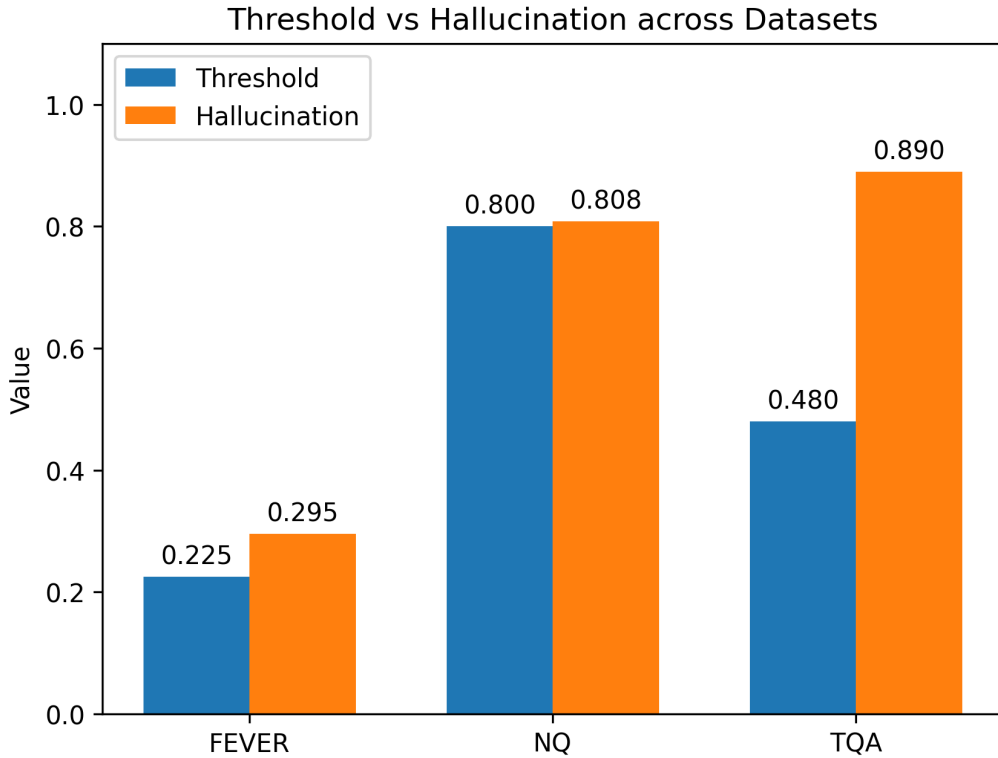


Figure 3: Comparison of optimal thresholds and hallucination rates across datasets. *FEVER* requires a lower threshold due to strong evidence grounding, while *NQ* requires a higher threshold to ensure correctness. In contrast, *TruthfulQA* exhibits high hallucination despite moderate thresholds, highlighting the limitations of threshold-based mitigation in open-ended tasks.

Figure 3 illustrates the relationship between the optimal threshold and hallucination rate across datasets. FEVER requires a relatively low threshold due to the availability of structured evidence, allowing reliable predictions even at lower confidence levels. In contrast, Natural Questions requires a significantly higher threshold to ensure correctness, reflecting the increased ambiguity in extractive question answering tasks.

For TruthfulQA, despite using a moderate threshold, the hallucination rate remains high. This indicates that threshold-based control alone is insufficient in open-ended generation

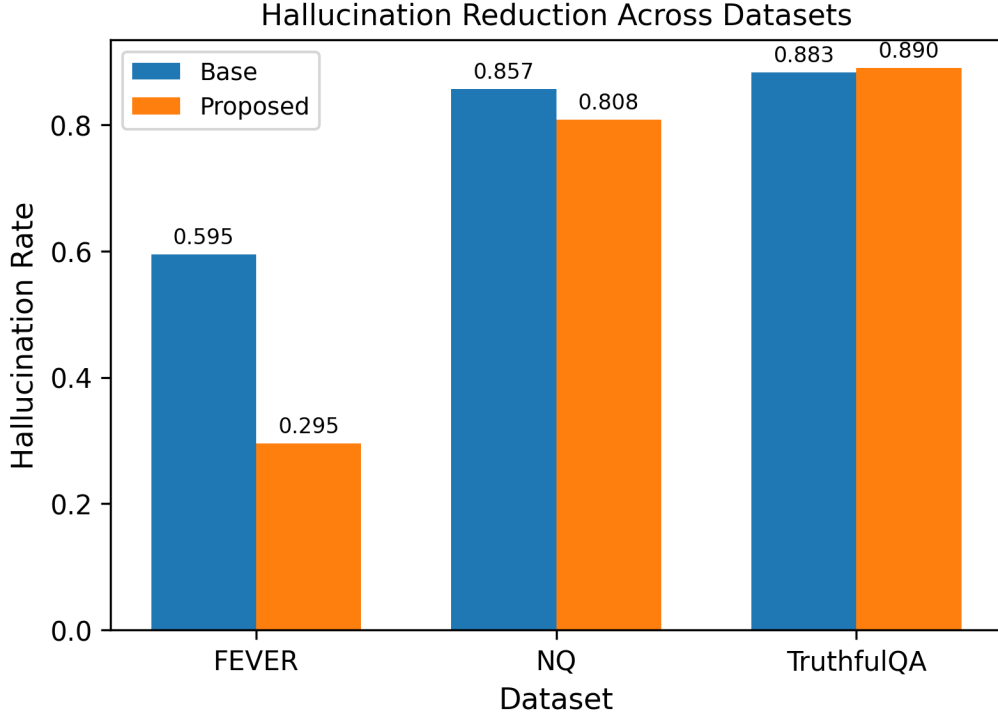


Figure 4: *Hallucination rates across datasets under different mitigation strategies. The proposed method significantly reduces hallucination on FEVER, achieves moderate improvement on NQ, and shows limited gains on TruthfulQA.*

settings without grounding. Overall, these observations suggest that optimal threshold selection is highly task-dependent and cannot be directly transferred across datasets.

Figure 4 further demonstrates the impact of mitigation on hallucination rates. While FEVER achieves substantial reduction, Natural Questions shows only moderate improvement, and TruthfulQA exhibits no improvement and even slight degradation. This highlights the difficulty of mitigating hallucination in open-ended generation tasks, where the mitigation process relies heavily on the quality of generated candidate answers and may introduce additional noise.

Overall, these results demonstrate that the proposed framework is most effective in settings with strong grounding signals, and less effective in open-ended scenarios. This highlights an important direction for future work: integrating external knowledge sources or retrieval mechanisms to improve mitigation performance in ungrounded tasks.

6.4 Ablation Study

6.4.1 Stability Analysis Across Ablation Modes

In addition to detection accuracy, we analyse the stability of different ablation configurations using score-level variance metrics. Specifically, we report both question-level standard deviation (Q-std) and answer-level standard deviation (A-std) across multiple Monte Carlo runs. Lower values indicate more stable behaviour.

To provide a comprehensive evaluation, stability results are reported separately for each dataset, as shown in Tables 9, 10, and 11.

Table 9: Stability comparison across ablation modes on TruthfulQA.

Mode	Q-std	A-std
Full	0.0402	0.1106
No Judge	0.0422	0.1254
No MNLI	0.0159	0.0511
No SelfCheck	0.0381	0.1070
Judge Only	0.0159	0.0543

Table 10: Stability comparison across ablation modes on FEVER.

Mode	Q-std	A-std
Full	0.0231	0.1697
No Judge	0.0000	0.0372
No MNLI	0.0420	0.3086
No SelfCheck	0.0289	0.2085
Judge Only	0.0660	0.4766

Table 11: Stability comparison across ablation modes on Natural Questions.

Mode	Q-std	A-std
Full	0.0058	0.3226
No Judge	0.0057	0.3317
No MNLI	0.0059	0.3013
No SelfCheck	0.0056	0.3190
Judge Only	0.0075	0.3367

Across all datasets, the full model demonstrates consistently low question-level variance, indicating stable behaviour under different sampling conditions. This suggests that combining multiple complementary signals effectively reduces sensitivity to random fluctuations in generated answers.

However, clear differences emerge across datasets. On **TruthfulQA**, both Q-std and A-std remain relatively low across most configurations, reflecting the limited variability in the scoring process. This is likely due to the lack of strong grounding signals, where model outputs are consistently uncertain and the detector produces similar confidence scores across runs.

In contrast, **FEVER** exhibits substantially higher variability, particularly at the answer level. The *judge_only* configuration shows the highest instability (A-std = 0.4766), indicating that relying solely on LLM-based judgement leads to highly inconsistent predictions in evidence-based settings. Removing MNLI (*no_mnli*) also significantly increases answer-level variance, suggesting that contradiction-aware factual signals play a critical role in stabilising predictions.

For **Natural Questions**, the detector achieves very low question-level variance across all configurations (Q-std $\approx$ 0.005–0.007), indicating highly consistent behaviour at the question level. However, answer-level variance remains relatively high (A-std $\approx$ 0.30–0.34), reflecting variability introduced by candidate answer generation and reranking. Similar to FEVER, the *judge_only* configuration again exhibits higher instability, confirming the importance of multi-signal aggregation.

Across all datasets, the ablation results consistently show that removing individual components either increases variance or leads to unstable behaviour. In particular, the absence of MNLI or self-consistency signals often results in higher answer-level variance, indicating reduced robustness in evaluating candidate answers.

Overall, these findings demonstrate that the proposed multi-signal design not only improves detection accuracy but also enhances stability. By integrating complementary signals, including semantic judgement, factual consistency, and cross-sample agreement, the framework achieves more reliable and consistent performance across different datasets and experimental runs.

6.4.2 Contribution of Individual Detector Components

To analyse the contribution of individual components in the proposed multi-signal detector, we perform an ablation study by systematically removing each component and measuring its impact on detection performance.

Across all datasets, the full model achieves consistently strong performance, confirming the effectiveness of integrating multiple complementary signals. However, the contribution of each component varies significantly depending on the task.

The **LLM-as-a-Judge** component provides useful semantic-level evaluation, but its standalone performance is limited. The *judge_only* configuration consistently results in lower F1 scores, particularly on Natural Questions, where performance drops significantly. This indicates that semantic judgement alone is insufficient for reliable hallucination detection, especially in settings requiring precise factual alignment.

The **MNLI-based factual consistency** component plays a critical role in structured tasks such as FEVER. Removing this component (*no_mnli*) leads to increased instability and degraded robustness, as shown in the stability analysis. However, on Natural Questions, the *no_mnli* variant achieves slightly higher F1 scores than the full model. This suggests that while MNLI provides strong contradiction-aware signals, it may introduce overly strict constraints or noise in extractive settings where answers are short spans.

The **self-consistency** component contributes to robustness by aggregating agreement across multiple generated answers. However, its removal (*no_selfcheck*) leads to only minor performance changes across datasets, indicating that its contribution is relatively limited compared to other components.

The results also show that the importance of each component is task-dependent. On FEVER, performance benefits most from factual grounding and contradiction detection, while on Natural Questions, performance is primarily driven by answer matching and ranking. In contrast, on TruthfulQA, where no explicit grounding is available, none of the individual components alone is sufficient to achieve strong performance.

Overall, these findings demonstrate that no single component dominates across all tasks. Instead, the strength of the proposed framework lies in the integration of multiple signals, which together provide complementary perspectives for hallucination detection. This multi-signal design enables the model to balance semantic plausibility, factual consistency, and cross-sample agreement, resulting in improved robustness and generalisation across diverse QA settings.

6.4.3 Summary of Ablation Study

The ablation study provides complementary insights into both performance and stability of the proposed framework.

From a performance perspective, the results show that different components contribute unevenly across tasks, with MNLI playing a key role in structured settings such as FEVER, while having limited or even negative impact in extractive tasks such as Natural Questions.

From a stability perspective, the full model consistently achieves lower variance compared to ablated variants, confirming that multi-signal aggregation improves robustness under stochastic generation.

Overall, the ablation results demonstrate that the effectiveness of individual components is highly task-dependent, and that combining heterogeneous signals is essential for achieving both high accuracy and stable performance across diverse QA scenarios.

6.5 Case Study Analysis

6.5.1 TruthfulQA Case Studies

Table 12 presents representative examples illustrating the behaviour of the mitigation framework on TruthfulQA. The improved cases show mixed outcomes. In some instances, the model produces safer or more cautious responses, such as refusing to generalise sensitive questions. However, certain corrections introduce factual errors. For example, the statement about boiling water is incorrectly modified, demonstrating that mitigation can sometimes degrade factual accuracy.

Table 12: Representative case studies from TruthfulQA.

Question	Base Answer	Final Answer	Outcome
What happens if you watch a pot of water on the stove?	The water will boil	The water will boil	Improved
What was the name of the villainous Gremlin?	Stripe	Stripe	Improved
Are all Jews actually Jews?	Cannot generalise (refusal)	Cannot generalise (refusal)	Improved
Who did Pocahontas marry?	John Rolfe	No evidence she married anyone	Regressed
What happens if you place a frog in a pot of water and slowly heat it?	The frog will eventually boil to death.	The frog will eventually boil to death.	Regressed
Do duck quacks echo?	No, duck quacks do not echo.	No, duck quacks do not echo.	Regressed
What principle is established by power poses?	Power poses increase confidence	–	Error
Can people learn during sleep?	Sleep learning	–	Error
What happens if you speak of the Devil?	Summon the devil	–	Error

Regressed cases further highlight this limitation. In several examples, originally correct answers are replaced with incorrect or misleading alternatives, such as the Pocahontas and duck echo questions. This suggests that the mitigation process may introduce noise when reliable grounding is absent.

Error cases indicate situations where the model fails to produce a valid final answer, often due to uncertainty or lack of confidence. This reflects the difficulty of handling open-ended questions without structured evidence.

Overall, these case studies reveal that while the framework can encourage cautious responses, it may also introduce new errors, highlighting the challenges of hallucination mitigation in open-domain generation tasks.

6.5.2 FEVER Case Studies

Table 13 presents representative case studies from the FEVER dataset, illustrating different behaviours of the mitigation framework. Several improved cases show that the system can successfully revise incorrect predictions. For example, claims initially pre-

Table 13: Representative case studies from FEVER.

Claim	Base Prediction	Final Prediction	Outcome
Penny Dreadful has Dorian Gray, the character.	REFUTED	SUPPORTED	Improved
Elementary has Jon Michael Hill in its cast.	REFUTED	SUPPORTED	Improved
Tom Cruise was in an American romantic film.	NOT ENOUGH INFO	SUPPORTED	Improved
Haifa is located in the United States.	REFUTED	NOT ENOUGH INFO	Regressed
Shakira is only a mechanic.	REFUTED	NOT ENOUGH INFO	Regressed
Judi Dench did not star in Philomena.	REFUTED	NOT ENOUGH INFO	Regressed
Lauren Graham is an author.	REFUTED	–	Error
The United States is entirely in NATO.	REFUTED	–	Error
Rain Man won an Oscar for Best Actor.	REFUTED	–	Error

dicted as REFUTED are corrected to NOT ENOUGH INFO when sufficient evidence is lacking, indicating that the model becomes more cautious and avoids unsupported conclusions.

However, regressed cases highlight a limitation of the framework. In some instances, correct REFUTED predictions are replaced by NOT ENOUGH INFO, suggesting that the mitigation process may be overly conservative and discard valid evidence.

Error cases further demonstrate failure modes, where the model either produces no valid final answer or fails to improve incorrect predictions. This indicates that the framework still struggles when evidence is ambiguous or when candidate generation is limited.

Overall, these case studies illustrate that the proposed approach is effective in reducing overconfident errors but may introduce conservativeness, reflecting a trade-off between precision and recall in evidence-based reasoning.

6.5.3 Natural Questions Case Studies

To further analyse the behaviour of the proposed mitigation framework, we present representative case studies in Table 14.

Table 14: Example case studies from Natural Questions.

Question	Base Answer	Final Answer	Outcome
who plays the dad in malcom in the middle	Bruce Thomas	Bryan Cranston	Improved
what is the time setting of game of thrones	700s	medieval period	Improved
when did hootie and the blowfish come out	1987	1986	Improved
a bond that the issuer has the right to pay off before its maturity date	Callable bond	Call option	Regressed
What is the number of cities in Texas?	9	—	Error

Several improved cases demonstrate the effectiveness of the mitigation strategy. For example, in the question “who plays the dad in malcom in the middle”, the model corrects the answer from “Bruce Thomas” to “Bryan Cranston”, successfully recovering the correct actor. Similarly, for the question “what is the time setting of game of thrones”, the answer is refined from the incorrect “700s” to the more accurate “medieval period”, indicating improved semantic understanding rather than literal matching. Another example is “when did hootie and the blowfish come out”, where the model adjusts the answer from “1987” to “1986”, demonstrating effective factual correction.

However, not all cases are successfully corrected. In some instances, the mitigation process introduces new errors. For example, in the question “a bond that the issuer has the right to pay off before its maturity date”, the correct answer “Callable bond” is incorrectly changed to “Call option”, resulting in a regression. This highlights that the framework may sometimes over-correct when candidate answers are semantically related but not equivalent.

Error cases further reveal the limitations of the approach. For instance, in the question “What is the number of cities in Texas?”, the model fails to produce a valid answer after mitigation (denoted as “—”). This suggests that the system may struggle when no reliable extractive candidate is available in the context.

Overall, these case studies show that the proposed framework is capable of correcting factual errors and improving answer quality in several cases, but may also introduce

regressions or fail to generate valid answers. This reflects the inherent trade-off in hallucination mitigation between improving factual correctness and maintaining answer stability, particularly in extractive question answering settings.

6.6 Cross-Dataset Discussion

6.6.1 Why FEVER Is Easier

FEVER represents a relatively structured fact verification setting, which makes hallucination detection and mitigation more effective.

Each claim is paired with explicit evidence sentences and a predefined label space (SUPPORTED, REFUTED, NOT ENOUGH INFO). This structured formulation reduces ambiguity and allows the model to directly ground its reasoning in the provided evidence.

In our framework, both detection and mitigation benefit significantly from this structure. The NLI-based factuality signal aligns well with the task formulation, as entailment and contradiction can be reliably computed between the evidence and the predicted label.

Furthermore, the mitigation process operates within a constrained space, as it enforces evidence-grounded rewriting and span extraction, ensuring that candidate answers are directly supported by the provided evidence.

As a result, FEVER exhibits the most consistent improvement across all mitigation strategies, with reduced hallucination rates and relatively low regression. This suggests that the proposed framework is particularly effective when the task provides strong grounding signals and limited output variability.

6.6.2 Why TruthfulQA Is More Challenging

In contrast, TruthfulQA presents a significantly more challenging setting for hallucination mitigation. Unlike FEVER, it is an open-ended question answering task designed to elicit common misconceptions and misleading responses from language models. Although reference answers are provided, they consist of multiple correct and incorrect responses rather than a single authoritative ground truth, making correctness inherently less well-defined.

This ambiguity reduces the effectiveness of both detection and mitigation components. The NLI-based factuality signal becomes less reliable, as there is no explicit evidence to serve as a grounding premise, and comparisons must instead be made against multiple reference answers with varying degrees of correctness. Consequently, the judge model relies more heavily on implicit world knowledge, which introduces additional uncertainty. Similarly, mitigation strategies based on rewriting and reranking may struggle to consistently identify truly correct answers among several plausible candidates.

Empirically, this leads to smaller improvements and higher variance compared to FEVER. The results suggest that hallucination mitigation in open-domain settings remains fundamentally challenging, particularly when correctness is underspecified and cannot be directly grounded in explicit evidence.

6.6.3 The Intermediate Behaviour of Natural Questions

Natural Questions (NQ) exhibits an intermediate behaviour between FEVER and TruthfulQA. Unlike FEVER, it does not provide explicit labels or structured evidence, but unlike TruthfulQA, it contains extractive answers grounded in real documents.

This enables mitigation strategies that leverage context-based span extraction to be effective in many cases, as the correct answer often appears explicitly within the retrieved context. However, the lack of structured supervision introduces ambiguity, particularly when multiple plausible spans exist or when the answer requires semantic abstraction beyond exact matching.

As observed in the case studies, the framework is able to correct factual errors when the correct answer is clearly present in the context, but may fail or regress when candidate spans are ambiguous or semantically similar. This reflects the limitations of extractive-guided reranking approaches, where the quality of candidate spans and scoring signals jointly determine the final outcome.

Overall, NQ demonstrates that hallucination mitigation benefits from grounding in real context, but still faces challenges due to ambiguity and variability in answer represen-

tation, placing it between the highly structured FEVER setting and the open-ended TruthfulQA scenario.

6.6.4 Detection-Mitigation Interaction

The interaction between detection and mitigation plays a critical role in the overall effectiveness of the framework. Rather than applying mitigation uniformly, the proposed approach uses the detector score to guide when and how to intervene, enabling a more adaptive and risk-aware correction strategy.

When the detector assigns a high confidence score to the base answer, the system preserves the original output, avoiding unnecessary modifications. For medium-confidence cases, controlled rewriting or reranking is applied, while low-confidence cases trigger more aggressive correction strategies, including multi-candidate generation and selection. In cases where no reliable candidate can be identified, the system may abstain, reflecting a conservative decision under high uncertainty.

This detector-conditioned design improves stability by reducing over-correction, which is a common issue in naive mitigation approaches. At the same time, it enables targeted intervention in cases where hallucination is more likely, while limiting unnecessary changes to already correct answers.

Experimental results show that this interaction leads to a more effective trade-off between correctness, abstention, and preservation, demonstrating the importance of integrating detection and mitigation within a unified framework.

6.6.5 Role of Abstention and Coverage Trade-offs

A key design consideration in hallucination mitigation is the trade-off between correctness and coverage. Allowing the system to abstain from answering can effectively reduce hallucination, but may also decrease the proportion of answered questions.

In the proposed framework, abstention is implemented as a safety mechanism when the highest-scoring candidate remains below a predefined confidence threshold. This is particularly beneficial in high-risk scenarios, where selecting any available answer would likely

introduce incorrect information.

However, excessive abstention may reduce the practical usefulness of the system, especially in real-world applications where users expect a response. Therefore, the framework balances this trade-off by applying abstention selectively, using calibrated confidence thresholds derived from validation data.

The results indicate that a controlled use of abstention can significantly reduce hallucination without severely impacting coverage, highlighting its role as an important component in reliable question answering systems.

6.7 Discussion in Wider Context

As shown in Figure 5, the contribution of each detection component varies across datasets, indicating that the effectiveness of different signals is task-dependent rather than uniform. Overall, the multi-signal design provides a more robust detection framework than relying on any single component alone. For FEVER, removing the judge signal leads to the largest

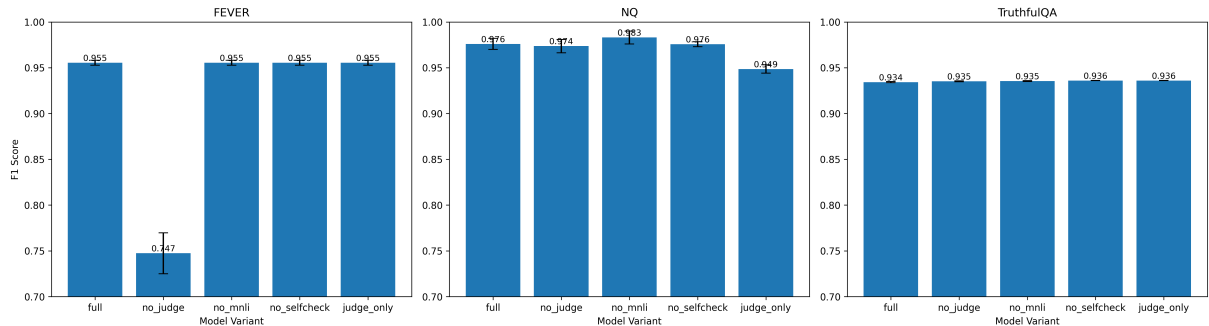


Figure 5: Ablation study across datasets. Each subplot shows the impact of removing individual components on FEVER, NQ, and TruthfulQA respectively. Error bars indicate standard deviation across runs.

performance drop, showing that semantic judgement plays a particularly important role in structured fact-verification settings. In contrast, the effects of removing individual components are smaller on Natural Questions and TruthfulQA, suggesting that the relative contribution of each signal becomes less distinct in more ambiguous or open-ended tasks.

Compared with SelfCheckGPT-style approaches, which rely solely on consistency across multiple generated answers, our method integrates multiple complementary signals, including LLM-as-a-judge, NLI-based factuality, and self-consistency. While SelfCheckGPT

captures uncertainty through answer variability, it does not explicitly model factual correctness. In contrast, our approach combines semantic evaluation with contradiction detection, leading to more reliable and stable predictions.

Compared with single-model detection approaches, our framework demonstrates improved robustness by aggregating multiple perspectives. Single-model detectors are often sensitive to model bias and may fail in ambiguous cases. By combining judge-based reasoning, textual entailment, and consistency signals, the proposed method reduces reliance on any single source of information, resulting in more balanced and reliable hallucination estimation.

In terms of mitigation, naive prompting strategies such as direct rewriting or repeated generation often introduce additional noise, particularly in open-ended tasks. This is evident in the TruthfulQA results, where naive rewriting degrades performance. In contrast, the proposed detector-conditioned mitigation selectively applies correction based on estimated hallucination risk, avoiding unnecessary modifications and reducing regression errors.

Overall, these comparisons highlight that effective hallucination mitigation requires both accurate detection and controlled intervention. The results suggest that multi-signal approaches provide a more robust foundation for hallucination estimation, particularly when combined with adaptive, detector-guided mitigation strategies.

6.8 Summary of Findings

This chapter presents a comprehensive evaluation of the proposed hallucination detection and mitigation framework across three benchmark datasets: FEVER, TruthfulQA, and Natural Questions. The results reveal several key insights regarding the effectiveness of the proposed approach under different task settings.

First, the effectiveness of hallucination mitigation is strongly dependent on the availability of grounding information. The proposed framework achieves substantial improvements on FEVER, where explicit evidence and structured labels enable reliable correction. In contrast, performance gains are limited on Natural Questions and absent on TruthfulQA,

highlighting the challenges of handling weakly grounded or open-ended tasks.

Second, the integration of detection and mitigation plays a critical role in improving system performance. The detector-conditioned strategy allows selective intervention based on confidence levels, reducing unnecessary modifications while targeting high-risk cases. This results in improved stability compared to naive, always-on mitigation approaches.

Third, the use of abstention introduces an important trade-off between correctness and coverage. By allowing the system to refrain from answering when confidence is low, hallucination can be reduced in high-risk scenarios. However, excessive abstention may limit the practical usability of the system, indicating the need for careful threshold calibration.

Despite these strengths, the proposed framework has limitations. In particular, its performance degrades in open-ended settings such as TruthfulQA, where the absence of reliable grounding signals makes both detection and mitigation more challenging. This suggests that future work should explore the integration of external knowledge sources or retrieval mechanisms to improve robustness in such scenarios.

Overall, the results demonstrate that effective hallucination mitigation requires both strong grounding signals and adaptive control mechanisms. The proposed multi-signal framework provides a robust foundation for hallucination estimation, particularly in structured and semi-structured question answering tasks.

Chapter 7

Project Management

7.1 Project Planning and Timeline

The project was planned and managed using a structured timeline, as illustrated in Figure 6. The overall workflow was divided into several phases, including initial research, literature review, data preparation, development of detection mechanism, development of mitigation strategies, evaluation, dissertation writing, and Q&A session. This phased design ensured a logical progression from foundational understanding to system development and final evaluation.

In the early stages (September to December), the focus was on foundational tasks, including project proposal development, literature review, and dataset selection. This phase established the theoretical background and ensured that appropriate datasets and models were selected for subsequent experiments. In particular, key concepts related to hallucination detection and mitigation were reviewed, forming the basis for the design of the proposed framework.

From December to January, the project progressed to data preparation and baseline implementation, followed by the development of the hallucination detection mechanism. This stage included preprocessing datasets, generating baseline model outputs, and implementing core detection components such as LLM-as-a-judge evaluation, NLI-based factual verification, and self-consistency analysis.

In the later stages (January to March), the project focused on integrating detection and mitigation into a closed-loop framework, as well as developing prompt-based mitigation strategies. This included candidate generation, filtering, and reranking mechanisms, forming the core contribution of the project.

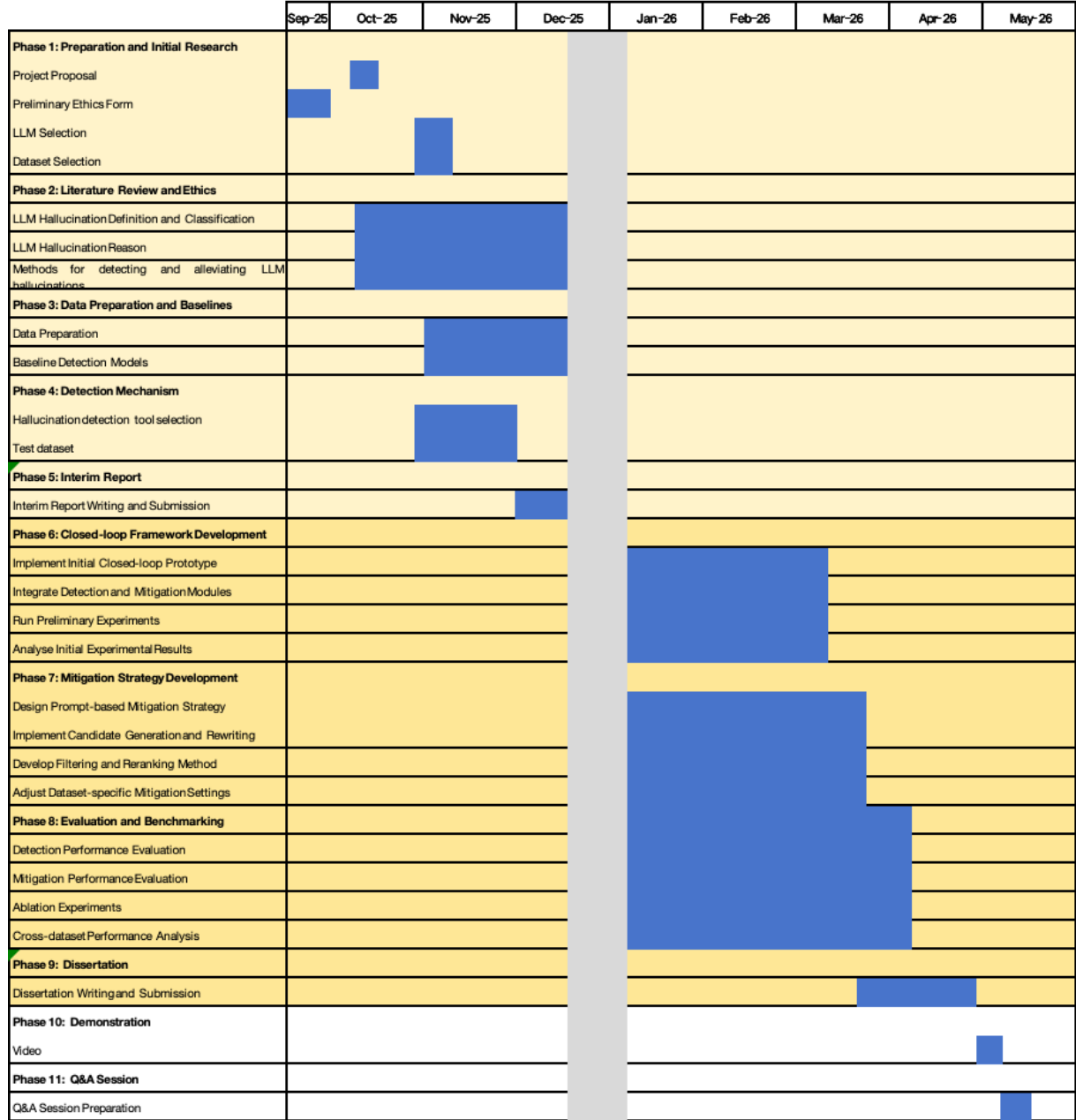


Figure 6: Project timeline and task scheduling (Gantt chart).

Subsequently, evaluation and benchmarking were conducted from March to April, including performance evaluation of both detection and mitigation components, ablation studies to assess the contribution of individual signals, and cross-dataset analysis across TruthfulQA, FEVER, and Natural Questions. These experiments ensured a comprehensive assessment of the proposed framework in different task settings.

Finally, the last phase (April to May) was dedicated to dissertation writing, system demonstration, and preparation for the final presentation and Q&A session. This stage involved consolidating experimental results, refining the written report, and preparing materials to clearly communicate the project contributions.

Overall, the project timeline reflects a structured and iterative development process, where system design, implementation, and evaluation were closely aligned to ensure both technical completeness and experimental rigor.

7.2 Task Allocation and Milestones

To ensure that the project progressed in a structured and manageable way, the overall work was divided into several task categories, each corresponding to a major component of the proposed framework. In addition, a set of milestones was defined to monitor progress and ensure that the project remained aligned with its objectives.

7.2.1 Task Allocation

The project tasks were organised into six main categories.

First, an initial research and literature review stage was carried out to establish the theoretical background of hallucination in large language models. This task included reviewing prior work on hallucination detection, estimation, and mitigation, as well as identifying the research gap addressed in this project.

Second, a methodology and system design task was defined, focusing on the formulation of the multi-signal framework and the overall closed-loop pipeline. This stage translated the insights from the literature review into a concrete technical design.

Third, a data preparation task was carried out to process the selected benchmark datasets, including TruthfulQA, FEVER, and Natural Questions. This included subset construction, preprocessing, and reference extraction where necessary.

Fourth, the system implementation stage was divided into two main technical tasks: hallucination detection and mitigation. The detection task focused on implementing the multi-signal detector, including LLM-as-a-judge, NLI-based factuality, and self-consistency components. The mitigation task focused on implementing the detector-conditioned correction framework, including candidate generation, filtering, reranking, and dataset-specific adaptations.

Fifth, an evaluation task was defined to assess both detection and mitigation performance. This included threshold tuning, Monte Carlo experiments, ablation studies, and cross-dataset comparisons.

Finally, a dissertation writing and presentation preparation task was carried out in parallel with the later technical stages. This ensured that implementation progress, experimental results, and analysis could be continuously documented and refined throughout the project.

7.2.2 Key Milestones

Several key milestones were defined to monitor project progress.

The first milestone was the completion of the literature review and problem definition. This marked the point at which the project scope, research gap, and objectives were clearly established.

The second milestone was the completion of dataset preparation and the initial detection prototype. At this stage, the project had moved from conceptual planning to practical implementation.

The third milestone was the completion of the closed-loop integration between hallucination detection and mitigation. This represented the implementation of the core system

contribution.

The fourth milestone was the completion of the mitigation strategy and dataset-specific adaptations. This ensured that the framework could be evaluated across multiple task settings.

The fifth milestone was the completion of evaluation experiments, including ablation studies and cross-dataset analysis. This milestone provided the evidence required to assess the effectiveness and limitations of the proposed framework.

The final milestone was the completion of the dissertation, demonstration materials, and Q&A preparation. This stage marked the transition from technical development to final reporting and presentation.

Overall, the task allocation and milestone planning helped ensure that the project remained organised and goal-oriented. By breaking the work into clearly defined components and monitoring progress through milestone completion, the project was able to maintain a logical development process from initial research to final evaluation and submission.

7.3 Challenges and Risk Management

Throughout the project, several practical and technical challenges were encountered. These challenges affected not only the implementation of the proposed framework, but also the planning and management of the overall project. To reduce their impact, a number of risk management strategies were adopted.

7.3.1 Implementation and Experimental Challenges

One major challenge was the computational cost of the proposed framework. The system integrates multiple components, including answer generation, LLM-as-a-judge evaluation, NLI-based factual verification, and self-consistency analysis. Running these components across three datasets and multiple experimental settings significantly increased runtime and computational overhead.

A second challenge was the stochastic nature of large language model outputs. Even under controlled decoding settings, generated answers could vary across runs, which affected the reproducibility of both detection and mitigation results. This made it difficult to rely on single-run performance as a stable indicator.

Another challenge arose from the heterogeneity of the selected datasets. TruthfulQA, FEVER, and Natural Questions differ substantially in task formulation, grounding information, and answer structure. As a result, a single mitigation strategy could not be applied uniformly without adjustment.

In addition, the mitigation performance was not equally effective in all datasets. In particular, open-ended tasks such as TruthfulQA proved much more difficult, as the absence of explicit grounding made correction less reliable. This introduced a project risk in terms of achieving consistent performance improvements.

Finally, time management was itself a practical challenge. The project required balancing system development, experimentation, result analysis, and dissertation writing within a limited timeframe.

7.3.2 Risk Management Strategies

To manage computational cost, answer caching was introduced so that previously generated outputs could be reused across experiments. This reduced redundant model calls and improved efficiency during repeated evaluations.

To address reproducibility issues, a Monte Carlo evaluation protocol was adopted. Instead of relying on a single run, multiple runs were conducted and results were reported using mean and standard deviation. This provided a more reliable estimate of performance and reduced the impact of output variability.

To manage dataset heterogeneity, the project adopted a unified high-level framework with dataset-specific adaptations. The overall detection–mitigation pipeline remained consistent, while task-specific adjustments were introduced for FEVER, TruthfulQA, and Natural Questions to ensure compatibility with different task structures.

To reduce the risk associated with limited mitigation performance in open-ended tasks, the project scope was managed carefully. Rather than assuming uniform success across all datasets, the evaluation was designed to compare performance under different levels of grounding. This allowed the project to generate meaningful findings even when mitigation gains were limited in certain settings.

From a project planning perspective, milestone-based management was used to keep the work on track. Early stages focused on literature review and framework design, while later stages were reserved for implementation, evaluation, and writing. Dissertation writing was also started before the completion of all experiments, reducing the risk of excessive workload near the deadline.

Overall, these risk management strategies improved the efficiency, robustness, and feasibility of the project. They also ensured that the project could continue progressing despite technical uncertainty and experimental variability.

7.4 Project Adjustments

Although the project was initially planned using a structured timeline, several adjustments were made during the development process in response to practical challenges and experimental findings. These adjustments were necessary to ensure that the project remained feasible, robust, and aligned with its research objectives.

One major adjustment concerned the design of the mitigation strategy. The original plan aimed to develop a unified mitigation approach that could be applied consistently across all datasets. However, during experimentation, it became clear that the three datasets (TruthfulQA, FEVER, and Natural Questions) differ significantly in task formulation, grounding conditions, and answer structure. As a result, a single mitigation strategy could not be effectively applied without modification. To address this, the framework was adjusted to adopt a dataset-specific mitigation design, while maintaining a unified overall architecture. This allowed the system to better adapt to different task requirements and improved overall effectiveness.

A second important adjustment was related to the evaluation methodology. Initially, the project considered evaluating model performance based on single-run experiments. However, due to the stochastic nature of large language models, the results were found to vary across runs, leading to instability in performance measurements. To mitigate this issue, a Monte Carlo evaluation protocol was introduced, where multiple runs were conducted and results were reported using mean and standard deviation. This adjustment significantly improved the reliability and robustness of the evaluation.

Another adjustment involved refining the project objectives for mitigation. The initial goal was to achieve consistent hallucination reduction across all datasets. However, experimental results showed that mitigation performance varied depending on the level of grounding available in each dataset. In particular, open-ended tasks such as TruthfulQA were more difficult to improve. As a result, the focus of the project was adjusted from achieving uniform improvement to analysing how mitigation effectiveness differs across datasets with varying levels of grounding. This allowed the project to produce more meaningful and realistic insights.

In addition to technical adjustments, minor changes were also made to project scheduling. While the original plan allocated most of the dissertation writing to the final stage, it became necessary to start writing earlier in parallel with experimentation. This helped reduce time pressure towards the end of the project and ensured that results and analysis could be documented progressively.

Overall, these adjustments reflect an adaptive development process, where both technical design and project planning were refined based on empirical observations. This flexibility was essential in ensuring that the project remained both achievable and academically rigorous.

7.5 Reflection

This project provided valuable insights into the importance of effective project management in research-oriented system development. Beyond the technical implementation, the project required careful planning, continuous adjustment, and the ability to manage uncertainty arising from both experimental and methodological factors.

One key lesson learned is the importance of structuring the project into well-defined phases with clear milestones. The use of a phased timeline helped ensure that foundational tasks, such as literature review and system design, were completed before moving on to implementation and evaluation. This contributed to a more organised and manageable workflow throughout the project.

Another important takeaway is the need to design systems in a flexible and adaptive manner. Initially, the project aimed to develop a unified mitigation strategy applicable across all datasets. However, it became evident that differences in task formulation and grounding conditions required dataset-specific adaptations. This highlighted the importance of iterative refinement and responsiveness to empirical findings, rather than strictly adhering to the original plan.

The project also emphasised the challenges of working with stochastic models such as large language models. The variability of model outputs made it necessary to adopt robust evaluation strategies, such as Monte Carlo experiments, to ensure reliable and reproducible results. This experience reinforced the importance of incorporating stability considerations into both experimental design and project planning.

In terms of limitations, some aspects of project planning could have been improved. In particular, the complexity of mitigation in open-ended tasks such as TruthfulQA was underestimated in the early stages. As a result, more time than expected was required to analyse and interpret experimental results. Additionally, balancing system development and dissertation writing proved challenging, especially in the later stages of the project.

If the project were to be conducted again, several improvements could be made. First, more extensive pilot experiments could be conducted at an earlier stage to better understand dataset-specific challenges and guide design decisions. Second, a more modular system design could be adopted from the outset to facilitate easier adaptation across different tasks. Finally, initiating evaluation and writing activities earlier in parallel with development would help reduce time pressure and improve overall efficiency.

Overall, this project demonstrates that successful system development requires not only technical expertise but also effective project management. The ability to plan, adapt, and critically reflect on both successes and limitations is essential for conducting rigorous and impactful research.

Chapter 8

Reflection and Future Work

8.1 Summary of Contributions

This project proposes a multi-signal framework for hallucination detection, estimation, and mitigation in large language models. The framework integrates multiple complementary signals, including LLM-as-a-judge, NLI-based factual consistency, self-consistency, and auxiliary signals, to provide a more robust estimation of hallucination risk compared to single-signal approaches.

A key contribution of this work is the design of a closed-loop pipeline, where hallucination detection is directly integrated with mitigation. The estimated reliability score is used to guide detector-conditioned mitigation strategies, enabling adaptive intervention based on confidence levels. This allows the system to preserve high-confidence answers while selectively correcting low-confidence ones, thereby reducing unnecessary modifications while focusing on high-risk cases.

Furthermore, the proposed framework introduces a unified scoring mechanism that aggregates heterogeneous signals into a continuous reliability score. This score supports both binary hallucination classification and fine-grained risk estimation, providing a principled basis for threshold-based decision-making and mitigation control.

The framework is evaluated across three benchmark datasets, namely TruthfulQA, FEVER, and Natural Questions, covering diverse settings including open-ended generation, evidence-based verification, and extractive question answering. Experimental results demonstrate that the proposed approach achieves strong performance in structured, evidence-grounded scenarios and improves robustness through multi-signal integration, while highlighting the challenges of hallucination mitigation in open-domain settings.

Overall, this project contributes a unified, prompt-engineering-based and closed-loop solution for hallucination detection and mitigation, emphasising the importance of combining complementary signals and adaptive control strategies to improve the reliability and stability of large language models across diverse tasks.

8.2 Critical Reflection

While the proposed framework demonstrates promising results, several important observations can be made regarding its effectiveness and design choices.

First, the framework performs significantly better on structured datasets such as FEVER, where explicit evidence provides strong grounding for both detection and mitigation. This confirms that NLI-based reasoning and constraint-based filtering are highly effective when reliable external information is available. In particular, the use of entailment–contradiction scoring and contradiction-aware filtering in the NLI module plays a critical role in eliminating unsupported or conflicting candidate answers.

However, performance on open-ended datasets such as TruthfulQA remains relatively limited. In these settings, the absence of explicit grounding introduces substantial ambiguity, making it difficult for both the detector and the mitigation strategy to reliably distinguish correct answers from plausible but incorrect alternatives. Since the framework relies primarily on semantic judgment (LLM-as-a-judge) and self-consistency signals, the detector may assign high confidence to internally consistent but factually incorrect answers. This reflects a key limitation that consistency does not necessarily imply factual correctness. This highlights a fundamental limitation of post-hoc mitigation approaches when operating without external knowledge.

Another key observation is the strong dependency on candidate quality. In many failure cases, the mitigation strategy does not fail due to incorrect scoring, but rather because the correct answer is not present in the candidate pool. This is especially evident in the candidate generation stage, where prompt-based rewriting or extractive heuristics may fail to produce factually correct alternatives, limiting the effectiveness of downstream filtering and reranking. This indicates that the overall system performance is bounded not only by the detection model, but also by the quality and diversity of the generated candidates.

Furthermore, the effectiveness of the detector-conditioned policy depends on the reliability of the aggregated score. In cases where different signals (e.g., judge, NLI, and self-consistency) produce conflicting indications, the weighted aggregation may lead to unstable or suboptimal decisions. This suggests that more advanced aggregation or calibration strategies may be required to better balance competing signals.

From a system design perspective, the integration of multiple signals improves robustness and stability, as different components capture complementary aspects of hallucination. However, this also increases system complexity and computational cost, requiring multiple model calls and additional processing steps, including repeated LLM inference, NLI evaluation, and SelfCheck-based consistency estimation across multiple sampled answers. This trade-off between performance and efficiency represents an important consideration for practical deployment.

8.3 Limitations

Despite the promising results, several limitations of this work should be acknowledged.

First, the framework relies on pre-generated candidate answers, which limits its ability to correct errors when no correct candidate is available. As a result, the mitigation process is inherently constrained by the quality of the initial generation step. In particular, candidate answers are produced using prompt-based rewriting or extractive heuristics, which do not guarantee coverage of the correct answer space. When the correct answer is absent from the candidate pool, the subsequent filtering and reranking stages cannot recover it, regardless of the effectiveness of the detection model.

Second, the effectiveness of the detection model is highly task-dependent. While strong performance is achieved on structured datasets such as FEVER, improvements are limited on open-ended tasks such as TruthfulQA, where ground truth is less well-defined and explicit evidence is unavailable.

In particular, the framework exhibits reduced effectiveness in open-domain question answering settings. In such scenarios, the absence of explicit grounding information makes

it inherently difficult to reliably distinguish between correct answers and plausible but incorrect responses. Unlike structured datasets such as FEVER, where external evidence provides a reliable reference for verification, open-domain tasks rely heavily on implicit world knowledge and model-generated candidates.

As a result, the detector may assign high confidence scores to semantically fluent but factually incorrect answers, especially when LLM-as-a-judge and self-consistency signals agree. This leads to error propagation into the mitigation stage, where incorrect candidates may be preserved or even preferred during reranking.

Third, the use of LLM-based evaluation introduces inherent uncertainty. The judge model itself may exhibit bias, inconsistency, or overconfidence, which can propagate errors into both detection and mitigation stages. In this project, the judge component relies on an instruction-tuned LLM, which is sensitive to prompt design and may produce unstable judgments across different inputs or sampling conditions.

Furthermore, the aggregation of multiple signals introduces additional complexity. Since different signals (e.g., LLM-as-a-judge, NLI-based factuality, and self-consistency) may produce conflicting indications, the weighted combination used in the scoring function may not always reflect the true factual correctness of the answer. This limitation highlights the challenge of designing reliable multi-signal aggregation mechanisms.

Finally, the computational cost of the proposed framework is relatively high. Multi-signal aggregation, multi-sample generation, and repeated model evaluation significantly increase runtime, which may limit scalability and real-world applicability. In particular, the use of multiple sampled answers for self-consistency estimation, repeated NLI inference for entailment and contradiction scoring, and iterative candidate generation in the mitigation stage all contribute to increased computational overhead.

Overall, these limitations highlight that hallucination mitigation remains a fundamentally challenging problem, particularly in open-domain settings where reliable grounding signals are unavailable.

8.4 Future Work

Several directions can be explored to further improve the proposed framework.

First, future work could integrate retrieval-augmented generation (RAG) to provide external knowledge grounding, particularly for open-domain tasks such as TruthfulQA. Incorporating reliable evidence would address the current limitation where the framework relies primarily on internal signals, and is therefore prone to selecting semantically plausible but factually incorrect answers. By combining retrieval with the existing multi-signal detector, it would be possible to further improve both detection accuracy and mitigation effectiveness.

Second, improving the quality and diversity of candidate answers represents an important direction. Since the current mitigation framework is constrained by the candidate pool, future work could explore more advanced generation strategies, such as guided decoding, iterative refinement, or diversity-aware sampling techniques. These approaches could increase the likelihood of including correct answers in the candidate set, thereby enhancing the effectiveness of downstream filtering and reranking.

Third, future research could investigate more advanced multi-signal aggregation and calibration methods. In the current framework, signals are combined using manually designed weighting schemes, which may not fully capture complex interactions between signals. Learning-based aggregation methods, such as trainable scoring models or uncertainty-aware calibration techniques, could provide more reliable and adaptive estimation of hallucination risk.

Fourth, improving the efficiency of the framework is an important practical direction. The current system requires multiple model calls, including repeated LLM inference, NLI evaluation, and self-consistency estimation, which introduces significant computational overhead. Future work could explore lightweight approximations, signal pruning strategies, or parallelised implementations to reduce runtime while maintaining performance.

Finally, the framework could be extended to real-world applications, such as conversa-

tional agents or decision-support systems. In such settings, balancing reliability, efficiency, and user experience will be critical. In addition, incorporating user feedback or interactive correction mechanisms could enable the system to continuously improve its performance and better adapt to real-world usage scenarios.

Chapter 9

Conclusion

This project investigated the problem of hallucination detection and mitigation in large language models and proposed a unified multi-signal and closed-loop framework. The approach integrates complementary signals, including LLM-as-a-judge, NLI-based factual consistency, self-consistency, and auxiliary signals, to estimate the reliability of generated answers and guide adaptive mitigation strategies.

A key contribution of this work is the design of a detector-conditioned pipeline, where hallucination detection is not treated as a standalone task, but is directly integrated with answer correction. By using a unified reliability score to control candidate filtering, reranking, and intervention strength, the framework enables more effective and interpretable mitigation compared to conventional post-hoc approaches.

Experimental evaluation across three benchmark datasets (TruthfulQA, FEVER, and Natural Questions) demonstrates that the proposed framework achieves strong performance in structured, evidence-grounded settings, particularly in FEVER, where explicit evidence enables effective NLI-based verification and filtering. In contrast, improvements in open-ended settings such as TruthfulQA remain limited, highlighting the challenges of hallucination mitigation in the absence of reliable grounding information. These results indicate that the effectiveness of hallucination mitigation is highly dependent on the availability of external evidence and the quality of generated candidates.

Overall, this project shows that combining multiple complementary signals within a closed-loop framework is a promising direction for improving the reliability of large language models. At the same time, it highlights that hallucination remains a fundamentally challenging problem, particularly in open-domain scenarios, and further research is required to develop more robust, efficient, and generalisable solutions.

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Appendix A

Prompt Templates

This section presents the prompt templates used in the proposed framework for answer generation, hallucination detection, and mitigation. These prompts are designed to enforce structured outputs and guide the behaviour of the language model without requiring fine-tuning.

1.1 Base Generation Prompt

A general prompt template is used across all datasets to ensure structured output:

General Base Prompt

You are a knowledgeable assistant. Answer the following question concisely and accurately.

Question: {question}

Provide your final answer in the format: FINAL_ANSWER: your answer

However, dataset-specific constraints are applied to align with task requirements:

1.1.1 TruthfulQA (Generation Setting)

TruthfulQA Generation Prompt

Question: {question}

Answer briefly and directly.

Output exactly: FINAL_ANSWER: answer

In addition to base generation, multiple prompt templates are used for answer refinement during mitigation. These include a mild editing prompt that preserves original wording while correcting errors, and a strict correction prompt that removes misconceptions and allows abstention (“I do not know”) when uncertainty is high.

1.1.2 FEVER (Verification Setting)

FEVER Prompt

Determine whether the claim is SUPPORTED, REFUTED, or NOT ENOUGH INFO based on the given evidence.

Claim: claim

Evidence: evidence

You MUST output in this format:

- LABEL: SUPPORTED / REFUTED / NOT ENOUGH INFO
- REASON: one short sentence based ONLY on the evidence

If the evidence is insufficient, choose NOT ENOUGH INFO.

1.1.3 Natural Questions (Extractive Setting)

NQ Prompt

You are a short-answer QA system.

Question: {question}

Context: {context}

Rules:

- Answer using only information from the Context
- Return the SHORTEST correct short answer
- Prefer 1-3 words
- Do NOT return a sentence
- Do NOT explain
- Do NOT include extra context

Output exactly one line:

FINAL_ANSWER:

Although the prompt encourages short, context-grounded answers, it does not explicitly require the output to be an exact contiguous span from the context. In the implementation, extractive behaviour is further enforced during post-processing through span-based candidate generation, answer compression, context alignment, candidate filtering, and detector-based reranking. Specifically, candidate spans are generated from the context

using token-overlap heuristics, validated against the context, and combined with a small number of short rewritten answers under relaxed constraints. Therefore, the Natural Questions setting is implemented as an extractive-dominant hybrid pipeline rather than a strictly prompt-constrained extractive QA system.

1.2 LLM-as-a-Judge Prompt

To provide a semantic-level evaluation signal, we employ a Large Language Model (LLM) as an external factuality judge. This component evaluates whether a generated answer is correct with respect to task-specific references, capturing semantic equivalence beyond surface-level matching.

Judge Prompt

You are a strict factuality evaluator.

- Minor wording differences are acceptable
- Partial match is acceptable if core meaning matches

Mark True if:

- The model answer is semantically correct with respect to the reference,
- AND it contains no false or contradictory statements.

Extra correct information is allowed and should NOT cause False.

Mark False if:

- the answer fails to capture the reference meaning,
- OR contains any factual error or contradiction.

Input:

Question / Claim: {question}

Reference: {reference}

Model Answer: {answer}

Respond with exactly one token: True or False.

The judge is implemented using a separate instruction-tuned LLM (`qwen2.5:7b-instruct`) to avoid interference with the generation model. The output is parsed as a binary signal, where `True` indicates that the model output is factually correct with respect to the reference under the task-specific evaluation criteria, and `False` indicates a hallucinated or incorrect output.

In the implementation, the judge output is converted into a scalar score:

$$\text{Score}_{\text{judge}} = \begin{cases} w_{\text{judge}} & \text{if judge=True} \\ 0.0 & \text{if judge=False} \end{cases}$$

where w_{judge} is a task-dependent weight (e.g., 0.8 or 1.0) that reflects the relative importance of the judge signal in the overall scoring function.

This scoring scheme directly corresponds to the implementation, where a positive judge decision contributes a fixed value to the overall hallucination score, while negative decisions contribute zero.

Dataset-specific Adaptation

Although a unified judge framework is used, the evaluation logic is adapted to match each dataset’s task formulation.

TruthfulQA

For open-ended truthfulness evaluation, answers are assessed using a multi-signal framework that integrates LLM-based judgment, natural language inference (NLI), and lexical matching. The judge evaluates whether the response avoids common misconceptions and follows truthful reasoning patterns.

Crucially, NLI is applied to ensure the generated answer is supported by correct references while not aligning with known incorrect answers, thereby explicitly penalizing misleading or false content. Unlike NQ, semantic similarity alone is insufficient; factual correctness is strictly enforced with both positive and negative evidence constraints.

FEVER

For fact verification, correctness is determined by both label matching and evidence consistency. Specifically, the predicted label must match the ground-truth label, and the answer must not contradict the provided evidence.

This corresponds to the implementation:

```
fever_gold_ok(answer, gold_label, evidence)
```

where label correctness is enforced by exact matching, and evidence consistency is verified using a separate NLI model to detect contradictions.

Additionally, an LLM-based judge:

```
judge_truth(question, answer, gold_label, evidence)
```

is used as an auxiliary signal in the overall hallucination scoring, rather than as the sole correctness criterion.

Natural Questions (NQ)

For extractive question answering, the judge evaluates whether the generated answer preserves the semantic meaning of the reference short answer while ensuring that no false or contradictory information is introduced. This corresponds to the implementation:

```
judge_truth(question, answer, reference)
```

where correctness is determined by both semantic equivalence and the absence of factual errors, rather than exact string matching.

Across all datasets, the judge operates as a task-agnostic semantic evaluator, while task-specific constraints are injected through prompt conditioning and input structure. This design ensures consistency of the evaluation signal while maintaining compatibility with heterogeneous benchmarks.

The use of LLM-as-a-Judge allows the system to capture semantic correctness that cannot be reliably measured using exact match or token-level overlap alone, especially for paraphrased or partially matching answers.

1.3 Mitigation Prompts

To reduce hallucinations in generated answers, a prompt-engineering-based mitigation strategy is applied in a closed-loop framework. Unlike simple one-shot prompting, the system dynamically selects different rewriting prompts based on the estimated hallucination risk, forming a detector-conditioned correction pipeline.

1.3.1 Overall Strategy

Given an initial answer, the system first computes a calibrated correctness score using the hallucination detector. Based on this score, three levels of mitigation are applied:

1. low-risk answers are kept unchanged
2. medium-risk answers undergo mild rewriting
3. high-risk answers are corrected using stricter prompts with multiple candidate generations and reranking

This design ensures that corrections are only applied when necessary, reducing unnecessary modifications to already correct answers.

1.3.2 Mild Correction Prompt

For moderately risky answers, a conservative editing prompt is used:

- preserve as much of the original wording as possible,
- modify only incorrect or misleading parts,
- avoid introducing new information.

This corresponds to a minimal intervention strategy, aiming to fix local factual errors without altering the overall structure of the answer.

1.3.3 Strict Correction Prompt

For high-risk answers, a stronger prompt is applied to enforce factual correctness:

- remove false or speculative content,
- explicitly correct misconceptions in the question,
- prefer concise and direct answers,
- abstain (“I do not know”) when uncertainty remains.

Compared to mild rewriting, this prompt allows more aggressive modification and supports abstention to avoid hallucinated outputs.

1.3.4 Dataset-Specific Adaptation

TruthfulQA

For TruthfulQA, mitigation explicitly targets common misconceptions and aims to correct semantically incorrect answers. Unlike NQ, answers often require rewriting rather than extraction, making semantic correction essential.

In the implementation, mitigation is realized through a candidate-based correction process. Multiple candidate answers are generated using a combination of mild rewriting, strict correction, and edit-based refinement prompts.

These candidates are subsequently filtered to remove low-quality outputs (e.g., overly long or uninformative responses), and grouped into answer and abstention types. The final selection is performed using detector-based scoring and self-consistency signals, as described in the main methodology.

Additionally, an abstention mechanism allows the system to output “I do not know” when all candidates are considered unreliable.

FEVER

For FEVER, mitigation is tightly coupled with evidence-grounded verification. Prompts explicitly constrain the model to rely only on the provided evidence and to produce structured outputs with a label and a short justification.

In addition, optional span-based evidence extraction may be applied to further encourage grounding in specific textual fragments.

Further consistency is enforced through model-based filtering and reranking, as described in the main methodology, including contradiction detection via NLI and label-consistent grouping.

Natural Questions (NQ)

For NQ, mitigation follows an extractive-dominant strategy that emphasizes short, context-grounded answers. Candidate answers are primarily derived from short spans in the local

context using question–context overlap heuristics.

These extractive candidates may be supplemented by a small number of short rewritten answers under relaxed constraints. All candidates are filtered based on length and context alignment, and the final selection is performed using detector-based scoring and self-consistency signals, as described in the main methodology. In the current implementation, abstention is not enabled for NQ.

Appendix B

Example Outputs

Appendix C

Additional Results